STORAGE, RETRIEVAL, AND EVIDENCE ACCESS FOR LONG-TERM CONVERSATIONAL MEMORY

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Chapter 1. Introduction

1.1 Research Background

1.1.1 From Single-Turn Question Answering to Long-Term Interactive Agents

Large language models (LLMs) are increasingly used as agents that perform tasks, call tools, and interact with users over time. In personal assistance, collaborative work, customer service, and similar settings, users reveal preferences, relationships, events, and plans across many sessions. An effective agent must therefore understand the current request and access information introduced much earlier. LoCoMo frames this challenge as long-term conversational understanding, while MemoryAgentBench emphasises incremental memory in multi-turn environments [1, 2].

Long-term interaction is difficult for more than one reason. Relevant facts may be scattered across sessions, expressed in different words, or updated by later events. A question may also require several pieces of evidence. For example, a user may propose a plan, revise its date, and later report the outcome. Retrieving only the first statement can produce an outdated answer. Retrieving all statements is still insufficient if the model cannot resolve entities, temporal order, and the current state. LoCoMo includes single-hop, multi-hop, and temporal questions, which capture these different requirements [1].

Long-term agents therefore need mechanisms to preserve, update, locate, and use historical information. Many systems provide these capabilities through external memory. Retrieval-augmented generation combines a parametric generator with retrievable non-parametric memory [3]. Filesystem-based memory instead treats files, memory-management agents, search agents, and tool interfaces as parts of a persistent memory system [4]. Reliable memory is important because an agent that cannot access past information may contradict earlier statements, ignore user preferences, or rely on outdated facts.

1.1.2 A Long Context Window Is Not a Long-Term Memory System

A direct approach is to place the entire conversation history in the model context. This preserves the raw source and avoids omissions caused by an explicit retrieval step. However, information that fits in the context window is not always used reliably. Long-context performance can vary with the position of relevant evidence, especially when it appears in the middle of the input [5]. Context capacity and effective memory are therefore not equivalent.

As the history grows, Full Context increases the amount of text that the Answerer must process on each query. This can raise token use, latency, and computation. Raw conversations contain repetition, irrelevant dialogue, similar entities, conflicts, and outdated states. A long-term memory system must address selection, organisation, updating, provenance, and auditability, not only input capacity. Filesystem-Based Memory and ReFind address these needs through maintainable file memory and controlled search over raw chat logs, respectively [4, 6].

This study therefore uses Full Context as a **diagnostic baseline without explicit retrieval selection**. It is not a storage representation, an additional experimental condition in the selective-retrieval matrix, or an oracle. It exposes the complete raw source to the answering model, referred to as the Answerer, but remains subject to long-context localisation, noise, conflict, and reasoning failures [1, 5].

1.1.3 External Memory as a Storage–Retrieval–Utilisation System

External memory has three interdependent stages. **Storage** determines how history is represented. The same conversations may be kept as flat raw records, organised into topic directories, or consolidated into entity-, event-, or topic-centred memory. These choices trade off source fidelity, navigation, compression, write cost, and maintenance. Their value also depends on the tools and agents used to access them [4].

Retrieval selects evidence for the current question. An agent may browse directories, search for terms, and inspect files. A lexical method such as BM25 may instead rank conversation units, supported by query reformulation, context expansion, temporal constraints, and iterative search. ReFind shows how an agent can use iterative lexical search over raw chat logs without first constructing a semantic memory structure [6]. Narrow retrieval may miss evidence, whereas broad retrieval may introduce noise or conflict.

Evidence utilisation occurs when the Answerer produces the final response. Relevant evidence does not guarantee a correct answer. The model may fail to resolve references, order events, combine facts, or identify the latest state. Recent diagnostic work distinguishes retrieval-stage failures from downstream utilisation failures, but groups missing stored detail and missed retrieval under the former category [7]. This study extends that separation by checking stored content, retrieved evidence, evidence packaging, and answer generation as distinct points in the end-to-end system.

1.2 Problem Statement

1.2.1 Storage: Fidelity, Organisation, and Compression

Raw logs preserve wording, temporal cues, and dialogue context, but distribute relevant facts across many files and turns. They reduce information loss at write time while increasing the search burden at query time. Topic directories can provide useful navigation cues. Organised stores may require more navigation steps while reducing the amount of text read, and they do not improve answer quality uniformly [4].

Curated memory can consolidate related facts and remove repetition. It can reduce the text read at query time, but curation is a lossy transformation. It may condense away details, flatten temporal updates, or leave stale states unreconciled [4]. The first problem is therefore whether storage organisation improves retrieval and answering when the source data are fixed, and whether the effect depends on the retrieval system.

1.2.2 Retrieval: Coverage, Noise, and Cost

Retrieval must collect sufficient evidence within a limited context and cost budget. Agent-led file search can adapt after each observation, but it requires repeated model calls and may follow an early wrong decision. Lexical retrieval is efficient and auditable, but depends on query formulation, index units, ranking, and context expansion. ReFind shows that these surrounding design choices affect retrieval from raw conversations [6].

More retrieved text may improve coverage but also add duplicates and conflicts. Less text may reduce noise but remove a critical condition. Retrieval should therefore be assessed through gold-source coverage and the evidence retained in the EvidenceBundle, not only through final answer quality. LoCoMo provides source turn identifiers that support this separation [1]. The broader question is whether retrieval-system effects remain stable across storage representations or form a storage × retrieval interaction.

1.2.3 Utilisation: Visible Evidence Does Not Guarantee a Correct Answer

An Answerer may still fail when the EvidenceBundle contains the required information. Common demands include entity resolution, temporal ordering, multi-hop composition, and separation of plans from completed events [1]. Retrieval sufficiency and evidence use must therefore be diagnosed separately [7].

Full Context supports the same distinction. If it repairs an error made by selective retrieval, the likely bottleneck may be access or evidence budget. If selective retrieval succeeds while Full Context fails, selection may have improved focus or reduced noise. If both fail, the cause may lie in reasoning, interpretation, or data ambiguity [5, 7]. Backbone models may differ in both search behaviour and evidence use, so an architecture observed with one backbone model may not transfer unchanged to another.

1.2.4 Evaluation: One Final Score Is Insufficient

LoCoMo evaluates question answering with normalised token-level partial-match F1 [1]. This study reports that official-style F1 together with project-defined, aligned token-overlap Precision and Recall as diagnostic measures. The aggregates are means of per-question values, so aggregate Precision and Recall should not be used to recompute aggregate F1. Exact normalisation and category-specific procedures are deferred to Chapter 4, where the evaluator implementation is documented. These metrics are reproducible, but remain sensitive to wording and answer length [1].

An LLM judge can complement lexical overlap with semantic assessment, but prior work documents position, verbosity, and self-enhancement biases, as well as limited reasoning ability [8]. This study therefore uses an independent LLM judge under a fixed rubric and reports Strict Accuracy, Partial Rate, and Soft Semantic Score alongside the deterministic metrics. None is treated as a complete measure of answer quality.

For retrieval, Evidence Recall is the fractional coverage of annotated gold source turns. Any-hit and All-hit capture binary source-access outcomes. These evidence measures apply only to the 58 evidence-eligible Main-60 questions and are not applicable to Full Context, which performs no retrieval. Process evaluation also includes model calls, input and output tokens, latency, construction cost, and maintenance burden.

The studies reviewed here do not provide one controlled comparison that jointly covers storage organisation, retrieval strategy, backbone model, dual-source fusion, and Full Context under shared data and evaluation [4, 6, 7]. This study addresses that combined gap and traces outcome differences to evidence access, evidence packaging, evidence utilisation, and cost.

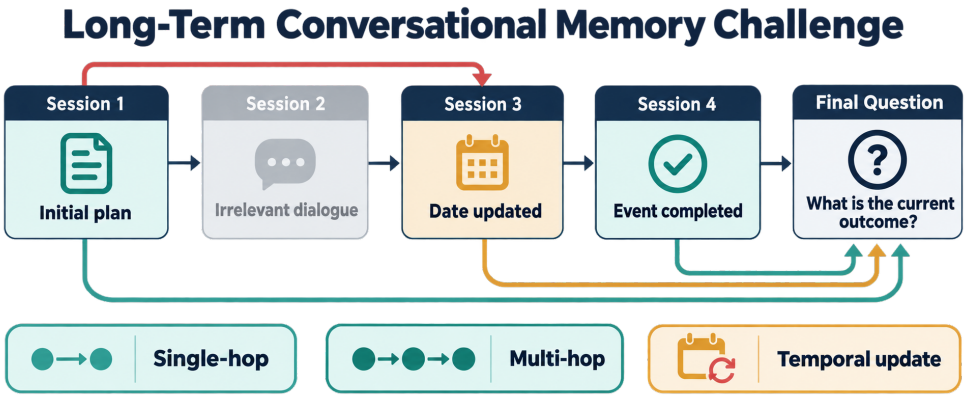


Figure 1.1. Long-term conversational memory challenge: relevant evidence may be distributed across sessions, revised over time, and surrounded by irrelevant dialogue.

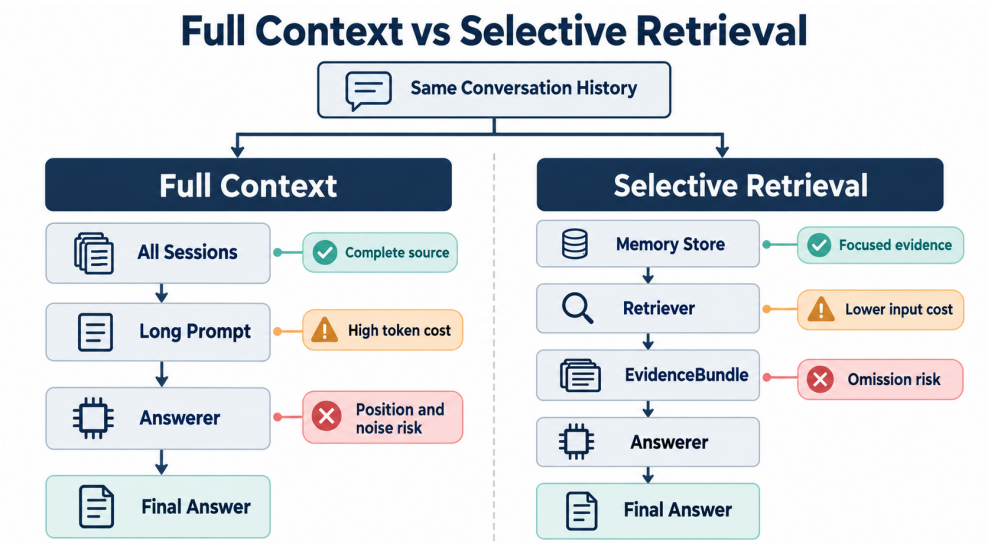


Figure 1.2. Full Context preserves complete source visibility but increases context length and noise, whereas selective retrieval provides focused evidence at the risk of omission.

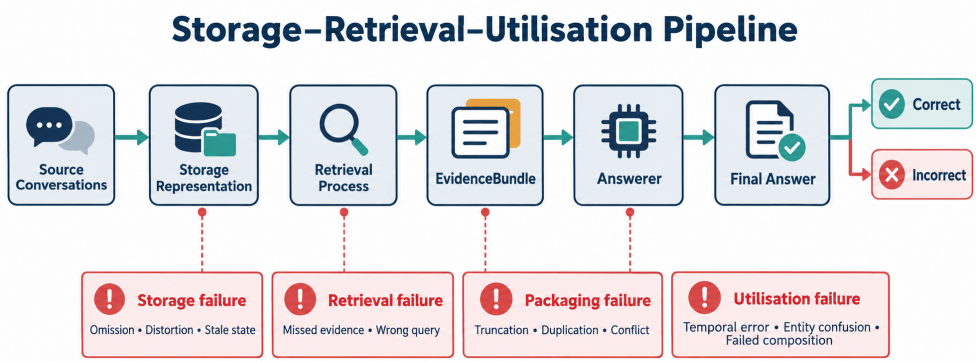


Figure 1.3. End-to-end storage–retrieval–utilisation pipeline and the principal failure points examined in this study.

1.3 Research Landscape and Gaps

1.3.1 Existing Memory Representations

Research on long-term agent memory has produced a wide range of representations. Retrieval-augmented generation introduced the general pattern of combining a parametric generator with externally retrievable, non-parametric memory [3]. Subsequent systems have represented memory as extracted facts, summaries, graph structures, linked notes, discourse units, embedding-organised indexes, raw interaction records, and hierarchical files [4]. Long-conversation benchmarks such as LoCoMo and incremental-agent evaluations further show why these representations matter: relevant information may be introduced across sessions, revised later, or require several pieces of evidence to answer one question [1, 2].

These representations make different trade-offs. Raw conversations preserve wording, provenance, temporal cues, and surrounding dialogue, but leave relevant facts distributed across many turns. Topic-based files add navigational structure without rewriting the source. Curated memory can consolidate related material and reduce repetition, but curation is a lossy operation that may omit qualifiers, weaken temporal relationships, or preserve an outdated state. Filesystem-Based Memory treats organisation, management agents, search agents, tool interfaces, cost, and store health as parts of one persistent-memory design space [4]. Its analysis also shows that a well-organised store should not automatically be assumed to produce better answers: the value of a representation depends on how it is searched and used.

1.3.2 Retrieval Paradigms

Memory representations are accessed through several retrieval paradigms. The simplest diagnostic is Full Context, in which the complete raw history is placed in the model input. This removes explicit retrieval selection but does not guarantee that the model will locate and use the relevant evidence. Long-context performance can depend on evidence position, with information in the middle of an input often used less reliably than information near its boundaries [5]. Full Context also increases input size and exposes the Answerer to irrelevant, repeated, or conflicting material.

Selective retrieval attempts to reduce this burden. Agent-led filesystem search allows a model to inspect directories, formulate search terms, open files, and revise its search after each observation [4]. ReFind instead uses an agent-controlled iterative lexical process over raw chat logs, combining query formulation, ranked retrieval, local context expansion, temporal constraints, and stopping decisions before a separate answering stage [6]. These approaches differ not only in search algorithm but also in the actions available to the agent, the units being ranked or viewed, the number of iterations, and the evidence finally passed to the Answerer. Retrieval quality must therefore be analysed as a process rather than treated as a single interchangeable component.

1.3.3 Evaluation Practices

Evaluation has likewise moved beyond one final accuracy number. LoCoMo uses normalised token-level partial-match F1 and provides answer-source turn identifiers for questions with annotated evidence [1]. Token overlap is transparent and reproducible, but it is sensitive to wording and answer length [1]. Semantic evaluation with an LLM judge can recognise meaning beyond exact token overlap, yet LLM judges may exhibit position, verbosity, and self-enhancement biases, as well as limited reasoning ability [8]. LLM judge scores should consequently complement rather than replace deterministic metrics.

A second distinction concerns where failure occurs. Recent diagnostic work distinguishes retrieval-stage failures from downstream utilisation failures and hallucinations, although its retrieval-stage category includes both missing stored detail and missed retrieval [7]. This study further separates storage fidelity, evidence access, evidence packaging, and answer utilisation. Filesystem-Based Memory additionally measures search effort, token use, content read, and store health, showing the importance of evaluating operational cost alongside answer quality [4]. Together, these practices motivate a multi-level evaluation comprising answer quality, gold-evidence coverage, process traces, and resource use.

1.3.4 Unresolved Gaps

Four gaps remain. First, emerging studies have begun to isolate memory shape [4] or cross write and retrieval strategies [7], but many end-to-end system comparisons still vary representation and access procedures jointly [6]. A result obtained with a curated store and a specialised retriever cannot reveal whether the gain came from representation, search procedure, their interaction, or the Answerer. Cross-paper comparisons are especially weak for this purpose because datasets, prompts, tools, budgets, and models also differ.

Second, the backbone model is itself a source of variation. Model capability can change query formulation, tool use, stopping behaviour, memory maintenance, and downstream outcomes [4]. An apparent property of a memory architecture may therefore depend on the Retrieval Agent or Answerer used to test it.

Third, the studies reviewed here do not establish the per-question paired diagnostic between selective retrieval and Full Context used in this study [1, 4, 6, 7]. Complete visibility may repair retrieval omissions, while selective evidence may outperform a longer context by reducing noise. If both fail despite access to the same source, the bottleneck may instead lie in evidence utilisation or answer generation [5, 7].

Fourth, final rankings provide limited causal explanation. Similar answer scores can conceal different evidence coverage, token cost, latency, and failure stages. Practical recommendations therefore require question-level comparisons, evidence measures, process records, and manual case analysis, not only aggregate F1 or LLM judge scores.

1.3.5 Response of This Study

This study addresses these gaps through one controlled framework built on a unified Main-60 question set. It compares three storage representations—flat raw conversations, topic-organised source files, and agent-curated memory—and three retrieval processes—agent-led filesystem search, ReFind-style iterative lexical retrieval over raw conversation units, and an adaptation of that iterative process to curated memory. All selective-retrieval paths produce a standardised EvidenceBundle for the Answerer. Matched raw-memory and topic-organised conditions support cleaner storage and retrieval contrasts; the curated-memory conditions are evaluated as complete systems and are not interpreted as pure single-component ablations.

The same selective-retrieval configurations are run with Qwen and DeepSeek Retrieval Agents and Answerers while the stored snapshots remain fixed, enabling matched analysis across backbone models. A DeepSeek Full Context run supplies a separate diagnostic comparison. Full Context performs no explicit retrieval selection and is therefore neither a storage representation, an eighth selective-retrieval condition, nor an oracle.

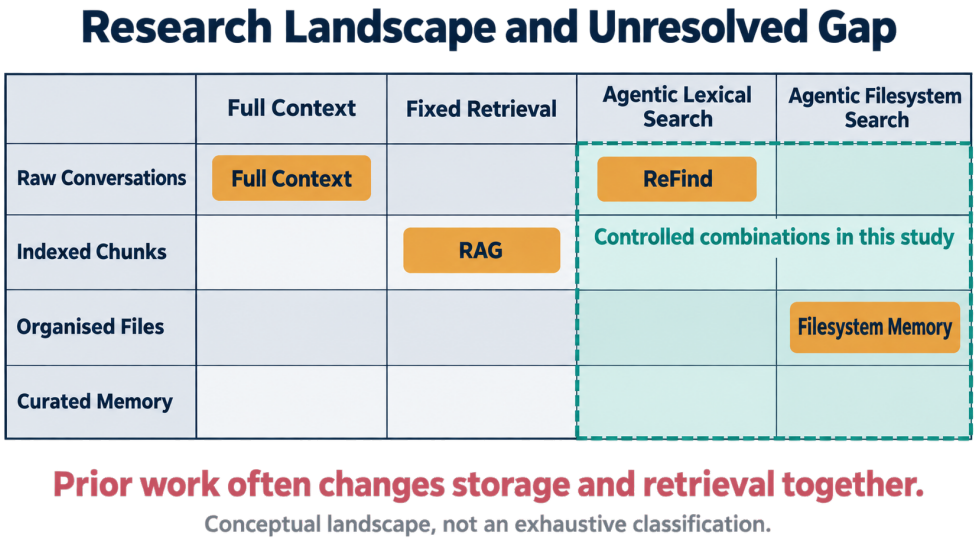


Figure 1.4. Conceptual synthesis of retrieval-augmented generation [3], organised and curated filesystem memory [4], ReFind-style raw-chat search [6], and controlled write–retrieval crossings [7]. The figure motivates cross-combination analysis rather than treating any cited system as one fixed storage–access pair.

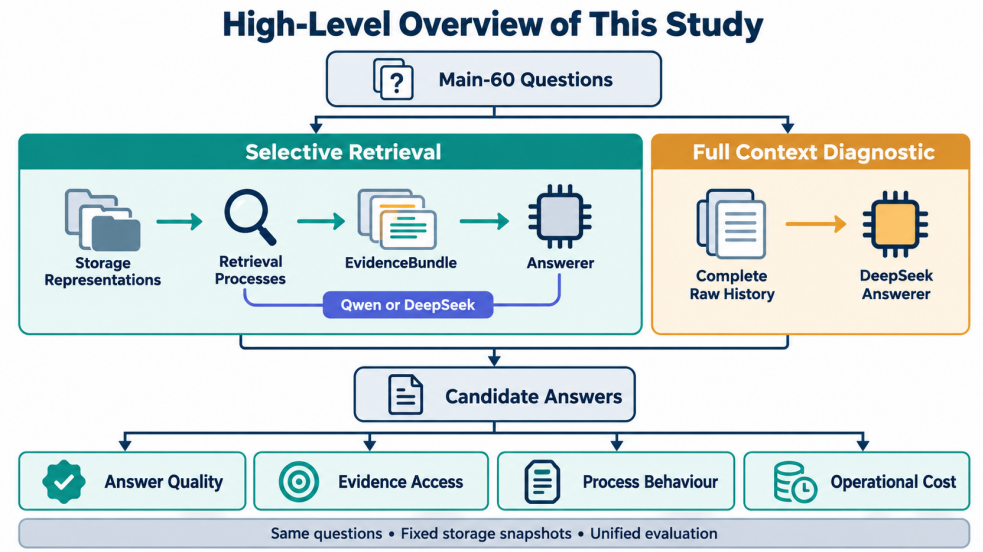


Figure 1.5. High-level overview of the study, including the selective-retrieval and Full Context paths and the four levels of evaluation.

Evaluation combines LoCoMo-style token F1 [1] with this study's aligned token Precision and Recall, semantic Strict Accuracy, Partial Rate, and Soft Semantic Score produced by an independent LLM judge under a fixed rubric [8], gold-source coverage for evidence-eligible questions, and process measures including model calls, tokens, latency, construction cost, and maintenance burden. Paired contrasts and manual review then trace likely failures to storage fidelity, retrieval access, evidence packaging, answer utilisation, or long-context processing. The purpose is not to identify one context-free winner, but to determine which conclusions are architecturally stable, which depend on the backbone model, and which system offers the most defensible quality–cost trade-off under different deployment priorities. Chapter 3 defines the system components and condition identifiers, while Chapter 4 specifies their implementation, controls, and evaluation protocol.

1.4 Research Objectives

1.4.1 Overall Objective

This study aims to evaluate long-term conversational memory under a controlled framework. It examines how storage representation, retrieval strategy, and backbone model capability affect answer quality, evidence coverage, and operational cost. Dual-source fusion and Full Context are used to diagnose system bottlenecks. The goal is to determine when each design is useful and whether its gains justify its cost.

1.4.2 Specific Objectives

The study has six objectives:

1. Construct three traceable storage representations from the same source data.
2. Connect agent-led file search, iterative lexical retrieval, and curated-memory retrieval to a common EvidenceBundle interface.
3. Compare the same selective-retrieval configurations with Qwen and DeepSeek Retrieval Agents and Answerers.
4. Use a DeepSeek Full Context diagnostic baseline to examine access and long-context utilisation.
5. Evaluate lexical quality, semantic quality, gold-evidence coverage, model calls, tokens, latency, and maintenance burden.
6. Develop a stage-based error analysis and derive bounded practical implications for different quality, cost, scalability, and auditability needs.

1.5 Research Questions

1.5.1 Architecture: RQ1

RQ1: What effects do storage representation and retrieval strategy have, and do they interact?

The experiment uses matched conditions that vary storage organisation while holding the retrieval approach fixed, and vary retrieval while holding storage fixed. These comparisons estimate storage effects, retrieval effects, and their interaction. Complete curated-memory systems are analysed separately because their outcomes cannot be attributed to one isolated component.

1.5.2 Robustness: RQ2

RQ2: Are these architectural effects stable across Qwen and DeepSeek?

The analysis compares the direction and magnitude of matched contrasts across the two backbone models. Consistency supports a more general architectural interpretation; a direct difference between model-specific effects indicates dependence on the backbone model.

1.5.3 Mechanisms: RQ3–RQ5

RQ3: When does dual-source fusion repair errors, and when does it introduce interference?

The dual-source condition reuses evidence from two existing retrieval paths and adds no new retrieval process. Its repair, preserve, harm, and unresolved outcomes are compared with those of the two single-source parent conditions.

RQ4: Does Full Context outperform selective retrieval, and what bottlenecks does each approach reveal?

The within-DeepSeek paired descriptive comparison tests whether complete source visibility repairs selective-retrieval failures and whether selected evidence can reduce noise. Full Context is not a storage representation, an additional selective-retrieval condition, or an oracle.

RQ5: Do errors mainly arise from storage fidelity, retrieval coverage, evidence packaging, answer utilisation, or long-context processing?

Stored content, retrieval traces, EvidenceBundles, gold evidence, candidate answers, and evaluation outputs are combined with manual review to locate the likely failure stage.

1.5.4 Decision-Making: RQ6

RQ6: Considering quality, cost, stability, and maintenance complexity, which conditions are most suitable for practical use?

RQ6 examines quality–cost trade-offs and stability across backbone models. It develops conditional guidance for quality-first, cost-sensitive, and audit-focused settings, while treating implications for continuously growing memory as tentative because sustained memory updates are outside the present experiment.

Research question	Main experimental material	Main evidence
RQ1	Controlled storage–retrieval comparisons and complete curated-memory systems	Paired quality differences, Evidence Recall, storage–retrieval interaction
RQ2	The same selective-retrieval configurations with two backbone models	Effect direction and magnitude across backbone models
RQ3	Two single-source conditions and their dual-source fusion condition	Repair, preserve, harm, unresolved, evidence complementarity and conflict
RQ4	DeepSeek selective retrieval and Full Context	Paired outcomes, answer quality, context size, and cost
RQ5	Storage, retrieval traces, EvidenceBundles, and answers	Stage-based errors, evidence conversion, and manual cases
RQ6	All formal conditions	Quality–cost Pareto analysis, cost amortisation, and decision matrix

Chapter 3 formally defines the system components and experimental condition identifiers. Chapter 4 presents the execution protocol, metric definitions, and statistical tests.

1.6 Research Contributions

1.6.1 Controlled Experimental Design

The study provides one storage–retrieval–answering framework with shared data, questions, storage snapshots, and evidence interfaces. Matched conditions support architecture contrasts, while separate integrated conditions evaluate complete curated-memory systems. This separation limits unsupported single-component claims and supports explicit analysis of storage–retrieval interaction.

1.6.2 Evidence Across Backbone Models and Full Context

The same selective-retrieval configurations are evaluated with Qwen and DeepSeek Retrieval Agents and Answerers while storage snapshots remain fixed. A DeepSeek-family Full Context diagnostic adds evidence about selective retrieval, complete source visibility, and long-context utilisation.

1.6.3 Multi-Level Evaluation and Error Diagnosis

The study combines LoCoMo-style token F1 [1] with this study's aligned token Precision and Recall, plus three semantic metrics defined for this study and produced by an independent LLM judge. The LLM judge follows the general semantic-evaluation paradigm described in prior work [8]. Gold-evidence measures and process and cost records provide further diagnostic evidence. Paired comparisons and manual case review distinguish failures to retrieve evidence from failures to use it correctly.

1.6.4 Practical Decision Support

Answer quality is considered together with model calls, tokens, latency, construction cost, maintenance burden, and stability across backbone models. The study provides bounded, conditional guidance rather than one unconditional system ranking.

1.7 Research Scope and Boundaries

1.7.1 Included Scope

The formal study uses the frozen Main-60 question set from LoCoMo `conv-50`. All 60 questions form one unified test set and cover the C1–C4 categories used in this work. The study uses only the question-answering material and relevant source annotations [1].

The experiments include three storage representations, three retrieval processes, seven selective-retrieval configurations, two backbone model configurations for the Retrieval Agent and Answerer, and one DeepSeek-family Full Context diagnostic. The evaluation covers token-level metrics and answer metrics produced by an LLM judge, evidence measures for the 58 eligible questions, model calls, tokens, latency, and maintenance cost. Retrieval traces, EvidenceBundles, and run audits support stage-based analysis.

1.7.2 Excluded Scope

This is not a full reproduction across all LoCoMo conversations. Other self-organising, fact-based, graph-based, and vector-RAG memory systems are not formal baselines. The study does not simulate months of continuous writing, active forgetting, conflict accumulation, or automatic reorganisation.

The backbone model comparison is not a full Retrieval-Agent × Answerer factorial design. The Retrieval Agent and Answerer change together, while the topic-organised and curated-memory snapshots remain fixed. It therefore measures dependence on the backbone model across the query-time pipeline, not the separate effect of each model role or the writing model. Full Context is evaluated only with DeepSeek, so it cannot support a conclusion across backbone models about Full Context prompting.

1.7.3 Limits on Interpretation

The matched raw-memory and topic-organised conditions provide the cleaner architecture contrasts, but retrieval-system comparisons still include their full search procedures. The integrated curated-memory conditions jointly change the memory representation and its corresponding retrieval process. They cannot support a claim about changing only storage or only retrieval.

Full Context performs no retrieval, but it is neither a new storage method nor an accuracy ceiling. Its comparison with selective retrieval is diagnostic rather than a strict causal ablation because the input mechanism and context scale differ. All conclusions are limited to the present question set, storage snapshots, models, and protocols. Broader generalisation requires further validation.

The remainder of this report is organised as follows. Chapter 2 reviews the relevant literature. Chapters 3 and 4 define the system and experimental methodology. Chapter 5 and Chapter 6 present and discuss the empirical findings. Chapters 7 and 8 address validity and future research directions, followed by the conclusion in Chapter 9.

Chapter 2. Background and Related Work

This chapter defines the concepts and prior work needed to study long-term conversational memory. It reviews memory representations, access methods, the two core system papers, the LoCoMo benchmark, and relevant evaluation methods. Exact system definitions, parameters, and formulas are deferred to Chapters 3 and 4.

2.1 Foundations of Long-Term Memory for LLM Agents

2.1.1 Parametric, Contextual, and External Memory

An LLM agent can obtain past information from three broad forms of memory. **Parametric memory** is encoded in model weights and is difficult to inspect or update with one local fact after deployment. **Contextual memory** is placed in the current input. It is immediately available, but bounded by the context window and the cost of long prompts. Models may also use different positions within a long input with unequal reliability [5].

External memory stores information outside the model as logs, files, summaries, indexes, vectors, or graphs. Relevant content can then be selected and returned to the model. This follows the retrieval-augmented principle of combining a parametric generator with an updateable, non-parametric source [3]. External memory can support provenance, local editing, and auditability, depending on its representation and tools [4, 9]. This study concerns text-centred external memory rather than weight editing or memory-policy training. Full Context uses the same external source but places it directly in the current input.

2.1.2 The Long-Term Memory Lifecycle

Long-term memory is a process rather than a static database. A conversation produces events, preferences, plans, corrections, and personal facts. A system must decide what to retain, how to represent it, and how to organise it by session, time, topic, entity, or relation. At query time, retrieval locates candidate evidence and an answering component uses it. Later maintenance may add entries, resolve conflicts, remove stale material, or reorganise the store. Incremental evaluation matters because memory agents receive information over time rather than as one fixed document [2].

The lifecycle therefore contains a feedback loop:

interaction → writing → organisation → retrieval → evidence use → answer → maintenance

An early error can propagate, while maintenance can change what later searches find. Chapter 3 maps this lifecycle to the study's concrete components.

2.1.3 Storage, Retrieval, and Utilisation as Distinct Problems

Three questions should be separated. **Storage fidelity** asks whether memory retains the required fact and its qualifiers. **Retrieval coverage** asks whether relevant material reaches the Answerer. **Utilisation** asks whether the Answerer correctly interprets and combines that material. A final score combines all three.

Consider an abstract question about the latest date of an event. The updated date may have been omitted during summarisation. This is a storage failure. It may be present in the store but absent from the retrieved evidence. This is a retrieval failure. It may also appear in the evidence while the Answerer still selects an older date. This is a utilisation failure. These cases require different remedies.

Recent diagnostic work crosses write strategies with retrieval methods and separately probes retrieval relevance and downstream memory utilisation [7]. Its failure taxonomy groups missing stored detail and missed retrieval under one retrieval-stage category; it does not independently measure write-time information loss. This study therefore uses a finer project-defined distinction among storage, retrieval, and utilisation while preserving the paper's retrieval-to-generation separation.

2.1.4 Principal Design Axes of Long-Term Memory Systems

Long-term memory systems can be organised along five design axes.

1. **Fidelity versus compression.** Raw records preserve detail, whereas summaries and extracted facts reduce size but may lose context.
2. **Flat versus hierarchical organisation.** A flat store is simple. A hierarchy may support navigation, but only if the search process can use it.
3. **Passive versus agent-controlled retrieval.** A fixed ranker returns one result list. An agent may reformulate queries, inspect paths, and stop when evidence is sufficient.
4. **Selective evidence versus Full Context.** Selection reduces noise and input size. Full Context avoids explicit search misses but creates a long-context utilisation problem.
5. **Quality versus total cost.** A system may move cost between memory construction, query-time search, answering, maintenance, and recovery.

These axes are not independent. Compression may lower query cost while increasing construction and fidelity risks; hierarchy helps only when access tools expose it; and Full Context removes explicit retrieval but not selection inside the model. Evaluation must therefore compare complete data paths as well as isolated components. The later chapters use these axes to distinguish clean contrasts from pipeline-level comparisons. This distinction also prevents a tidy store from being treated as evidence of better memory.

2.2 Main Representations of External Memory

2.2.1 Raw Dialogue and Log-Based Storage

Raw dialogue retains wording, speaker labels, session boundaries, dates, and turn order. It therefore preserves provenance and allows later reinterpretation without relying on an earlier summary. This matters when a qualifier, correction, or temporal cue becomes important.

Its weakness is dispersion. An answer may depend on turns from several sessions, while repetition, pronouns, and omitted context make isolated matches difficult to interpret. Raw logs still have minimal structure through session, turn, time, and speaker metadata, but they require effective search or full-archive processing.

2.2.2 Filesystem-Based Memory

A filesystem adds addressable structure to textual memory. Paths, filenames, folders, headings, and descriptions can act as retrieval cues. An agent can list directories, search text, inspect files, and follow new leads. Files are also editable and compatible with standard tools.

Filesystem-Based Memory defines a memory store as files under a rooted hierarchy and a tool harness as the operations through which agents access that store [4]. Store shape and access interface can therefore jointly influence what the agent finds.

Hierarchy is not automatically useful. A taxonomy can add maintenance work without improving answers if the searcher ignores it. Filesystem quality must be evaluated with navigation behaviour, tool constraints, and model capability.

2.2.3 Agent-Curated or Summarised Memory

Agent-curated memory rewrites interaction history into a compact form. A manager may extract facts, group events, merge repetition, or maintain topic notes. Mem0 uses extraction and consolidation, while A-MEM adds linked, evolving notes [10, 11].

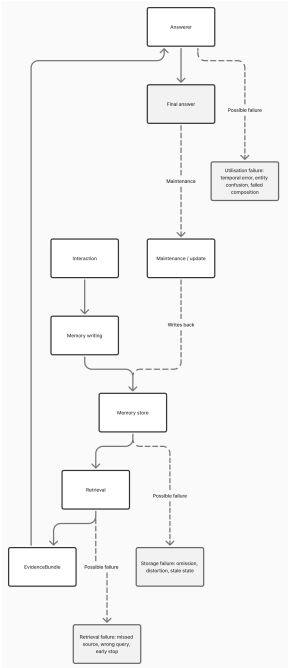


Figure 2.1. Long-term memory lifecycle and the project-defined separation of storage, retrieval, and utilisation failures. The retrieval-to-generation distinction is informed by [7], which does not independently measure write-time information loss.

Curation introduces another interpretation layer. It can surface stable facts, but may omit qualifiers, flatten temporal order, merge entities, or preserve stale states. Its value depends on the writing model, prompt, update policy, and construction and maintenance costs.

The study's curated store belongs to this broad family but remains a fixed experimental snapshot. Chapter 3 defines its construction and audit rules.

2.2.4 Vector, Graph, and Other Structured Memories

Vector memory encodes source units as embeddings and retrieves by semantic similarity. This extends the retrieval-augmented pattern of selecting information from non-parametric memory [3]. In practice, semantic similarity is not identical to task relevance, and a vector system also depends on choices such as source units, embedding model, query, and return depth. Graph memory represents entities, events, or notes as connected nodes. It supports relation traversal, but construction and conflict maintenance add assumptions and cost. Mem0 includes a graph-enhanced variant, while A-MEM links evolving notes [10, 11].

Hybrid systems combine several forms. MemGPT, for example, manages information across memory tiers [9]. Such systems define the wider design space but are not formal baselines here because their writing, indexing, and update protocols introduce further variables.

2.2.5 Comparison of Representations and Scope of This Study

Table 2.1 compares the main representation families through observable design properties rather than an ordinal ranking.

Table 2.1. Conceptual comparison of external-memory representations.

Representation	Primary stored unit	Source-text preservation	Added organisation or typical access	Typical update	Principal risk
Raw dialogue logs	Turns or exchanges with session metadata	Verbatim	Session and turn chronology; lexical or long-context access	Append new turns	Relevant facts are dispersed in long histories
Foldered raw files	Verbatim turns grouped into session or topic files	Verbatim, subject to grouping	Paths, filenames, and headings; file navigation and text search	Append or reorganise files	The taxonomy may not help the searcher
Curated textual memory	Extracted facts, summaries, or topic notes	Selective and rewritten; size may decrease or increase	Explicit thematic or entity structure; file or ranked access	Rewrite, merge, and reconcile notes	Omission, distortion, or stale consolidation
Vector index	Embeddings linked to source chunks	Depends on whether the source chunks are retained; the index is additive	Similarity ranking over indexed units	Encode, delete, or re-index chunks	Semantic similarity may miss task relevance
Graph or linked memory	Entities, events, or notes plus relations, often linked to source text	Depends on node construction and provenance	Relation traversal or hybrid access	Add, merge, or revise nodes and edges	Construction, linkage, and conflict-maintenance errors

Note. This table is a project-defined conceptual synthesis, not an empirical ranking. The representation families are not mutually exclusive; vector and graph indexes can be layered over textual sources. Sources: [3, 4], [9–11].

This study selects the first three representations because they remain human-readable and auditable while spanning different levels of organisation and compression. Adding vector or graph systems would also change the retrieval infrastructure.

2.3 Memory Retrieval Methods

2.3.1 Agentic File Navigation and Text Search

In agentic filesystem retrieval, an LLM controls a sequence of tool actions. It may inspect directories, formulate keywords, search text, open a matching region, and continue when evidence is incomplete. `grep` is an access tool, not a retrieval strategy; the agent determines the query, scope, inspection sequence, and stopping point.

This approach can use both lexical matches and filesystem structure. Its visible actions support audit and query-specific adjustment, but add model rounds, tool calls, context growth, and behavioural variation. *Filesystem-Based Memory* treats the harness as part of the system rather than a neutral interface [4].

This study adopts the paradigm but separates evidence collection from final answering. Exact controls appear later.

2.3.2 Lexical Ranking and Iterative Retrieval

Lexical rankers score stored units against a query. BM25 rewards informative term overlap while accounting for frequency and document length [12]. The query may be the original question or an LLM reformulation, and ranked units may be aggregated at a higher level. Reciprocal Rank Fusion combines multiple ranked views by favouring items that appear near the top of those lists [13].

ReFind applies lexical ranking to dialogue turns and adds session-level signals, local expansion, temporal filtering, and duplicate control [6]. An LLM controller can search again, retain evidence, or stop. Iteration may repair a poor first query, but adds state and cost. The ranker, controller, actions, and final evidence should therefore be distinguished.

2.3.3 Multi-Source Retrieval and Evidence Fusion

A system may retrieve from raw logs, curated notes, vector indexes, or other sources. Sources can be complementary: one may preserve exact wording and time, while another offers a cross-session summary. Fusion may also add repetition, conflicting states, or excessive context.

Multi-source evidence fusion is not necessarily a new joint retriever. It may combine outputs from existing paths without another search. The study's dual-source condition follows this design: it reuses two fixed evidence bundles and applies deterministic fusion rules before answering. Chapter 3 defines its provenance and budget handling.

2.3.4 Full Context and Long-Context Answering

Full Context places the complete available conversation in the Answerer input. It avoids explicit search misses, but does not guarantee correct localisation or use. Long-context models can be sensitive to evidence position [5], while repetition, similar speakers, and changing facts add interference. Search calls disappear, but input cost and attention burden grow with the history.

Here, Full Context is a no-retrieval diagnostic. It is neither a storage representation, an eighth selective condition, nor an oracle. It tests whether complete visibility repairs omissions or whether selected evidence improves focus.

2.3.5 Framework for Comparing Retrieval Methods

Table 2.2 separates four access paradigms. It focuses on control flow rather than implementation details.

Table 2.2. Conceptual comparison of memory-access paradigms.

Access paradigm	Control and selection mechanism	Main unit	Iteration	Use of hierarchy	Principal cost or risk
Agentic filesystem retrieval	An LLM selects paths, queries, tool actions, and the stopping point	Files, sections, and text regions	Multi-round	When hierarchy is exposed and used	Model rounds, tool calls, and behavioural variation
Ranked retrieval	A user or LLM formulates a query; a ranker scores indexed units	Exchange, passage, or fact	Optional controller loop	Not required; grouping or reranking may use metadata	Query mismatch and ranker error
Multi-source evidence fusion	Existing retrieval outputs are combined by fixed or learned fusion rules	Multiple evidence bundles	Inherits the parent paths; no additional retrieval in this study	Inherits the parent paths	Duplication, conflict, and context expansion
Full Context	No external selector; the Answerer receives the complete history	Complete history	No explicit retrieval loop in this study	Not applicable	Long input, attention dispersion, and scaling cost

Note. These access patterns can be composed; the rows identify where explicit selection occurs rather than mutually exclusive algorithms. The filesystem and ReFind-style rows draw on [4] and [6]. The fusion and Full Context operationalisations are project-defined.

2.4 Filesystem-Based Memory

2.4.1 Memory Store, Tool Harness, and Agent Roles

Filesystem-Based Memory for LLM Agents treats the filesystem as an external memory medium [4]. Its **memory store** is a rooted hierarchy of files and folders, while the **tool harness** defines the role-specific operations through which agents access it. Paths, descriptions, content, and headings expose structure.

The paper distinguishes three roles. A management agent writes and maintains memory. In the conversational setting, a search agent answers queries with cited sources. In the skill setting, an execution agent completes tasks using retrieved skills. The same store may therefore behave differently under different tools, prompts, or models.

The filesystem should therefore not be treated as one indivisible algorithm: store shape, access behaviour, and downstream use remain distinct.

2.4.2 Experimental Axes and Control Logic

The paper varies memory builder or shape, stream scale, tool harness, and model capability, while identifying what is fixed within each comparison [4, Sec. 2.4]. It organises the analysis around questions of organisation, retrieval, growth, cost, and sustainability. This controlled logic is more informative than a simple system ranking. The present study follows it where possible, while treating inseparable configurations as complete pipelines rather than false single-component ablations.

2.4.3 Main Findings and Open Questions

The paper reports that no memory shape gives the best answer quality in every setting [4]. At larger scales, organised variants can reduce the material read and search cost relative to verbatim storage, while heavier curation is not universally superior. Effects depend on the task, scale, model, and harness.

The evaluation also covers cost, growth, and store health, exposing trade-offs between management work and later search. Open questions include horizons beyond those measured, whether useful hierarchy can be maintained, and longer-horizon store health under continued updates.

These findings are conditional on the paper's tasks and systems. Its numerical results are not directly comparable with this study.

2.4.4 What This Study Inherits, Changes, and Extends

The relationship is summarised in Table 2.3.

Table 2.3. Relationship of this study to *Filesystem-Based Memory for LLM Agents* [4].

Relationship	Elements in this study
Adapted from [4]	Filesystem as an external store; raw, foldered, and curated textual memory as broad comparison families; role separation as a design principle; agentic navigation through tools
Project-specific modifications	A common evidence interface before final answering; the composite frozen Main-60 set from one LoCoMo conversation; audited memory snapshots; independently logged retrieval and answer stages
Additional components beyond [4]	ReFind-style retrieval; two query-time backbone models; fixed dual-source fusion; a DeepSeek-family Full Context diagnostic; LoCoMo category-aware F1 following the pinned official evaluator [14]; project-defined Precision/Recall, semantic, evidence-coverage, and stage-diagnostic measures

Note. Only the first row describes elements adapted from [4]. The remaining rows document this study's own design choices and evaluation scope; they do not imply that these additions were evaluated in [4].

This is not a line-by-line reproduction. The paper supplies a framework for representation, roles, controlled variation, and operational evaluation; the next section introduces iterative search over unmodified dialogue.

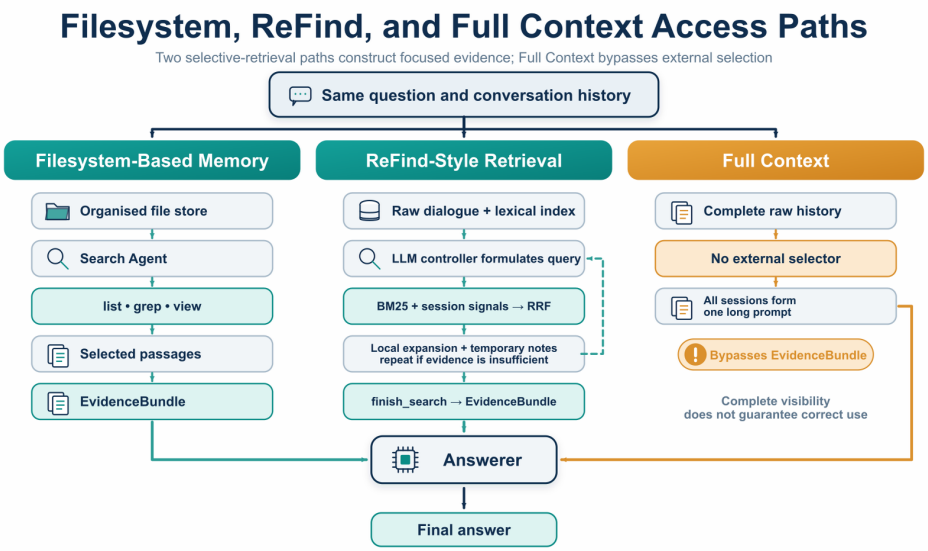


Figure 2.2. Conceptual access paths for filesystem-based memory, ReFind-style retrieval, and Full Context. The two selective paths construct an EvidenceBundle before answering, whereas Full Context passes the complete history directly to the Answerer. Adapted conceptually from [4] and [6]; the common interface and Full Context path are project-defined.

2.5 ReFind and Iterative Retrieval over Raw Dialogue

2.5.1 Motivation for Searching Raw Dialogue

Preprocessing can ease search but change evidence by removing a name, negation, or event order. Raw chat avoids this write-time transformation and retains speaker and local context, while leaving facts dispersed across the archive. ReFind therefore shifts effort from LLM-based memory construction to LLM-controlled query-time lexical search [6]. The controller must formulate effective searches and recognise sufficient evidence.

2.5.2 Two-Level Ranking and Iterative Search

ReFind indexes and ranks dialogue turns with BM25 and derives session-level signals [6, 12]. Reciprocal Rank Fusion combines the turn and session views [13]. High-ranking turns are expanded with local context. Temporal filtering narrows candidates by date, while seen-session filtering prevents later rounds from revisiting returned sessions.

Search and final answering are separate. The controller can query, inspect results, and save question-local notes without rewriting long-term memory. `finish_search` ends evidence collection; a later answering stage uses the notes. Stopping search is therefore not answering the question.

ReFind limits search iterations. This study's exact parameters appear in Chapters 3 and 4.

2.5.3 Empirical Findings and Cost Perspective

ReFind reports that agent-controlled lexical search can rival structured-memory methods in its evaluated settings [6]. Its controls and ablations show that single-shot BM25, or BM25 without the iterative and chat-native controls, is insufficient to explain the gains. Local expansion, session-level fusion, temporal filtering, seen-session control, and multi-round control interact to shape the evidence.

The design avoids LLM-based memory construction but uses an LLM controller at query time. Its value therefore requires joint analysis of answer quality and resource use.

The claim remains setting-specific. Relative performance may change with implicit matches, language, benchmark, or controller model.

2.5.4 Boundaries of the ReFind-Style Adaptation in This Study

This study adapts ReFind's turn-level idea to exchange-level lexical units while retaining higher-level grouping, rank fusion, local context, iterative control, temporary notes, and separate final answering. The topic-organised raw store adds project-specific path handling.

The study also applies the iterative principle to curated memory, where retrieval units and grouping differ from ReFind's turns and sessions. The common evidence interface, LoCoMo mapping, and curated adapter are project designs.

The report therefore uses **ReFind-style** or **ReFind-inspired**, not exact reproduction, and does not attribute the curated variant to ReFind.

2.5.5 Complementarity of Filesystem Memory and ReFind

The papers address complementary questions. *Filesystem-Based Memory* studies store shape, maintenance, and tool access [4], whereas ReFind studies iterative retrieval over raw chat [6]. This study combines their ideas to examine storage–retrieval interaction under shared data, questions, evidence interfaces, and Answerer protocols, not to rank the papers globally.

2.6 Long-Term Dialogue Evaluation and LoCoMo

2.6.1 LoCoMo Data Structure

LoCoMo is a benchmark for very long-term conversational memory [1]. Its conversations span multiple sessions and retain speakers, order, dates, and image captions where relevant. Human verification and editing were used to reduce long-range inconsistencies in the generated conversations.

Each question has a reference answer, and samples include turn identifiers for supporting dialogue evidence. This supports both answer evaluation and source-retrieval analysis.

LoCoMo also contains event-summarisation and dialogue-generation tasks. This study uses question answering; Chapter 4 describes its frozen sample.

2.6.2 Question Categories and Required Capabilities

LoCoMo defines five question categories [1]. **Single-hop** questions use a fact from one session; **multi-hop** questions combine sessions; **temporal** questions require ordering or tracking change; **open-domain** questions combine dialogue with commonsense or world knowledge; and **adversarial** questions use misleading, unanswerable premises.

This study uses one unified Main-60 set from the four non-adversarial categories. Category analysis is secondary to paired overall comparisons; Chapter 4 gives the selection rationale and distribution.

2.6.3 Gold Answers, Gold Evidence, and Evaluation Boundaries

LoCoMo favours reference wording drawn directly from conversations where feasible and uses partial-match F1 after normalisation [1]. Lexical scoring is reproducible, but can penalise valid paraphrases.

Evidence annotations identify source turns containing the answer and enable retrieval measures. They are not a perfect account of every valid reasoning path: some questions use background knowledge, and alternative passages may support the answer. Evidence metrics should therefore be limited to eligible questions and treated as diagnostics.

Question answering also does not test the full lifecycle, including prolonged updates, active forgetting, conflict repair, or autonomous reorganisation.

2.6.4 Why LoCoMo Is Used

LoCoMo offers dispersed multi-session history, varied reasoning demands, and source-turn annotations. It also provides a common reference point for long-context and memory-system research [1, 7].

The experiment nevertheless uses one frozen conversation and 60 questions. It supports controlled, traceable comparison rather than universal coverage.

2.7 Evaluating Long-Term Memory Systems

This section reviews general evaluation perspectives and explains why they are adopted in this study. It does not define the final experimental protocol. The exact metrics, formulas, prompts, eligibility rules, and aggregation procedures are specified in Chapter 4.

2.7.1 Token Precision, Recall, and F1

Token metrics compare a candidate with the reference after normalisation. **Precision** is the fraction of candidate tokens overlapping the reference; **Recall** is the fraction of reference tokens recovered by the candidate; and **F1** balances the two for each question. Low precision often reflects extra or non-matching text, while low recall often reflects omission.

The F1 measure aligns with LoCoMo's paper-described QA protocol [1], and its exact category-aware implementation follows the pinned official evaluator [14]. Precision and Recall are project-defined diagnostics that use the same matching basis. These token metrics are deterministic but sensitive to wording and answer length. A valid paraphrase may share few tokens, while a wrong answer may repeat key terms.

Chapter 4 defines normalisation and aggregation. Aggregate F1 is the mean of per-question values, not a recomputation from aggregate Precision and Recall.

2.7.2 Semantic Evaluation with an LLM judge

An LLM judge compares meaning rather than exact wording. Under a fixed rubric, **Strict Accuracy** counts fully correct answers, **Partial Rate** records partly correct responses, and **Soft Semantic Score** assigns partial credit.

Judging can recognise meaning beyond exact token overlap, but remains sensitive to position, verbosity, self-enhancement bias, prompt wording, and limited reasoning ability [8]. Accordingly, this study keeps the LLM judge independent of the answering backbone model, blind to condition names, and fixed to one rubric and prompt. It complements rather than replaces token metrics.

The rubric and label mapping are project-specific and specified in Chapter 4.

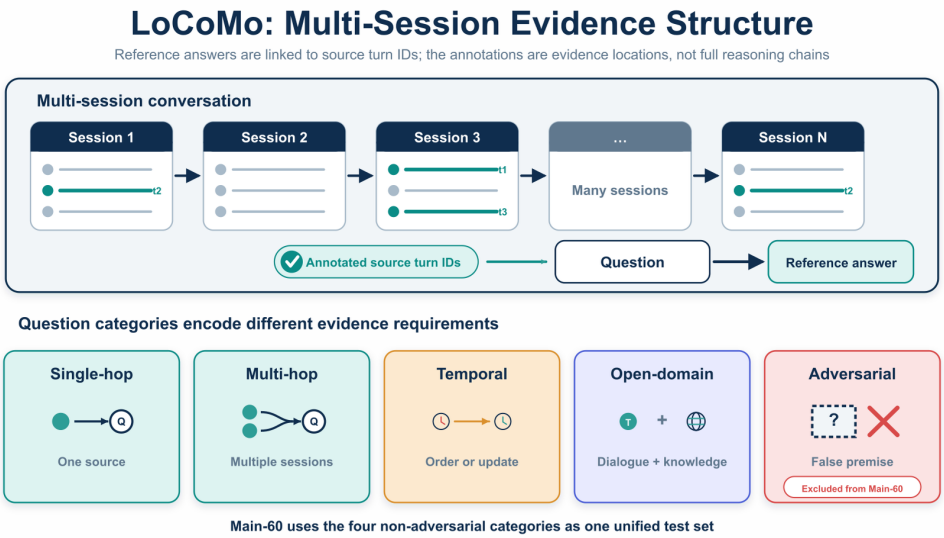


Figure 2.3. LoCoMo's multi-session question-answering structure, question categories, and source-turn annotations. Source-turn identifiers locate annotated evidence but do not constitute complete reasoning chains. The Main-60 set uses the four non-adversarial categories as one unified test set. Adapted conceptually from [1].

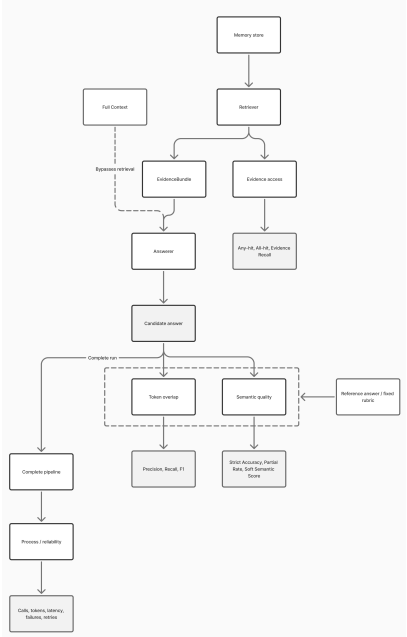


Figure 2.4. Observation points of answer-level, evidence-level, and process-level evaluation. Full Context bypasses explicit retrieval, so evidence-retrieval metrics are not applicable to that path.

2.7.3 Evidence-Level Evaluation

Evidence evaluation asks whether retrieval surfaced annotated sources. **Any-hit** records at least one match, **All-hit** requires all annotated sources, and **Evidence Recall** is the fraction covered.

These measures help locate failure, but are not causal. Retrieved text may be noisy, annotations incomplete, and the Answerer may use unannotated support. High recall therefore does not imply precise or correctly used evidence.

They apply only to selective retrieval and eligible questions. Full Context performs no retrieval selection, so complete visibility should not be labelled perfect retrieval recall.

2.7.4 Process, Efficiency, and Reliability Metrics

Quality must be interpreted with process cost. Query-time measures include model and tool calls, tokens, latency, failures, retries, and recovery. Store-level measures include size, hierarchy, and construction cost. Failed attempts remain part of the record. A filesystem round and an iterative lexical action expose different tools, so their counts are not equivalent. Chapter 4 separates construction, retrieval, answering, judging, and recovery costs.

2.7.5 Why Multi-Perspective Evaluation Is Needed

No single metric explains answer quality and system behaviour. Table 2.4 summarises the four complementary views.

Table 2.4. Evaluation perspectives, scope, and limits.

Metric family	Main question answered	What it cannot answer alone
Official LoCoMo F1 and project-defined token Precision/Recall diagnostics	How much normalised token or stem overlap does the candidate share with the reference?	Whether a valid paraphrase is semantically complete or factually correct
LLM judge measures: Strict Accuracy, Partial Rate, and Soft Semantic Score	Is the answer fully correct, partly correct, or incorrect in meaning under the fixed rubric?	Why the system failed or whether required evidence was retrieved
Gold-source coverage: Any-hit, All-hit, and Evidence Recall	Did selective retrieval cover the eligible annotated sources?	Evidence precision, source uniqueness, or whether the Answerer used the evidence correctly
Process, efficiency, and reliability measures	How many resources, actions, failures, retries, and recovery steps were observed?	Whether output quality is acceptable, or monetary cost without a stated pricing rule

Note. LoCoMo describes normalised partial-match F1 in the paper [1]; the exact category-aware implementation, including its normalisation and stemming rules, follows the pinned official evaluator [14]. Precision and Recall are project-defined diagnostics computed on the same matching basis. The judge-derived and evidence-coverage measures, including label mapping, source matching, and eligibility rules, are also project-defined. LoCoMo supplies the reference answers and source-turn identifiers [1], while [8] motivates the complementary use of an LLM judge.

Together, the four views support mechanism analysis and system choice rather than only a leaderboard.

2.8 Positioning and Differences of This Study

2.8.1 Related-Work Comparison Matrix

Table 2.5 locates the study relative to its three main foundations. Two panels separate system design from evaluation so that the comparison remains readable in a standard report layout.

Table 2.5. Panel (a). System-design comparison of the principal related work and this study.

Work	Primary focus	Memory representation or benchmark source	Access method	Model scope
Filesystem-Based Memory [4]	Effects of organisation, tools, scale, and model capability on persistent filesystem memory	Chunk-retrieval control; verbatim dump; foldered sessions; reorganised and agent-curated stores	Agentic filesystem search; BM25 chunk-retrieval control; targeted harness variants	Model capability is an experimental axis
ReFind [6]	Whether agent-controlled search over raw chat can rival structured memory	Raw chat logs	Turn-level BM25; session-rank RRF; ±2-turn local context expansion; temporal and seen-session filtering; iterative ReAct controller	GPT-4o-mini in the main suite; GPT-5-mini on LongMemEval subsets rather than a full cross-backbone replication
LoCoMo [1]	Evaluation of very long-term conversational memory	Multimodal multi-session conversations plus QA and source-turn annotations; no prescribed system memory store	The paper evaluates base or truncated context, long-context, and RAG variants; the benchmark itself is method-agnostic	Multiple models and methods are compared; backbone is not isolated under one shared pipeline
This study	Interaction among storage, retrieval, backbone model, fusion, and context scope under shared controls	Three auditable textual representations built from one frozen conversation source	Agentic file search; ReFind-style iterative retrieval; fixed evidence fusion; Full Context	Two selective-retrieval backbone pipelines under one scientific protocol; one DeepSeek-family Full Context diagnostic

Table 2.5 (continued). Panel (b). Evaluation and diagnostic comparison of the principal related work and this study.

Work	Long-context or complete-history comparator	Outcome evaluation	Process or diagnostic analysis
Filesystem-Based Memory [4]	No matched complete-history, no-retrieval diagnostic equivalent to this study's Full Context condition	Task quality, cost, and store properties	Growth, search effort, cost, and store health
ReFind [6]	Long-context comparator baselines are reported or reused, but not as a matched complete-history diagnostic	Answer quality, retrieval behaviour, ablations, and cost	Component ablations and search-process analysis
LoCoMo [1]	Includes long-context baselines	QA, event summarisation, dialogue generation, and RAG context-retrieval recall@k	Category-level QA analysis and RAG recall@k; manual error taxonomy for event summarisation
This study	Completed DeepSeek-family, no-explicit-retrieval Full Context diagnostic over the same Main-60 source	Official LoCoMo F1 plus project-defined token, semantic, evidence, process, and cost measures	Stage-based diagnosis, paired contrasts, and manual review

Note. The two panels compare scope and method, not reported scores. Direct numerical comparison would be misleading because the works use different tasks, models, prompts, and system boundaries. RAG = retrieval-augmented generation; RRF = reciprocal rank fusion.

2.8.2 From Literature Gaps to the Study Design

The methodological limitations identified in prior work and in this project's synthesis lead to the six design responses in Table 2.6.

Table 2.6. Methodological limitations motivating this study and project-defined responses.

Methodological limitation	Evidence basis	Response in this study
Storage and retrieval effects are not always isolated	Store shape and tool harness interact in [4]; [7] explicitly crosses write and retrieval strategies	Use clean storage–retrieval contrasts where possible; treat inseparable configurations as complete pipelines rather than pure ablations
Evidence from one backbone model has limited generalisability	[7] identifies single-backbone evaluation as a limitation	Repeat the selective-retrieval conditions with Qwen and DeepSeek backbone models
Retrieval-to-answer interfaces differ across systems	The tool-mediated filesystem pipeline in [4] and iterative retrieval pipeline in [6] expose different intermediate states	Use a common EvidenceBundle interface and a common Answerer prompt within each backbone pipeline
Final-answer scores can hide retrieval and utilisation failures	[7] separates retrieval relevance, memory utilisation, and downstream correctness	Add gold-source coverage, retrieval traces, stage-based diagnosis, and manual review
Complete visibility does not guarantee reliable use of long context	[5] shows that long-context utilisation can depend on evidence position	Add a DeepSeek-family Full Context condition to contrast explicit selection with complete visibility
Accuracy alone does not describe operational cost	[4] and [6] both report process or cost dimensions alongside answer quality	Record model calls, tokens, latency, construction cost, failures, and recovery

Note. The limitations are a project-defined synthesis rather than claims from a systematic literature review. The response column records this study's design choices; it does not attribute those choices to the cited papers.

These responses define the conceptual design rather than its implementation. Chapter 3 introduces the formal components and conditions; Chapter 4 defines the exact controls, statistical contrasts, and error rules.

2.8.3 Boundaries Relative to the Two Core Papers

The study inherits the filesystem store, role separation, and harness perspective from *Filesystem-Based Memory* [4], but requires auditable evidence before final answering through a common interface. From ReFind it inherits raw-chat, lexical, session-aware, iterative search [6], while adapting turn-level retrieval to project units. The adapters, evidence schema, source mapping, curated variant, prompts, and action limits are project choices. The objective is a controlled synthesis, not reproduction of either full leaderboard.

2.8.4 Transition to Chapter 3

Prior work shows that representation, search control, tools, model capability, context scale, and evaluation perspective all affect memory performance. A final answer score cannot identify the failing stage. Chapter 3 therefore defines the study's stores, retrieval processes, selective conditions, common EvidenceBundle and Answerer interfaces, fusion path, and Full Context diagnostic; Chapter 4 specifies execution and evaluation.

Chapter 3. System Design and Methodology

Chapter 2 identified several unresolved problems in long-term conversational memory. Storage and retrieval are often changed together, which makes their effects difficult to separate. Curated memory can improve organisation but may also omit or alter information. Selective retrieval can reduce context length, but it can fail to surface required evidence. Complete context avoids explicit retrieval omission, yet it may introduce noise and long-context utilisation problems. A final answer score alone does not reveal which stage caused an error.

This chapter translates these general problems into a concrete system design. The study separates memory representation, evidence retrieval, evidence packaging, and answer generation. It then derives a set of conditions that supports controlled component comparisons, complete-system comparisons, dual-source fusion, and comparison with a no-retrieval Full Context path.

The chapter explains why the system has this structure and how its components operate. Method-defining constants are introduced where their mechanisms are specified. Chapter 4 consolidates the experimental settings and defines the frozen question set, backbone models, remaining run controls, execution protocol, evaluation metrics, and statistical analyses. Chapter 5 presents and interprets the results. The purpose of this chapter is therefore to define the system and justify the condition structure, rather than to determine which condition performs best.

3.1 From Research Gaps to Methodological Requirements

3.1.1 Separating Storage, Retrieval, and Answering

A long-term memory system should not be treated as one indivisible component. Its end-to-end process can be divided into four stages:

$$\begin{aligned}\text{Conversation history} &\rightarrow \text{Memory representation}\\ &\rightarrow \text{Retrieval and evidence selection}\\ &\rightarrow \text{Answer generation}.\end{aligned}$$

Each stage can produce a different type of failure. The required information may not be retained when the memory is constructed. It may be present in the memory but remain unretrieved. Relevant material may appear in search results but not be selected for the final evidence input. Evidence may also be removed by a fixed budget. Finally, complete evidence may reach the Answerer but still be interpreted incorrectly.

These cases require different remedies. A write omission requires a better memory-construction process. A retrieval omission requires a better search strategy. Evidence loss may require a different packaging or budget policy. A utilisation error instead concerns the Answerer's ability to interpret the available evidence. One end-to-end answer score cannot distinguish among these mechanisms.

Building on the retrieval-to-generation separation in [7], the system therefore makes storage, retrieval, evidence packaging, and answering explicit. This four-stage distinction is a project-specific extension: [7] groups missing stored detail and missed retrieval under one retrieval-stage category. Each retriever produces an inspectable retrieval trace and a standardised EvidenceBundle. The Answerer operates only after retrieval has ended. This separation supports later analysis of where relevant information was preserved, found, retained, and used.

3.1.2 Requirement 1: Isolating Storage Organisation

Prior studies compare flat or raw archives with structured, curated, or summarised memory variants [4, 7]. Such a comparison can change both content and organisation. Any observed difference may result from rewritten text, removed detail, new semantic labels, or a different path structure.

The present study introduces two raw memory representations to isolate directory organisation. S1 stores the original sessions in one flat directory. S2 places the same complete session files into topic directories. The intended content relationship is:

$$\text{content}(S1) = \text{content}(S2),$$

while their path structures differ:

$$\text{paths}(S1) \neq \text{paths}(S2).$$

S2 does not summarise, split, merge, or rewrite a session. It only adds a directory hierarchy. This creates a controlled way to ask whether navigational structure helps memory access when the underlying dialogue remains unchanged.

The comparison does not measure semantic compression or fact extraction. It is restricted to the incremental effect of directory organisation and the way a retriever responds to that organisation.

3.1.3 Requirement 2: Comparing Retrieval Strategies on the Same Memory

A storage representation cannot be assessed independently if every store is paired with a different retriever. The study therefore applies two retrieval families to both raw stores. On S1, R1 defines E1 and R2 defines E2. On S2, the corresponding conditions are E3 and E4. These four conditions form the controlled raw-memory core.

Holding the memory representation constant supports two retrieval comparisons. E2 versus E1 compares R2 with R1 on flat raw sessions. E4 versus E3 compares the same retrieval families on foldered raw sessions. Holding the retrieval family constant supports two storage comparisons. E3 versus E1 examines how directory organisation affects R1, while E4 versus E2 examines whether the same organisation also affects R2.

The four conditions also support an interaction question: does directory organisation help one retrieval family more than the other? R1 can inspect and navigate paths directly. R2 instead begins from lexical ranking over raw conversation units. The R2 index excludes directory names, so the same query and date filter produce the same first-stage ranking over S1 and S2. However, result provenance can expose an S2 path to the Retrieval Agent. That path may influence a later query. Directory structure can therefore affect adaptive retrieval even when it is excluded from lexical scoring.

This is a controlled partial-factorial core rather than a claim that all retrieval interfaces are identical. Each retrieval family requires representation-appropriate observations and instructions. Chapter 4 defines the paired contrasts and controls used when interpreting them.

3.1.4 Requirement 3: Evaluating Curated Memory as a Complete System

S3 differs fundamentally from S1 and S2. It is an agent-curated memory rather than a relocated copy of the raw dialogue. Its construction process can rewrite statements, combine evidence from several sessions, group facts by entity and topic, reconcile later updates, and remove repetition. It can also omit qualifiers, preserve an outdated state, or join facts incorrectly.

S3 must therefore be evaluated as a complete representation rather than as a third directory layout. The study provides two ways to access the same frozen S3 store:

- E5 combines S3 with R1, the general filesystem retriever.
- E6 combines S3 with R3, a retrieval adapter designed for factual units and topic groups.

This pair addresses two questions. First, can a general filesystem Retrieval Agent use a curated store effectively? Second, does a retriever adapted to the internal structure of curated memory access the same store differently?

The conditions do not form an exhaustive 3×3 storage–retrieval matrix. R2 and R3 are not two independent generic retrievers. R2 operates over exchanges grouped by raw sessions. R3 adapts the same broad iterative retrieval principle to curated facts grouped by topic sections. R3 is therefore representation-specific. By contrast, R1 can browse S1, S2, and S3 through a common filesystem interface.

The condition set includes theoretically meaningful and operationally valid pairings rather than every nominal combination. Comparisons involving S3 must also remain system-level comparisons. Since S3 changes content, granularity, organisation, and provenance presentation together, a difference between S3 and a raw store cannot be attributed to one isolated storage feature.

3.1.5 Requirement 4: Testing Raw–Curated Complementarity

Raw dialogue and curated memory provide different forms of evidence. Raw dialogue retains exact wording, local context, temporal order, and complete source history. Its information may nevertheless be dispersed across many sessions and surrounded by irrelevant material. Curated memory can consolidate related facts and organise them by entity and topic. However, curation may omit detail, alter wording, or combine inconsistent states.

The two sources may therefore be complementary. Raw evidence may repair a write omission or distortion in curated memory. Curated evidence may connect facts that remain separated in raw sessions. Conversely, combining the sources may introduce duplication, conflicting descriptions, or additional distraction.

E7 tests this relationship by combining the EvidenceBundles already produced by E2 and E6. E2 represents raw dialogue accessed through the raw-chat retrieval path. E6 represents curated memory accessed through its adapted retrieval path. This pairing is fixed by the system design and is not selected after observing performance.

E7 does not formulate a new query or execute a third retriever. It performs deterministic post-retrieval fusion and then invokes the shared Answerer once. The condition therefore examines whether already retrieved raw and curated evidence are complementary or interfering.

3.1.6 Requirement 5: Distinguishing Retrieval Failure from Long-Context Failure

A failed selective-retrieval answer does not show whether the Answerer lacked the required information or failed to use it. To support this distinction, the study adds a Full Context diagnostic path. Full Context sends the complete ordered raw conversation directly to the Answerer without an external retrieval stage.

This path supports several diagnostic questions. If Full Context repairs an error made by selective retrieval, the selective path may have omitted necessary evidence. If selective retrieval succeeds while Full Context fails, evidence selection may have reduced distracting material. If both paths receive the required material but still fail, the result indicates a downstream utilisation or reasoning problem [7]. Subsequent analysis can then test question-specific mechanisms such as temporal ordering, entity resolution, multi-source composition, and long-context position effects [5].

Full Context is not a fourth memory representation, a fourth retriever, or an eighth selective condition. It is also not an oracle. Complete source visibility does not guarantee correct use. It is a separate no-retrieval path used to examine the consequences of removing explicit evidence selection.

3.1.7 Requirement 6: Controlling the Answering Stage

A comparison of memory access becomes difficult to interpret if every retriever uses a different evidence format or answering prompt. The study therefore requires all selective conditions to follow the same downstream structure:

$$\begin{aligned}\text{Condition-specific retrieval} &\rightarrow \text{Common EvidenceBundle}\\ &\rightarrow \text{Shared Answerer}\\ &\rightarrow \text{Common output schema}.\end{aligned}$$

The Answerer receives the question, an ordered EvidenceBundle, and shared answering instructions. It does not receive the condition identifier, retriever name, retrieval score, retrieval trace, reference answer, gold evidence, or evaluation label. It also has no retrieval tools. Its task begins only after the evidence input has been finalised.

The same component graph and interface definitions are used when the experiment is instantiated with different backbone models. Within each backbone pipeline, the selective conditions use the same Answerer protocol. Chapter 4 describes the backbone deployments and identifies which factors remain fixed across them.

3.1.8 Derivation of the Final Condition Set

The preceding requirements lead to four related parts of the study. E1–E4 form the controlled raw-memory core. E5–E6 extend the design to agent-curated memory. E7 tests dual-source evidence fusion. Full Context removes explicit retrieval as a diagnostic comparison.

Table 3.1. Derivation of the system conditions from the research problems.

Research problem	Design response	Relevant conditions or path
Storage organisation is confounded with content transformation	Compare flat and foldered stores with identical raw session contents	E1/E3 and E2/E4
Retrieval effects are confounded with storage effects	Apply R1 and R2 to the same raw stores	E1–E4
Curated memory changes both content and structure	Evaluate one S3 store through general and adapted retrieval	E5 and E6
Raw and curated evidence may be complementary	Fuse the existing E2 and E6 EvidenceBundles	E7
Retrieval omission is difficult to separate from utilisation failure	Remove selective retrieval and expose the complete raw history	Full Context
Answer generation may confound memory comparisons	Use a shared EvidenceBundle and Answerer protocol	E1–E7

The resulting design is theory-driven rather than an exhaustive Cartesian product. Each condition has a defined role in isolating raw-store organisation, comparing retrieval behaviour, examining curated-memory access, testing evidence fusion, or removing explicit retrieval.

3.2 Task Formalisation

3.2.1 Conversation, Session, and Source Turn

Let a conversation be an ordered sequence of sessions:

$$\mathcal{C} = (\sigma_1, \sigma_2, \ldots, \sigma_M).$$

Each session σ_i is an ordered sequence of source turns:

$$\sigma_i = (u_{i,1}, u_{i,2}, \ldots, u_{i,n_i}).$$

LoCoMo supplies ordered sessions, speaker contributions, session dates, dialogue text, optional image captions, and answer-source turn identifiers [1]. In this study, a **source turn** is one such speaker contribution. The local canonicaliser preserves the source fields and adds a stable project locator and content identity.

The report uses *source turn* rather than the unqualified term *turn*. This avoids confusion with the larger exchange units used by R2 and with the reasoning rounds used by R1.

3.2.2 Question and Evaluation-Only Information

Let q denote the question presented to the retrieval and answering pipeline. During online execution, the system may access q and the permitted memory representation. It cannot access the reference answer or the benchmark's annotated evidence.

Let a^* denote the reference answer. Let G_q denote the set of annotated gold source identifiers when such evidence is available. The question category and evaluation scores are also evaluation-only information. These fields are introduced only after the candidate answer has been generated.

This boundary applies to every online component. The memory-construction process, Retrieval Agent, deterministic retrieval backend, fusion procedure, and Answerer cannot use a^*, G_q, or any evaluation label.

3.2.3 Memory Representation

A memory representation is a transformation of the same canonical conversation:

$$\mathcal{M}_s = T_s(\mathcal{C}), \qquad s \in \{1, 2, 3\}.$$

Here, $\mathcal{M}_1$, $\mathcal{M}_2$, and $\mathcal{M}_3$ correspond to S1, S2, and S3. A representation includes both its text and its access-relevant structure. This structure may include filenames, directory paths, headings, topic groups, session boundaries, and source locators.

The transformation T_s does not imply that every store rewrites the source. S1 and S2 serialise the raw sessions without rewriting their contents. S3 instead produces curated factual memory through an agent-managed construction process.

3.2.4 Retrieval Trace and Evidence Selection

For a compatible memory representation and retrieval system, retrieval is defined as:

$$\mathcal{R}_r(q, \mathcal{M}_s) \rightarrow (\tau_{s,r,q}, \mathcal{B}_{s,r,q}),$$

where $\mathcal{R}_r$ is the retrieval system, $\tau_{s,r,q}$ is the retrieval trace, and $\mathcal{B}_{s,r,q}$ is the resulting EvidenceBundle.

A retrieval trace is an ordered sequence of observations and actions:

$$\tau = ((o_1, a_1), (o_2, a_2), \ldots, (o_L, a_L)).$$

Depending on the retriever, a trace may include directory observations, file reads, search expressions, generated lexical queries, ranked results, evidence-selection decisions, seen-state, and stopping decisions. R1 measures interaction through provider rounds, whereas R2 and R3 use bounded controller actions. These units are not assumed to be equivalent and are defined separately in Section 3.6.

3.2.5 EvidenceBundle

An EvidenceBundle is an ordered sequence of validated EvidenceItems:

$$\mathcal{B} = (e_1, e_2, \ldots, e_K).$$

Each item contains exact evidence text and the metadata required to identify its origin. Conceptually, this includes an evidence identifier, source kind, file path or structural group, source locators, canonical source identifiers, dates, speakers where applicable, content identity, and selection order.

The model does not provide arbitrary text as formal evidence. It selects material that has appeared in a valid observation. The host then re-reads the declared source and resolves the final text and provenance. This distinction separates model-directed selection from evidence authenticity.

3.2.6 Answer Generation

Let A_θ denote the Answerer instantiated with backbone model parameters θ. For a selective-retrieval condition, answer generation is:

$$\hat{a} = A_\theta(q, \mathcal{B}),$$

where $\hat{a}$ is the candidate answer.

The Answerer receives only the question, the final ordered evidence, and the shared answering instructions. It does not inspect the retrieval trace or access the memory store directly.

For Full Context, the evidence-access path changes:

$$\hat{a}_{FC} = A_\theta(q, \mathcal{C}).$$

Conceptually, the complete ordered raw conversation replaces a selectively constructed EvidenceBundle. In the shared Answerer interface, the complete session bodies are still wrapped as non-selective EvidenceItems so that provenance and citation validation remain available. The Answerer remains responsible for producing the candidate response, but no external retriever selects the preceding context.

3.2.7 RunRecord and Offline Evaluation Record

An online RunRecord can be represented as:

$$r = (q, c, \tau, \mathcal{B}, \hat{a}, \rho),$$

where c identifies the system condition and ρ contains operational metadata such as model-call and token measurements. For Full Context, the retrieval trace and selective EvidenceBundle are absent. The record instead contains the non-selective full-context bundle and identifies its complete source input.

An offline evaluation record is created after answer generation:

$$z = (r, a^*, G_q, \mathbf{s}),$$

where $\mathbf{S}$ is the vector of answer, evidence, and process measures defined in Chapter 4. Separating r from z makes the evaluation boundary explicit. Gold information is attached to an existing answer record rather than supplied to the online pipeline.

Table 3.2. Principal notation used in the system definition.

Symbol	Meaning
$\mathcal{C}$	Complete ordered conversation
σ_i	Session i
$u_{i,j}$	Source turn j in session i
q	Online question

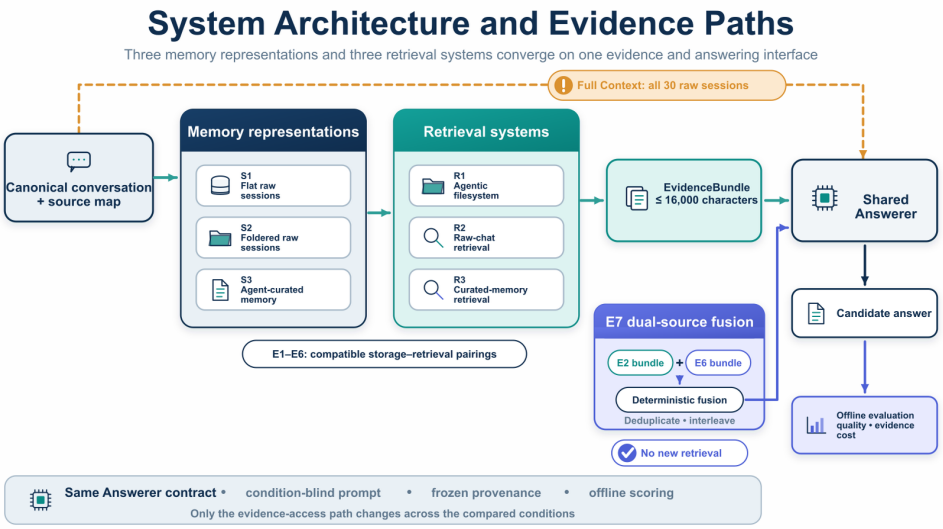


Figure 3.1. Overall system architecture and evidence-access paths. The canonical conversation is transformed into three memory representations and accessed through compatible retrieval systems. Selective conditions converge on a common EvidenceBundle and shared Answerer; E7 fuses the completed E2 and E6 bundles, whereas Full Context bypasses question-dependent retrieval.

Continued from the previous page

Symbol	Meaning
a^*	Reference answer used only for evaluation
G_q	Annotated gold source set, where applicable
T_s	Transformation that constructs memory representation s
$\mathcal{M}_s$	Memory representation S1, S2, or S3
$\mathcal{R}_r$	Retrieval system R1, R2, or R3
τ	Retrieval trace
e_k	One validated EvidenceItem
$\mathcal{B}$	Ordered EvidenceBundle
A_θ	Answerer using backbone model parameters θ
$\hat{a}$	Candidate answer
r	Online RunRecord
z	Offline evaluation record
$\mathbf{s}$	Evaluation metric vector

3.3 Overall System Architecture

3.3.1 Layered System Structure

The system contains six logical layers. The first layer is the canonical source conversation. It preserves the original session order, source turns, dates, speakers, dialogue identifiers, and available captions.

The second layer performs normalisation and establishes provenance. It converts the source into canonical records, raw session files, exchange units, and a common source map. These artefacts provide the shared foundation for all later paths.

The third layer contains the three memory representations. S1 preserves raw sessions in a flat directory. S2 preserves the same session contents under a topic hierarchy. S3 stores agent-curated facts organised by entity and topic.

The fourth layer performs evidence access. R1 navigates the filesystem directly. R2 performs iterative lexical retrieval over raw exchanges and sessions. R3 adapts this iterative approach to curated facts and topic groups. The Full Context path bypasses this layer.

The fifth layer standardises the retrieved information. All selective paths produce the same conceptual EvidenceBundle before answer generation. E7 operates at this layer by fusing the completed bundles from E2 and E6.

The sixth layer contains the shared Answerer and the offline evaluation boundary. The Answerer produces a candidate answer from either a selective EvidenceBundle or the complete raw history. Reference answers, gold evidence, and evaluation procedures are introduced only after this stage.

3.3.2 Selective-Retrieval Paths

The six single-source selective conditions follow the same high-level process:

$$\begin{aligned}\text{Canonical conversation} &\longrightarrow \text{Memory representation}\\ &\longrightarrow \text{Retriever}\\ &\longrightarrow \text{EvidenceBundle}\\ &\longrightarrow \text{Answerer}\\ &\longrightarrow \text{Candidate answer}.\end{aligned}$$

The internal retrieval process differs by condition, but the boundary before answering remains common. R1 produces evidence from observed file ranges. R2 produces evidence from raw exchange windows. R3 produces evidence from curated fact windows. These outputs are converted into the same evidence interface before they reach the Answerer.

This structure prevents a retriever from answering the benchmark question directly. Search decisions and answer generation remain separate model roles, even when they use the same backbone model within one pipeline.

3.3.3 Fusion Path

E7 begins only after E2 and E6 have completed retrieval for the same question. Their validated EvidenceBundles become the two parent inputs to a deterministic fusion procedure:

$$(\mathcal{B}_{E2}, \mathcal{B}_{E6}) \rightarrow F \rightarrow \mathcal{B}_{E7},$$

where F is the fixed fusion function.

The fused bundle is then passed to the shared Answerer. Fusion does not access S1 or S3 directly and does not produce a new retrieval trace. Its role is limited to validating, ordering, and deduplicating the existing parent evidence.

3.3.4 Full Context Path

Full Context follows a separate route. The canonical raw sessions are assembled into one complete, ordered context package. This package is supplied directly to the Answerer together with the question.

The path does not invoke S2, S3, R1, R2, R3, or E7. It does not convert the complete conversation into a selective EvidenceBundle; it only wraps the complete session bodies in the shared evidence interface. Its purpose is to remove external evidence selection while retaining the same source conversation.

3.3.5 Online and Offline Boundaries

The architecture maintains a strict distinction between online execution and offline evaluation. Online execution includes memory access, retrieval, evidence construction, fusion where applicable, and answer generation. Offline evaluation loads the reference answer and annotated sources only after the candidate answer has been finalised.

This boundary prevents gold information from shaping queries, stopping decisions, selected evidence, citations, or answer wording. It also permits one completed answer record to be assessed through several evaluation views without rerunning the online system.

Figure 3.1 summarises how the system layers and evidence-access paths converge on the shared Answerer.

3.4 Data Normalisation and Traceability

3.4.1 Canonical Source Records

The study uses one complete conversation selected from the released LoCoMo benchmark [1]. Local preprocessing verification identifies 30 sessions and converts their contents into one canonical sequence of 568 source-turn records before any memory representation is constructed.

Each canonical record contains:

- the conversation identifier;

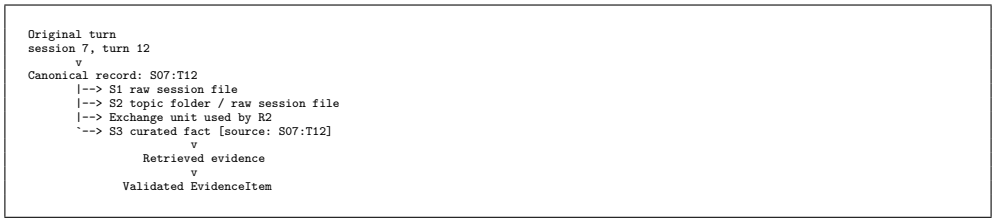


Figure 3.2. One canonical source turn remains traceable across raw, foldered, exchange-based, and curated representations.

- session identifier and session date;
- position within the session;
- global position within the conversation;
- speaker;
- original dialogue text;
- original LoCoMo dialogue identifier;
- a stable project locator;
- an available image caption;
- a content identity.

A locator follows the form $[SxTy]$, where x identifies the session and y identifies the source turn within that session. For example, $[S3T4]$ denotes the fourth source turn in the third session.

The canonical layer standardises identity and ordering without intentionally rewriting dialogue content. S1, S2, and S3 are derived from this shared layer. Differences among the stores therefore do not result from separate source parsing procedures.

The source contains 30 sessions and 568 source turns. Existing image captions are retained as fixed text when available. The system does not reopen image URLs or run a new vision model. The experiment is therefore based on the textual dialogue and the captions already present in the benchmark source.

3.4.2 Raw Session Serialisation

The canonical records are serialised into one Markdown file per natural session. Within each file, source turns remain in their original order. The serialisation preserves speaker labels, session date, source text, available captions, and stable locators.

These session files form the common textual basis of S1 and S2. S1 places them in one flat directory. S2 moves each complete file into a topic directory. Since the file bodies remain unchanged, any difference between the two raw representations arises from their navigational structure rather than rewritten dialogue.

Session serialisation also preserves natural temporal boundaries. No source turn is moved from one session body to another, and no source turn is duplicated or deleted during construction of the raw stores.

3.4.3 Exchange Construction

R2 requires retrieval units smaller than a full session but broader than one isolated source turn. The system therefore constructs deterministic exchange units within each session.

For session σ_i, each complete adjacent pair forms an exchange:

$$x_{i,j} = (u_{i,2j-1}, u_{i,2j}), \qquad j \leq \left\lfloor \frac{n_i}{2} \right\rfloor .$$

Pairing follows the original order:

$$(T_1, T_2),\ (T_3, T_4),\ (T_5, T_6), \ldots$$

If n_i is odd, the final exchange is the singleton:

$$x_{i,\lceil n_i/2 \rceil} = (u_{i,n_i}).$$

Exchanges never cross a session boundary. Every source turn appears in exactly one exchange.

The 568 source turns produce 292 exchange units across 30 sessions. An exchange is a retrieval unit rather than a summary. Its indexed content is derived directly from the corresponding speakers, source text, and available captions.

This unit provides more local conversational context than one source turn while remaining small enough for lexical ranking and neighbourhood expansion.

3.4.4 Source Map and Stable Provenance

The source map connects the project's canonical identifiers to the original benchmark records. For each locator, it resolves:

- the canonical source-turn record;
- the original LoCoMo dialogue identifier;
- session and date;
- speaker;
- source text;
- available caption.

This map creates a common provenance layer for all memory and retrieval paths. It is not supplied to a Retrieval Agent as hidden benchmark information. It is used by the host to verify selected sources and to support later evidence evaluation.

Stable provenance is important because the same source may appear in different forms. In S1 and S2, the source text appears directly in a session file. In S3, one curated fact may restate or combine several source turns. The attached locators preserve a link from that rewritten fact to the canonical source.

3.4.5 Traceability across S1, S2, and S3

Traceability is implemented differently for each representation.

S1 contains the original session text and locator directly. Evidence selected by R1 can therefore be resolved from the observed file and line range. Evidence selected by R2 is resolved from exchange windows whose source turns already have canonical identifiers.

S2 preserves the same file bodies. Its additional directory path changes navigation but not source identity. A source turn therefore retains the same locator and original dialogue identifier in S1 and S2.

S3 contains curated factual units rather than verbatim session records. Each formal fact retains one or more source locators. R1 may select a range of curated text, while R3 selects fact windows. In either case, the locators are parsed and resolved through the common source map.

This design does not imply that S3 contains every source turn. Curation may omit source material. It means that every formal S3 factual unit used as evidence must identify the source records on which it is based. Provenance validity and source coverage are separate properties.

Figure 3.2 illustrates how one original source turn remains traceable across the raw, foldered, exchange-based, and curated representations before becoming a validated EvidenceItem.

3.4.6 Conceptual Integrity Requirements

The normalised data and provenance layer must satisfy several requirements.

First, every raw source turn must resolve to one canonical record. Second, S2 must preserve the contents and source identities of the corresponding S1 session files. Third, every formal factual unit in S3 must contain valid provenance. Fourth, every EvidenceItem must resolve to the declared memory representation and its underlying canonical records. Fifth, missing or invalid provenance cannot be silently reconstructed from model output. Finally, reference answers, gold evidence, question categories, and evaluation labels remain outside the online memory pipeline.

These requirements make it possible to compare representations that differ substantially in structure while retaining a common source basis. They also support the later distinction between information that was absent from a store, information that was present but unretrieved, and information that reached the Answerer but was not used correctly.

3.5 Three Storage Representations

The three memory stores originate from the same canonical conversation, but they expose the information in different forms. S1 preserves the original sessions in a flat directory. S2 preserves the same session contents but introduces a topic-based directory hierarchy. S3 rewrites the source dialogue into entity- and topic-centred facts. The three stores therefore differ in both the amount and the type of transformation applied before retrieval.

3.5.1 S1: Flat Raw Sessions

S1 is the least transformed memory representation. It contains one Markdown file for each of the 30 conversation sessions. All session files are stored in a single directory. Within each file, source turns remain in their original chronological order. Speaker names, session dates, dialogue text, image captions, and source locators are preserved.

S1 does not add topic folders, summaries, extracted facts, or cross-session consolidation. Its main advantage is source fidelity. A retriever can inspect the original wording and surrounding dialogue without depending on an earlier summarisation decision. Direct source locators also make retrieved evidence straightforward to trace back to the canonical records.

However, source fidelity does not guarantee easy access. Information about the same person, event, or plan may be distributed across many sessions. Repeated descriptions and later corrections can also make the relevant state difficult to identify. S1 therefore provides a raw-filesystem reference representation rather than an assumption that unmodified dialogue is the best memory format.

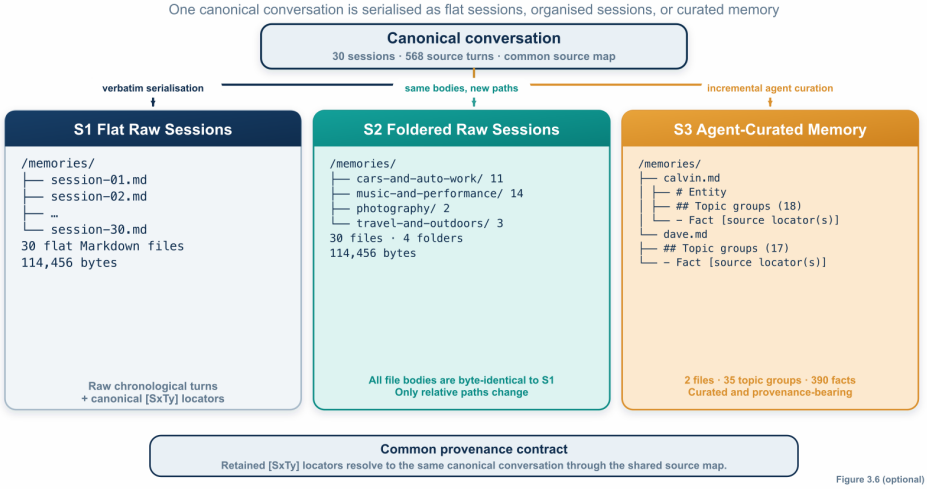


Figure 3.3. Organisation of the three memory representations. S1 stores 30 raw session files in a flat directory; S2 places byte-identical session bodies under four topic folders; S3 reorganises the conversation into two entity files containing 35 topic groups and 390 provenance-bearing facts.

3.5.2 S2: Foldered Raw Sessions

S2 is constructed from a verified copy of S1. It contains the same 30 session files, but places each complete file under one of four topic folders: `music-and-performance`, `photography`, `travel-and-outdoors`, and `cars-and-auto-work`. The body of every session file remains byte-identical to its S1 counterpart. Only the relative path changes.

This design isolates the navigation structure as closely as possible. S2 does not summarise a session, split it into new semantic units, or rewrite its dialogue. It instead adds topic labels that a filesystem Retrieval Agent may use when deciding where to search. The contrast between S1 and S2 therefore asks whether directory organisation adds value when the underlying source text remains fixed.

The path information affects the retrievers in different ways. R1 can inspect and navigate the folder hierarchy directly. R2 does not include folder names in its first-stage BM25 text, so the initial lexical ranking remains based on the same dialogue content as in S1. However, relative paths are visible in the returned observations. They may therefore influence a later query produced by the Retrieval Agent.

S2 also has a limitation. A complete session may contain several themes even though it must occupy one directory. Its folder assignment is therefore a navigation cue rather than a complete semantic description of every source turn.

3.5.3 S3: Agent-Curated Memory

S3 is an agent-curated filesystem. Unlike S2, it changes both the organisation and the wording of the stored information. A Management Agent processes the conversation incrementally and maintains persistent Markdown files. It may create files, add or revise facts, group related information, reconcile later updates, and introduce cross-references. This design follows the broad management-agent paradigm of *Filesystem-Based Memory* [4], while the local provenance and validation rules are project-specific.

The frozen S3 store contains two entity-oriented files, one for each main speaker. Each file uses an H1 heading for the entity and H2 headings for topic groups. Factual list items appear under these topics. The store contains 390 locator-bearing fact units under 35 H2 groups. Every formal fact contains one or more source locators that link it back to the canonical conversation records.

A curated fact may combine information from several source turns or sessions. S3 can therefore place facts that were distant in the raw history into one thematic section. It may also express an updated state more directly than the original dialogue. These properties can reduce local dispersion and provide stronger entity and topic cues.

Curation also introduces a new failure surface. The Management Agent may omit a fact, remove a qualifier, merge different entities, flatten a temporal sequence, or retain a stale state. A later retriever cannot recover information that was never written into S3. Furthermore, curation does not necessarily reduce storage size. In this implementation, detailed consolidation and repeated provenance can make curated facts longer than their source fragments. S3 should therefore be understood as a complete agent-curated memory representation, not as a guaranteed compression method.

Figure 3.3 makes the structural differences among S1, S2, and S3 explicit while retaining their common canonical source basis.

Table 3.3. Comparison of the three memory representations.

Property	S1: Flat Raw Sessions	S2: Foldered Raw Sessions	S3: Agent-Curated Memory
Primary stored unit	Raw session file	Raw session file	Curated factual item
Source wording rewritten	No	No	Yes
Main structure	One flat directory	Four topic folders	Entity files and H2 topic groups
Chronology	Preserved directly	Preserved directly	Represented through facts, dates, and locators
Provenance	Direct source locators	Direct source locators	Inline source locators
Main intended benefit	Maximum source fidelity	Additional navigation cues	Thematic grouping and cross-session consolidation
Principal risk	Relevant facts remain dispersed	Folder taxonomy may be imperfect	Omission, distortion, or stale consolidation

3.6 Three Retrieval Systems

R1, R2, and R3 are evidence-retrieval systems. They do not generate the final benchmark answer. Each retriever observes a question, searches one declared memory representation, and selects evidence. The selected material is then validated and converted into the common EvidenceBundle described in Section 3.7.

3.6.1 R1: Agentic Filesystem Retrieval

R1 is an LLM-controlled filesystem Retrieval Agent based on the filesystem-search paradigm in [4]. It receives the question, store-specific search instructions, and read-only access to one mounted memory store. Its role is to locate supporting source material rather than answer the question.

The Agent can use four filesystem tools:

- `view`, which inspects a file or directory;
- `grep`, which finds text matches across files;
- `toc`, which exposes headings and document structure;
- `section_read`, which reads a selected document section.

These tools provide access but do not determine the search strategy. In particular, `grep` performs deterministic text matching. The Agent must decide which words or expressions to search, which files require inspection, and whether the returned context is sufficient.

A typical R1 episode follows an observation–action loop. The Agent first surveys the available store. It then formulates a search direction, calls one or more read-only tools, inspects the returned observations, and refines its search when evidence remains incomplete. This process continues until the Agent saves sufficient evidence or reaches a stopping condition.

One accepted action-bearing provider completion is defined as one R1 retrieval round. A completion may request several independent filesystem reads, so one round is not equivalent to one tool call. However, every accepted round must use exactly one action mode: filesystem operations, `take_note`, or `finish_search`. These modes cannot be mixed in one completion. Evidence selection and search termination therefore consume rounds of their own. Protocol-correction attempts and context-compaction calls are counted separately and do not create retrieval rounds. Store-specific round limits are reported in Chapter 4.

R1 adds two host-side orchestration actions to the four filesystem tools. These actions are project-specific and are not tools proposed by *Filesystem-Based Memory* [4].

The first action, `take_note`, records an evidence selection. The Agent may identify only line ranges that have already appeared in a successful observation. It cannot supply a rewritten quotation or model-generated summary as evidence. The host reopens the frozen file, verifies that the selected range was observed, extracts the exact text, resolves its source locators, and creates an EvidenceItem.

The second action, `finish_search`, terminates the retrieval episode. It carries a structured stopping reason, but it cannot contain an answer or introduce new evidence. Its purpose is to close retrieval and hand the validated EvidenceBundle to the downstream Answerer. Thus:

$$\texttt{finish_search} \neq \text{answer generation}.$$

R1 uses different routing guidance for the three store shapes. On S1, the Agent searches a flat collection of raw sessions. On S2, it can use topic folders before inspecting the original session text. On S3, it navigates entity files and topic sections. The underlying filesystem capability remains read-only in all three cases.

3.6.2 R2: ReFind-Style Raw-Chat Retrieval

R2 is an iterative lexical retrieval system for raw dialogue. It adapts the ReFind search principle [6] to the canonical exchange units defined in Section 3.4. The local exchange is the functional counterpart of ReFind's paired dialogue turn, not the single-speaker LoCoMo source-turn unit. Its retrieval hierarchy is:

$$\text{exchange} \longrightarrow \text{session}.$$

An exchange is the first-stage scoring unit, while a session is the higher-level grouping unit. R2 operates over 292 exchanges derived from the 568 source turns in the 30-session conversation.

At each search step, the Retrieval Agent generates one lexical query and may also specify a date range. The query is tokenised and scored jointly against every eligible exchange. R2 does not automatically execute a separate search for each query word.

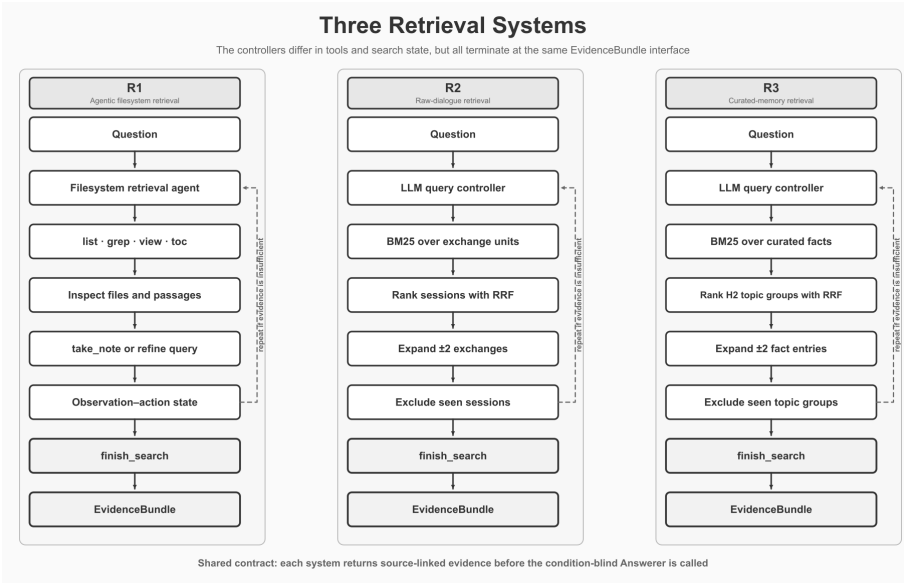


Figure 3.4. Control flow of R1, R2, and R3. R1 navigates a read-only filesystem through iterative observations and actions, R2 ranks raw exchanges and expands within sessions, and R3 ranks curated facts and expands within H2 topic groups. Each path terminates retrieval through `finish_search` and returns a validated EvidenceBundle rather than a benchmark answer.

The indexed exchange text includes speaker names, original dialogue, and available image captions. It excludes gold information and experimental labels. Source locators and folder paths are retained for provenance but do not contribute lexical ranking terms.

R2 first assigns a BM25 score to each eligible exchange [12]. Positive exchange scores are then aggregated within each session to produce a session-level score. Each exchange consequently receives both an exchange rank and the rank of its parent session. The two views are combined by Reciprocal Rank Fusion [13]:

$$\mathrm{RRF}(x) = \frac{1}{60 + r_x} + \frac{1}{60 + r_{\sigma(x)}},$$

where r_x is the exchange rank and $r_{\sigma(x)}$ is the rank of its parent session. The system returns the five highest-ranked centre exchanges.

A centre exchange is not presented in isolation. It is expanded with up to two preceding and two following exchanges from the same session. Expansion never crosses a session boundary. This local window provides the Retrieval Agent with nearby speaker and event context that may not share the query terms.

R2 also maintains a seen-session state. Once a session appears in a search result, that session is excluded from later searches in the same question episode. This rule encourages the Agent to inspect new parts of the conversation rather than repeatedly retrieve the same session.

The controller supports three actions:

1. `search_chatrecord`;
2. `take_note`;
3. `finish_search`.

Each of the four permitted provider completions may produce one valid action. Search, evidence selection, and stopping all consume this same cap. A valid four-completion episode therefore does not necessarily contain four searches. A malformed or empty completion also consumes one completion even though it produces no accepted action.

When the Agent selects a returned context block, the host reopens the raw store and resolves the block to canonical source turns. Overlapping selections are merged deterministically, duplicated source turns are removed, and provenance is validated before an EvidenceItem enters the EvidenceBundle. Model-written text cannot replace the source evidence.

R2 uses the same dialogue text for S1 and S2. Their initial backend ranking is therefore identical for the same query. Later behaviour may differ because S2 paths are visible in observations and can influence subsequent query reformulation. This distinction allows E2 and E4 to test whether folder cues affect an iterative retrieval process even when they are excluded from first-stage lexical scoring.

3.6.3 R3: Retrieval Adapted to Curated Memory

R3 adapts the main R2 retrieval pattern [6] to the structure of S3. It is not a third algorithm proposed by ReFind. The appropriate description is **ReFind-inspired retrieval over agent-curated memory**.

R2 cannot be applied unchanged to S3 because S3 does not consist of raw exchanges grouped by sessions. Its retrieval hierarchy is instead:

$$\text{curated fact} \longrightarrow \text{H2 topic group}.$$

The frozen S3 index contains 390 locator-bearing fact units under 35 H2 topic groups. YAML front matter is excluded because it provides file-level description rather than factual evidence.

Each fact is indexed using three components: the entity name, the H2 topic heading, and the fact text with source-locator syntax removed. Entity and topic terms allow S3's organisation to contribute to lexical retrieval. The separately stored relative-path and source-locator metadata do not become ranking terms. However, a literal cross-reference path written inside the curated fact remains part of the fact text and can affect lexical scoring.

R3 assigns a BM25 score to each eligible fact [12]. Positive fact scores are aggregated within their H2 groups. Fact rank and topic-group rank are then combined using the same Reciprocal Rank Fusion formula as R2 [13]. The five highest-ranked centre facts are returned.

Each centre fact is expanded with up to two neighbouring facts in the same H2 section. Expansion cannot cross an H2 or file boundary. This restriction preserves the thematic context created during memory construction.

R3 tracks seen H2 groups rather than seen sessions. Once a topic group appears in a result, that group is excluded from later searches in the same episode. Other topics for the same entity remain eligible. This is important because one entity file may contain many distinct topic groups.

Temporal filtering also requires an adaptation. A raw exchange belongs to one session date, whereas one curated fact may cite several source turns from different dates. R3 resolves all source dates through the fact's locators. A centre fact is eligible for ranking when at least one of these dates overlaps the requested interval. The subsequent ± 2 neighbourhood expansion preserves local topic context and does not reapply the date predicate, so a neighbouring fact may lie outside the requested interval.

The controller retains the same three-action interface and four-completion cap as R2: `search_chatrecord`, `take_note`, and `finish_search`. Invalid completions also consume the cap. When the Agent selects a result, the host reopens the frozen S3 file, verifies the fact range and H2 boundary, parses its locators, resolves the corresponding canonical records, and constructs the EvidenceItem.

R3 can improve access to information that exists in S3, but it cannot repair a write omission. It is also sensitive to the structure created during curation. Very large topic groups may receive stronger aggregate scores, and a long curated fact may introduce more material than the question requires. These properties form part of the retrieval mechanism rather than evidence of superior or inferior performance.

Figure 3.4 compares the control flow of the three retrieval systems and their common handoff boundary.

Table 3.4. Comparison of the three retrieval systems.

Property	R1	R2	R3
Formal name	Agentic Filesystem Retrieval	ReFind-Style Raw-Chat Retrieval	Retrieval Adapted to Curated Memory
Compatible stores	S1, S2, and S3	S1 and S2	S3
Main retrieval unit	File region or section	Raw exchange	Curated fact
Higher-level group	Filesystem path or section	Session	H2 topic group
Search mechanism	Agent-controlled filesystem tools	BM25, session aggregation, and RRF	BM25, topic aggregation, and RRF
Local context	Agent-selected reads	Up to ± 2 exchanges within one session	Up to ± 2 facts within one H2 group
Seen-state	No explicit backend exclusion state; prior observations remain in the trace	Previously returned sessions	Previously returned H2 groups
Evidence selection	Observed line ranges	Returned exchange windows	Returned fact windows
Stopping mechanism	`finish_search` or round limit	`finish_search` or action limit	`finish_search` or action limit
Principal limitation	Tool-use cost and behavioural variation	Query mismatch or missed implicit relations	Dependence on S3 content and topic structure

3.7 Shared EvidenceBundle and Answerer

The retrieval systems differ internally, but their outputs must reach the Answerer through a common interface. Otherwise, an apparent retrieval effect could instead be caused by a different evidence format or answering prompt. All selective conditions therefore convert their selected material into the same EvidenceBundle schema.

3.7.1 Retrieval-Neutral Evidence Interface

An EvidenceBundle is an ordered collection of validated EvidenceItems associated with one question and one condition. It hides retriever-specific control details from the Answerer while preserving the text and provenance required for answering.

R1 contributes observed filesystem ranges, R2 contributes raw exchange windows, and R3 contributes curated fact windows. These internal units remain visible in the retrieval record, but the Answerer receives them through one common item format. Retrieval scores, search queries, tool observations, and stopping reasons are not part of the answering input.

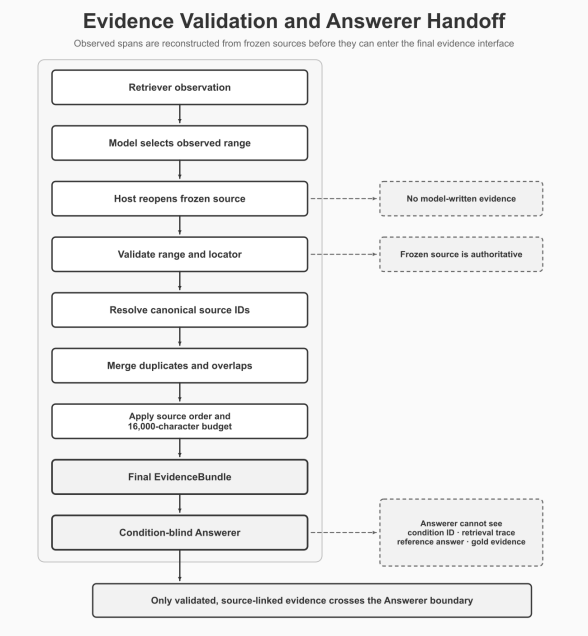


Figure 3.5. Evidence validation and Answerer handoff. Model-selected evidence references are resolved against frozen source bytes, validated, ordered, merged, and budgeted before the condition-blind Answerer generates a structured candidate response.

3.7.2 EvidenceItem Structure

Each EvidenceItem contains:

- a stable evidence identifier;
- the exact selected text;
- the source kind;
- the source path or structural group;
- source locators;
- canonical source-record identifiers;
- source dates and speakers where applicable;
- selection order; and
- a deterministic content identity.

This schema preserves two levels of provenance. The representation-level fields identify where the evidence was read, such as a raw session path or a curated topic group. The canonical fields identify the underlying source turns. Both are required because a curated fact may differ in wording from the source records on which it is based.

3.7.3 Evidence Validation

The Retrieval Agent chooses evidence references rather than authoring evidence text. For R1, it selects previously observed file ranges. For R2 and R3, it selects returned result blocks. The host then reopens the declared frozen store and reconstructs the evidence from source bytes.

This validation boundary prevents a model-generated paraphrase from being treated as retrieved evidence. It also ensures that every source locator resolves through the common provenance layer. Invalid, unobserved, or unresolvable selections are rejected rather than silently repaired.

3.7.4 Evidence Ordering, Merging, and Budget

EvidenceItems retain a deterministic order derived from their accepted selections. Same-path selections with identical, contained, overlapping, or adjacent source ranges may be rejected or merged when the provenance permits it. Repeated anchors are also rejected in R2 and R3. Identical text at two distinct provenance locations is not globally deduplicated.

Each single-source condition from E1 to E6 uses a maximum evidence budget of:

$$\beta = 16{,}000 \text{ characters}.$$

The budget is applied after deterministic evidence merging. Complete EvidenceItems are retained or skipped; their text is not silently cut in the middle. This common limit prevents one selective retriever from obtaining a larger Answerer input simply because it emits more evidence. Chapter 4 treats the budget as a fixed experimental control.

3.7.5 Shared Answerer Input

The Answerer receives the question, the ordered EvidenceBundle, and the shared answering instructions. Its prompt presents evidence through stable identifiers so that the generated answer can cite the supplied material. The Answerer has no retrieval tools and cannot request more memory after the bundle has been finalised.

Full Context uses the same Answerer renderer, answer-generation objective, and structured response contract. To reuse this interface, the 30 complete S1 session bodies are wrapped as 30 content-addressed EvidenceItems in a non-selective full-context bundle. The Answerer therefore cites ordinary evidence identifiers. What Full Context bypasses is question-dependent retrieval and the selective 16,000-character budget, not evidence packaging or citation validation.

3.7.6 Condition-Blind Answering

The Answerer does not receive the condition identifier, store label, retriever name, retrieval trace, retrieval scores, gold answer, gold evidence, or evaluation labels. It is therefore blind to the explicit condition label. It is not blind to evidence provenance: the rendered items expose source kind, text, timestamps, and locators. In E7, the presence of both raw and curated items is consequently visible even though the label "E7" is not.

This condition-label-blind boundary keeps answer generation comparable. E1–E7 may change the memory path and the evidence delivered, but they do not change the role of the Answerer. Any remaining difference enters through the evidence content and provenance or through stochastic model behaviour, not through an explicit condition instruction.

3.7.7 Output Contract

The Answerer returns a structured object with three fields:

- `answer`;
- `citations`;
- `insufficient_evidence`.

A citation may refer only to an evidence identifier present in the input bundle. The `insufficient_evidence` field records whether the Answerer judges the supplied evidence to be inadequate. It does not trigger another retrieval attempt.

If the initial response violates the output schema, the system may perform a constrained format repair. The repair cannot retrieve new material or change the EvidenceBundle. Its prompt instructs the model to preserve the answer content, use only allowed evidence identifiers, and add no new facts or citations. The purpose is to recover a valid representation of the original response; the repair is not treated as another retrieval or answering stage.

Figure 3.5 summarises the validation, packaging, and answering boundary shared by the selective conditions.

3.8 Mapping of E1–E7 Conditions

3.8.1 Complete Condition Map

The seven experimental conditions describe complete paths from a memory representation to the shared Answerer. They are not seven independent retrieval algorithms.

Table 3.5. Storage, retrieval, and evidence composition of E1–E7.

Condition	Memory representation or source	Retrieval or processing	Evidence supplied to the Answerer	Primary methodological role
E1	S1: Flat Raw Sessions	R1	Raw filesystem spans	Flat raw-filesystem reference path
E2	S1: Flat Raw Sessions	R2	Ranked raw exchange windows	Enhanced retrieval over flat raw dialogue
E3	S2: Foldered Raw Sessions	R1	Raw filesystem spans	Effect of directory organisation on R1
E4	S2: Foldered Raw Sessions	R2	Ranked raw exchange windows	Interaction between folder cues and R2
E5	S3: Agent-Curated Memory	R1	Curated filesystem spans	General filesystem retrieval over curated memory
E6	S3: Agent-Curated Memory	R3	Ranked curated fact windows	Retrieval adapted to curated structure
E7	E2 and E6 EvidenceBundles	Deterministic fusion	Interleaved raw and curated evidence	Dual-source complementarity and interference

3.8.2 Controlled Raw-Memory Core

E1–E4 form the controlled raw-memory core. E1 and E2 share S1 but use different retrieval systems. E3 and E4 share S2 but use different retrieval systems. E1 and E3 use R1 over the same raw session contents under different directory structures. E2 and E4 apply R2 to the same raw exchange corpus while exposing different provenance paths in the observations.

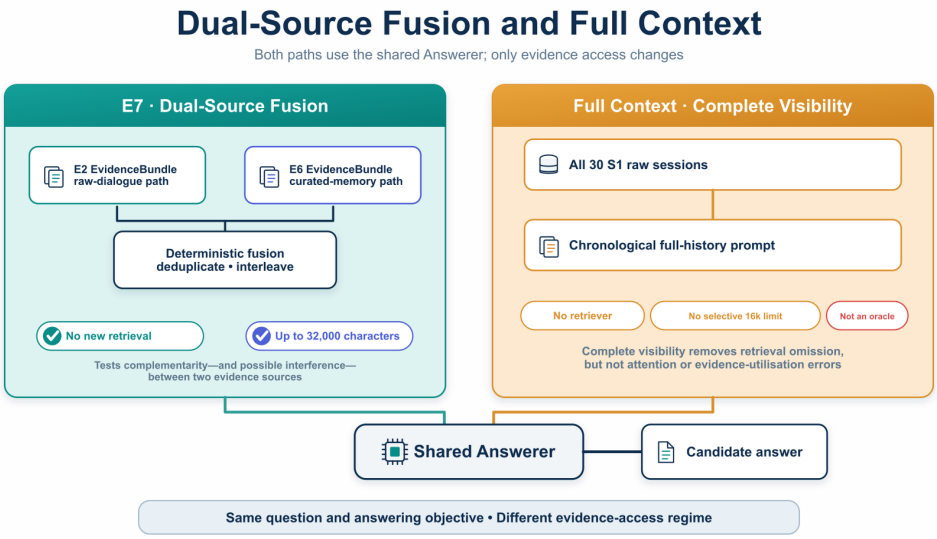


Figure 3.6. Comparison of E7 dual-source fusion and the Full Context diagnostic path. E7 deterministically combines the completed E2 and E6 EvidenceBundles without new retrieval, whereas Full Context exposes the complete ordered raw history without the selective 16,000-character limit. Both paths use the shared Answerer.

These relations support the closest component-level comparisons in the study. They distinguish the effect of directory organisation from the effect of the retrieval family without changing the underlying dialogue.

3.8.3 Curated-Memory Extension

E5 and E6 use the same frozen S3 snapshot. E5 asks a general filesystem Agent to navigate curated files and sections. E6 applies retrieval that is explicitly adapted to S3's fact and topic structure.

These conditions compare access methods over the same curated store. Comparisons between E5 or E6 and raw-store conditions are broader system comparisons. S3 changes wording, granularity, organisation, and the presentation of provenance, so such comparisons cannot be interpreted as a single-feature storage ablation.

3.8.4 Dual-Source Condition

E7 is constructed from the completed E2 and E6 EvidenceBundles. It does not add another memory representation or retrieval system. Its purpose is to test whether already selected raw and curated evidence provide complementary information or create interference when presented together.

The parent pair is methodologically meaningful. E2 supplies evidence from raw dialogue through the retrieval system designed for raw chat. E6 supplies evidence from curated memory through the adapter designed for curated facts. Their fusion therefore contrasts two complete, representation-appropriate evidence paths.

3.8.5 Interpretation Boundaries

The condition set is theory-driven rather than an exhaustive Cartesian product. R3 is a representation-specific adaptation of the R2 retrieval principle, not a universal retriever that must be paired with every store. E7 is a fusion condition, not a new retrieval algorithm. Full Context is a diagnostic path outside E1–E7.

The strength of an inference also depends on the comparison. E1 versus E3 and E2 versus E4 isolate directory organisation most directly because the raw file contents are held constant. E1 versus E2, E3 versus E4, and E5 versus E6 compare retrieval systems on one fixed store. Comparisons between raw and curated conditions concern complete system paths because curation changes several properties at once.

Replicating the same condition graph with another backbone model does not create new E-condition definitions. It tests whether the observed effects remain stable when the model changes.

3.9 Alternative Evidence-Access Paths

E7 and Full Context depart from the standard single-store selective path in different ways. E7 retains selective retrieval but combines two completed evidence sources. Full Context removes external evidence selection and sends the complete raw conversation to the Answerer.

3.9.1 E7 Dual-Source Fusion

E7 requires a completed E2 EvidenceBundle and a completed E6 EvidenceBundle for the same question. The two parent bundles represent different evidence sources:

- E2 provides raw dialogue evidence selected through R2;
- E6 provides curated facts selected through R3.

Before fusion, the system verifies that both bundles correspond to the same question and the expected parent conditions. Their provenance and single-source budgets must also remain valid. If either parent is missing or mismatched, E7 is not produced.

Fusion is deterministic. The order inside each parent bundle is preserved, while the two streams are interleaved:

$$\text{raw}_1,\ \text{curated}_1,\ \text{raw}_2,\ \text{curated}_2,\ \ldots$$

This rule prevents one source from occupying the entire beginning of the Answerer input.

Exact duplicates are identified using the combination of source kind, canonical source identifiers, and evidence-text identity:

$$\big(\text{source kind},\ \text{source identifiers},\ \text{text hash}\big).$$

Source kind remains part of the key because raw and curated items may cite the same source turns while expressing them differently. Such items are retained when their text differs. This allows E7 to preserve the distinction between original dialogue and its curated representation.

Each parent bundle retains the single-source budget:

$$|\mathcal{B}_{E2}| \le \beta, \qquad |\mathcal{B}_{E6}| \le \beta.$$

The fused evidence is therefore bounded by:

$$|\mathcal{B}_{E7}| \le 2\beta = 32{,}000 \text{ characters}.$$

The fused bundle is then passed to the shared Answerer once. Fusion does not formulate a query, rerun R1, R2, or R3, invoke an LLM, or create an intermediate summary. E7 therefore measures post-retrieval complementarity and interference rather than the performance of a new joint retriever.

3.9.2 Full Context Diagnostic Path

Full Context uses a separate evidence-access path. For each question, it constructs one deterministic input containing all 30 raw sessions and all 568 source turns. Sessions and turns remain in fixed chronological order. Original dialogue and available image captions are retained. No question-dependent filtering, summarisation, or reordering is performed.

The Full Context answer can be represented as:

$$\hat{a}_{FC} = A_\theta(q, \mathcal{C}),$$

where $\mathcal{C}$ is the complete canonical raw conversation. In contrast, a selective condition uses:

$$\hat{a} = A_\theta(q, \mathcal{B}),$$

where $\mathcal{B}$ is a question-specific EvidenceBundle.

Full Context does not use S2, S3, R1, R2, R3, or E7 fusion. It packages the 30 complete S1 session bodies as a non-selective full-context bundle so that it can use the shared Answerer interface. It does not apply the 16,000-character selective-evidence limit. Applying that limit would convert it into a truncated-context condition rather than Full Context.

The Answerer retains the same answer-generation objective and structured output contract where applicable. The principal change is the information-access path: the model must locate and combine relevant information inside the complete history instead of receiving externally selected evidence.

Full Context is a diagnostic path, not a fourth memory store, a fourth retriever, or an eighth E-condition. It is also not a theoretical oracle. Complete visibility does not ensure correct use, and long-context performance can depend on evidence position [5]. The Answerer may still be affected by repeated facts, changing states, entity similarity, temporal confusion, or failed multi-session composition.

The comparison is informative in both directions. If Full Context repairs a selective-retrieval error, missing evidence is a plausible explanation. If selective retrieval succeeds while Full Context fails, evidence reduction may have improved focus. If both paths receive the required information but fail, the remaining difficulty is more likely to involve evidence utilisation or reasoning [7]. Chapter 4 defines how these paths are evaluated, and Chapter 5 examines their observed behaviour.

Figure 3.6 contrasts the two alternative evidence-access paths while holding the answering objective constant.

This chapter has defined the complete system graph. E1–E4 form the controlled raw-memory core, E5–E6 extend the design to agent-curated memory, E7 combines two completed evidence sources, and Full Context removes explicit retrieval as a diagnostic comparison. All selective conditions converge on the same EvidenceBundle and Answerer boundary.

Chapter 4 instantiates this architecture as a controlled experiment. It consolidates the method constants defined here and specifies the frozen question set, backbone models, remaining run controls, execution protocol, evaluation metrics, and paired statistical analysis.

Chapter 4. Experimental Design and Evaluation Methodology

Chapter 3 defined the memory representations, retrieval systems, EvidenceBundle interface, Answerer, dual-source fusion mechanism, and Full Context path. This chapter explains how those components are instantiated as a controlled experiment. It defines the frozen test set, experimental matrix, model roles, controlled conditions, evaluation metrics, paired analysis, and error-review method.

The chapter separates design from outcome interpretation. It specifies which comparisons are valid and what each comparison can identify. It does not report final scores or select a winning condition. Those results are presented in Chapter 5.

4.1 Research Questions, Variables, and Identification Strategy

4.1.1 Mapping Research Questions to Experimental Comparisons

The six research questions introduced in Chapter 1 require different experimental evidence. RQ1 concerns storage, retrieval, and their interaction. RQ2 asks whether the same architectural effects remain stable across backbone models. RQ3 examines dual-source fusion. RQ4 compares selective retrieval with Full Context. RQ5 locates likely failures along the information path. RQ6 combines quality, stability, cost, and maintenance considerations.

No single condition answers all six questions. The study therefore assigns each research question a defined comparison family. This mapping prevents an overall condition ranking from replacing the analysis required by each question.

Table 4.1. Mapping from research questions to experimental comparisons and evidence.

Research question	Main experimental material	Principal comparisons	Main evidence
RQ1: storage and retrieval effects	E1–E6	E3−E1, E4−E2, E2−E1, E4−E3, E6−E5, and the storage–retrieval interaction	Paired answer-quality differences, evidence coverage, and retrieval traces
RQ2: cross-backbone stability	Qwen and DeepSeek E1–E7	The same condition contrast under both backbone models	Effect direction, magnitude, question-level agreement, and contrast-of-contrasts
RQ3: dual-source fusion	E2, E6, and E7	E7−E2 and E7−E6	Repair, preserve, harm, unresolved cases, evidence overlap, and cost
RQ4: Full Context versus selective retrieval	DeepSeek selective conditions and DeepSeek-family Full Context	Paired Full Context comparisons, with the raw-memory conditions as the principal diagnostic family	Answer quality, context size, latency, and paired success or failure states
RQ5: failure location	Stores, retrieval traces, EvidenceBundles, candidate answers, and evaluation outputs	Stage-based review within and across conditions	Storage fidelity, evidence coverage, budget records, answer utilisation, and manual cases
RQ6: practical choice	All formal conditions	Quality–cost and stability comparisons	Answer quality, calls, tokens, latency, construction cost, recovery cost, and maintenance implications

The comparisons in Table 4.1 are paired by question whenever both conditions produce an answer for the same Main-60 item. The pair, rather than the aggregate condition mean alone, is the basic comparative unit.

Figure 4.1 summarises the complete experimental and evaluation workflow. It also makes the evaluation boundary explicit: gold answers and gold evidence enter only after candidate-answer generation.

4.1.2 Independent, Dependent, and Controlled Variables

The experiment varies five main factors. The first is the **memory representation**: S1, S2, or S3. The second is the **retrieval system**: R1, R2, or R3, subject to the compatible pairings defined in Chapter 3. The third is the **backbone model** used by the Retrieval Agent and Answerer. The fourth is the **evidence-access regime**, which distinguishes selective retrieval from Full Context. The fifth is **evidence composition**, which distinguishes a single EvidenceBundle from E7 dual-source fusion.

The dependent variables cover four levels. Answer-level lexical measures are token Precision, Recall, and F1. Answer-level semantic measures are Strict Accuracy, Partial Rate, and Soft Semantic Score. Evidence-level measures are Evidence Recall, Any-hit, and All-hit. Process measures include calls, tokens, latency, retrieval rounds or actions, failed attempts, recovery work, and construction cost.

The principal controls are the unified Main-60 test set, its fixed order, the canonical conversation, frozen S1–S3 snapshots, condition definitions, the EvidenceBundle contract, the Answerer contract, and the offline evaluation rules. These controls are applied wherever they are meaningful. Full Context is not subject to a selective-evidence budget because doing so would change it into a truncated-context condition.

4.1.3 Identifiable and Non-Identifiable Effects

The raw-memory conditions provide the cleanest component comparisons. S1 and S2 contain the same raw session bodies. E3−E1 and E4−E2 therefore examine directory organisation under R1 and R2, respectively. E2−E1 and E4−E3 compare the two retrieval systems while holding the raw store fixed. Together, E1–E4 support a storage–retrieval interaction analysis.

These contrasts still have explicit limits. R1 and R2 are complete retrieval systems with different observation and action interfaces. Their difference cannot be reduced to one ranking algorithm. In E2 and E4, directory names do not enter the initial lexical score, but returned provenance paths may affect a later Retrieval Agent action. E4−E2 can therefore capture adaptive use of folder cues, not a direct effect on first-stage BM25 scoring.

E5 and E6 use the same frozen S3 store. E6−E5 compares a general filesystem retriever with retrieval adapted to curated facts and topic groups. Comparisons between S3 and a raw store are broader. S3 changes wording, information granularity, organisation, and provenance presentation together. A raw-versus-curated difference is therefore a complete-system difference rather than a single-component effect.

E7 evaluates fusion. It reuses the completed E2 and E6 EvidenceBundles and performs no new retrieval. Its comparison with either parent tests whether a second evidence source repairs, preserves, or harms the answer. It does not estimate the value of a third retrieval algorithm.

The backbone comparison also changes more than one role. Within an E1–E7 pipeline, the same backbone model serves as both Retrieval Agent and Answerer. A Qwen–DeepSeek difference is therefore an end-to-end query-time pipeline difference. It does not isolate retrieval capability from answer-generation capability.

Full Context is diagnostic. It removes explicit retrieval and expands the information visible to the Answerer. It also changes input scale. It is not a strict one-factor ablation, an eighth E-condition, or an oracle. In addition, Full Context uses a later served identifier within the same DeepSeek model family. Comparisons with the DeepSeek selective conditions must therefore be interpreted as strong descriptive diagnostics rather than strict single-factor causal estimates.

4.1.4 Primary and Exploratory Analyses

The primary analyses use all 60 questions and retain same-question pairing. They cover the raw-memory storage and retrieval contrasts, the S3 access contrast, the E7 parent comparisons, matched architectural effects across backbone models, and the Full Context diagnostic comparisons. The analysis emphasises effect direction, effect magnitude, and the distribution of question-level changes.

Category-level analyses use the four non-adversarial LoCoMo categories. They are exploratory mechanism analyses. They indicate where an effect may be concentrated, but they do not replace the full-set result. Correlation analyses, anomaly inspection, and individual cases are also exploratory.

The term **primary comparison** is used to describe analytical priority. It does not imply that the full report was preregistered before every run. Claims in Chapter 5 must therefore distinguish frozen design choices from later exploratory interpretation.

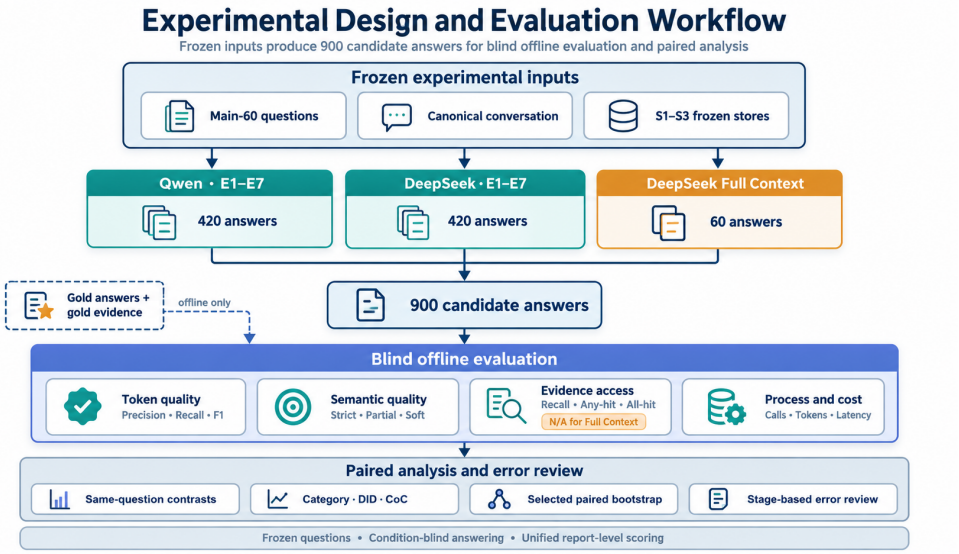


Figure 4.1. Experimental design and evaluation workflow. Frozen inputs are processed through the Qwen E1–E7 matrix, the DeepSeek E1–E7 matrix, and the DeepSeek Full Context path. Their 900 candidate answers then enter condition-blind offline evaluation, paired analysis, and stage-based error review. Gold answers and gold evidence are restricted to offline evaluation.

4.2 Dataset and the Frozen Main-60 Test Set

4.2.1 LoCoMo Source and Conversation Scope

The experiment uses the question-answering material from LoCoMo [1]. The frozen source file was obtained from the official LoCoMo repository at commit `cbfbc1dba6bc53d00625212a0f22d55ffee7c1fc` and canonicalised under the local data-contract version `locomo-canonical-v1`. The study focuses on the released conversation identified locally as `conv-50`. This conversation contains 30 ordered sessions and 568 source turns. The source preserves speaker identity, session order, dates, dialogue text, and the released image captions where present.

The study does not rerun a vision model. Existing captions are treated as part of the released textual record. All storage representations and retrieval indices originate from the same canonicalised conversation described in Chapter 3.

Using one conversation supports detailed provenance checks and exact same-question comparisons. It also limits external validity. The results describe controlled variation within this conversation and question set. They do not estimate variation across LoCoMo conversations, users, or datasets.

The source-file fingerprint, licence record, ordered question identifiers, source manifest, and evidence-eligibility fields are recorded in Appendix B.

4.2.2 Composition of Main-60

Main-60 is one frozen test set of 60 questions. It contains only the four non-adversarial categories used in this study. The fixed distribution is shown in Table 4.2.

Table 4.2. Composition and evidence eligibility of Main-60.

Category	Label used in this report	Questions	Evidence-eligible questions	Role in analysis
C1	Multi-hop	13	13	Requires information from more than one part of the history
C2	Temporal	13	13	Requires time, order, update, or state tracking
C3	Open-domain	7	5	Combines dialogue information with wider knowledge or inference
C4	Single-hop	27	27	Uses a more localised fact or relation
Total	**Unified Main-60**	**60**	**58**	**Primary full-set analysis unit**

The question order is fixed across conditions and backbone models. Selection did not use candidate answers, evaluation scores, or model outputs. The 60 questions are analysed as one set. The report does not divide its main results by when a question was added to the experimental suite.

The category counts are unequal. C3 contains only seven questions. Category comparisons are therefore descriptive and should be interpreted with particular care for small groups.

4.2.3 Gold Answers and Gold Evidence

Each Main-60 question has a reference answer used for offline answer evaluation. LoCoMo also supplies source-turn annotations for many questions [1]. The two forms of gold information have different roles. The reference answer supports token and semantic evaluation. The source identifiers support evidence-access evaluation.

Fifty-eight Main-60 questions have usable annotated evidence for the evidence metrics. Two questions do not. The denominator for Evidence Recall, Any-hit, and All-hit is therefore 58, not 60. This eligibility rule is fixed before condition-level evidence aggregation.

Annotated source turns are evidence locations, not complete reasoning chains. A valid answer may use an alternative supporting passage, especially for open-domain questions. Evidence measures are therefore diagnostic indicators of annotated-source coverage. They are not a complete definition of retrieval correctness.

Gold answers, gold evidence, question categories, and evaluation labels are unavailable to the online memory pipeline. They are attached only after candidate-answer generation. The Retrieval Agent, deterministic retrieval backend, fusion procedure, and Answerer cannot use them.

4.3 Experimental Matrix and Core Comparisons

4.3.1 Complete Experimental Matrix

The formal study contains three answer sets. Qwen produces one answer for every Main-60 question under E1–E7. DeepSeek produces the same E1–E7 matrix. A DeepSeek-family Full Context path produces one further answer per question. The complete scale is shown in Table 4.3.

Table 4.3. Formal experimental matrix.

Answer set	Conditions or path	Questions per condition	Candidate answers	Query-time model scope
Model 1	Qwen E1–E7	60	420	Qwen Retrieval Agent and Qwen Answerer
Model 2	DeepSeek E1–E7	60	420	DeepSeek Retrieval Agent and DeepSeek Answerer
Diagnostic baseline	DeepSeek-family Full Context	60	60	DeepSeek-family Answerer; no Retrieval Agent
Total	**Two E1–E7 matrices plus Full Context**	—	**900**	—

The 900 answers are not 900 independent test questions. They are repeated observations over the same 60 questions. Statistical comparisons must preserve this structure.

Every E1–E7 answer is associated with its storage path, retrieval trace, EvidenceBundle, Answerer record, and process measurements. Full Context has no retrieval trace. It records the complete source input, Answerer output, and applicable process measurements.

4.3.2 E1–E4: Controlled Storage–Retrieval Comparisons

E1–E4 form the controlled raw-memory core. As defined in Table 3.5, E1 and E2 use S1 with R1 and R2, respectively, while E3 and E4 use S2 with the same two retrieval systems.

The storage-organisation contrasts are E3−E1 and E4−E2. The retrieval-system contrasts are E2−E1 and E4−E3. The same four cells also support a difference-in-differences description of whether directory organisation affects R1 and R2 differently.

For a per-question metric m_i, the storage–retrieval interaction is defined as:

$$DID_i^{(m)} = \big(m_i(E4) - m_i(E2)\big) - \big(m_i(E3) - m_i(E1)\big)\,.$$

The raw interaction is the mean of these paired question-level values. For cross-metric directional interpretation, it is oriented using the metric-direction term o_m defined in Section 4.7.2:

$$DID_{i,+}^{(m)} = o_m\, DID_i^{(m)}.$$

A positive oriented value means that the S1-to-S2 change is more favourable under R2 than under R1. The interaction does not identify which internal R2 action caused the difference.

4.3.3 E5–E6: Curated-Memory Systems

E5 and E6 share the same frozen S3 Agent-Curated Memory. E5 uses R1, while E6 uses R3. The contrast E6−E5 therefore compares two access methods on one curated representation.

This contrast is narrower than a raw-versus-curated comparison, but it is still a complete retrieval-system comparison. R1 navigates files and sections through filesystem tools. R3 ranks curated facts and topic groups, expands local context, and uses topic-level seen-state. Their output is standardised only at the EvidenceBundle boundary.

Comparisons between E5 or E6 and E1–E4 are retained as complete-system comparisons. They may show whether a curated-memory path is useful as deployed. They cannot isolate the effects of summarisation, fact consolidation, directory structure, or provenance formatting.

4.3.4 E7: Dual-Source Fusion

E7 reuses the completed E2 and E6 EvidenceBundles for the same question. E2 supplies selected raw-dialogue evidence. E6 supplies selected curated-memory evidence. Fusion validates the parent identities, removes exact provenance-aware duplicates, preserves within-source order, and interleaves the two evidence streams.

The two principal comparisons are E7−E2 and E7−E6. They test whether adding the second evidence source repairs an error, preserves a correct answer, introduces harm, or leaves an error unresolved. E7 does not make a new search call and does not generate an intermediate summary.

Its cost has two interpretations. The recorded E7 operation measures the post-retrieval fusion and fused Answerer call because the parent bundles already exist in the experiment. A deployable online E7 system would need to execute both parent retrieval paths before the fused Answerer call. Chapter 5 reports these two cost views separately.

4.3.5 Full Context Diagnostic

For each question, Full Context supplies all 30 raw S1 sessions in fixed chronological order. It performs no explicit retrieval, filtering, summarisation, or evidence reordering. It also does not apply the selective EvidenceBundle limit.

Full Context is compared with the DeepSeek selective conditions on the same questions. The raw-memory E1–E4 family provides the principal diagnostic anchor because it avoids the additional content transformation introduced by S3. Results against E5–E7 are also reported, but they answer broader system-level questions.

Full Context has complete source visibility by construction, but this property is not Evidence Recall because no selection process exists. Section 4.6.3 defines the resulting metric-applicability rule.

The diagnostic comparison is bidirectional. A Full Context success paired with a selective-retrieval failure is consistent with retrieval omission or evidence-budget loss. A selective-retrieval success paired with a Full Context failure is consistent with a benefit from evidence selection or reduced noise. Neither pattern establishes a cause without trace and case review.

4.4 Models and Evaluation Roles

4.4.1 Backbone Models and Query-Time Roles

The experiment uses two pipelines with different backbone models for E1–E7. Model 1 uses a Qwen served model. Model 2 uses a DeepSeek served model. Within each pipeline, the same backbone model performs both Retrieval Agent and Answerer roles. The prompts, stores, condition graph, EvidenceBundle interface, and evaluation rules remain fixed.

Full Context uses the DeepSeek V4.1 Flash family but a later served identifier. It contains only the Answerer role because explicit retrieval is absent. Table 4.4 records the model identities returned by the service.

Table 4.4. Backbone models and evaluation roles.

Report role	Requested service alias	Recorded served identity	Applicable conditions	Role in the pipeline
Model 1: Qwen	`coding`	`qwen3.8:27b`	E1–E7	Retrieval Agent and Answerer
Model 2: DeepSeek	`x-test-2`	`deepseek-v4.1-flash`	E1–E7	Retrieval Agent and Answerer
DeepSeek-family Full Context	`x-test-2`	`deepseek-v4.1-flash-1017`	Full Context	Answerer only
Independent LLM judge	`x-test-1`	`glm-5.3-flash`	All 900 candidate answers	Offline semantic evaluator

The served identities are deployment records, not independently verified descriptions of model weights or training data. The Full Context comparison is described at the model-family level because its served identity is not identical to that of Model 2 E1–E7.

Detailed model metadata and fixed configuration values belong in Appendix F.

4.4.2 Independent LLM Judge

Semantic evaluation uses one independent LLM judge from a different model family. The judge receives only the question, reference answer, and candidate answer. It does not receive the condition identifier, backbone model, storage representation, retrieval system, trace, EvidenceBundle, cost, token score, or prior judgement.

The judge applies one fixed three-label rubric. A response is `correct` when it contains all required information and has no material contradiction. It is `partial` when it contains a relevant but incomplete answer, or when correct material is mixed with a material unsupported claim. It is `incorrect` when the required answer is missing, contradicted, or replaced by a substantively wrong response.

One judgement produces all three semantic measures defined in Section 4.6.2. The study does not use one model for strict scoring and another for partial scoring. The LLM judge supplements deterministic token evaluation because semantic equivalence may not preserve lexical overlap. It is not treated as error-free ground truth, since LLM-based evaluation can exhibit position, verbosity, and reasoning biases [8].

The complete judge prompt, rubric, output schema, and prompt identity belong in Appendix A. Judge identity, blindness controls, parsing rules, and evaluation configuration belong in Appendices F and G.

4.5 Controlled Experimental Conditions

Fair comparison requires both shared controls and explicit exceptions. Table 4.5 separates factors that are held fixed, deliberately varied, or not applicable.

Table 4.5. Held-fixed, varied, and non-applicable experimental factors.

Factor	Held-fixed component	Intended variation	Non-applicable or bounded exception
Questions	Unified Main-60 membership and order	None	None
Source conversation	Canonical `conv-50` records	None	None
Memory stores	Frozen S1–S3 snapshots across query-time runs	S1, S2, or S3 according to condition	Full Context reads complete S1 session bodies rather than a selective store path
Retrieval	Frozen R1–R3 definitions and condition pairing	Retrieval system according to E1–E6	E7 reuses parent bundles; Full Context has no retriever
Evidence input	EvidenceItem schema, provenance validation, deterministic ordering, and shared renderer	Retrieved content and E7 source composition	Full Context is not subject to the selective-evidence limit
Answer generation	Shared Answerer objective and structured output contract	Backbone model and supplied evidence	Full Context uses the same objective with complete raw history
Backbone model	One model is used consistently within each E1–E7 pipeline	Qwen versus DeepSeek	Full Context uses only the DeepSeek family
Evaluation	Frozen reference data, metric definitions, eligibility rules, and LLM judge rubric	None	Retrieval metrics are `N/A` for Full Context

Selective conditions use the fixed evidence budget and retriever-specific search limits introduced in Chapter 3. These values were frozen before formal evaluation and were not tuned from Main-60 outcomes. The same condition definitions are used under Qwen and DeepSeek.

The controls do not make every comparison a single-factor experiment. S3 changes several representation properties together. R1, R2, and R3 expose different retrieval mechanisms. Changing the backbone model changes both retrieval and answering. Full Context changes the access mechanism and context scale. These limits are part of the interpretation, not implementation noise.

Exact budgets, caps, decoding settings, model identities, and condition contracts belong in Appendix F. Complete prompts belong in Appendix A. Local process-management details are not part of the scientific comparison and are omitted from the report.

4.6 Evaluation Metrics and Cost Accounting

The evaluation uses complementary answer, evidence, and process measures. Each metric has a defined unit, denominator, aggregation rule, and applicability range. No single score is used as a complete measure of system quality.

4.6.1 Token Precision, Recall, and F1

Token evaluation follows the category-aware LoCoMo scoring basis [1, 14]. Category-specific alignment is performed where required. Each aligned candidate–reference pair is then normalised before token comparison. The implementation converts text to lowercase, removes punctuation and commas, removes the articles and conjunctions `a`, `an`, `the`, and `and`, collapses whitespace, and applies Porter stemming.

All 900 candidate answers are rescored offline with this single evaluator before report-level aggregation. Scores emitted by individual generation runners are retained only as execution diagnostics and are not mixed into the reported aggregates. This common rescoring step is required for direct Precision and Recall comparisons between E1–E7 and Full Context.

Let P_i be the normalised candidate-token multiset for question i, and let G_i be the aligned reference-token multiset. Let c_i be their multiset overlap count. For the ordinary token comparison, per-question Precision and Recall are:

$$\text{Precision}_i = \frac{c_i}{|P_i|}, \qquad \text{Recall}_i = \frac{c_i}{|G_i|}.$$

If either aligned multiset is empty, Precision and Recall are both defined as zero for that comparison. This rule prevents division by zero and is applied before F1 is calculated.

The per-question F1 score is:

$$\text{F1}_i = \begin{cases} \dfrac{2\,\text{Precision}_i\,\text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}, & \text{Precision}_i + \text{Recall}_i > 0, \\ 0, & \text{otherwise.} \end{cases}$$

The implementation retains LoCoMo's category-aware alignment. For C1, let g_{ik} be reference item k and let p_{ij} be candidate item j, after splitting both strings on commas. Each reference item is matched to the candidate item that gives the highest item-level F1:

$$j^*(k) = \arg\max_j \text{F1}(p_{ij}, g_{ik}).$$

The question-level C1 measures average the matched item-level measures:

$$m_i = \frac{1}{K_i} \sum_{k=1}^{K_i} m\Big(p_{i,j^*(k)}, g_{ik}\Big), \qquad m \in \{\text{Precision}, \text{Recall}, \text{F1}\}.$$

If no valid reference item remains after parsing, all three C1 measures are defined as zero.

For C3, only the part of the reference before the first semicolon is used. C2, C3, and C4 otherwise use the ordinary normalised token comparison. Exact parsing and edge-case rules are recorded in Appendix G.

For any per-question metric m_i, the condition-level macro-average is:

$$\overline{m} = \frac{1}{N} \sum_{i=1}^{N} m_i.$$

Thus, reported aggregate F1 is the mean of the 60 per-question F1 values. It is not recomputed from aggregate Precision and aggregate Recall.

Precision and Recall are project-defined diagnostics on the same matching basis. Precision helps identify additional or non-matching answer text. Recall helps identify omitted reference content. F1 provides a balanced lexical score and supports alignment with LoCoMo-style evaluation. All three remain sensitive to wording and answer length.

4.6.2 LLM Judge Metrics

Let the LLM judge label for question i be:

$$y_i \in \{\texttt{correct}, \texttt{partial}, \texttt{incorrect}\}.$$

Strict Accuracy gives credit only to `correct` responses:

$$\text{StrictAccuracy} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{I}(y_i = \texttt{correct}).$$

Partial Rate reports the proportion of answers labelled `partial`:

$$\text{PartialRate} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{I}(y_i = \texttt{partial}).$$

Soft Semantic Score assigns full credit to `correct`, half credit to `partial`, and zero credit to `incorrect`:

$$s_i^{\text{soft}} = \begin{cases} 1, & y_i = \texttt{correct}, \\ 0.5, & y_i = \texttt{partial}, \\ 0, & y_i = \texttt{incorrect}, \end{cases}$$

and:

$$\text{SoftSemanticScore} = \frac{1}{N}\sum_{i=1}^{N} s_i^{\text{soft}}.$$

Partial Rate is descriptive. A high Partial Rate is not necessarily desirable because it may reflect movement from `incorrect` to `partial` or from `correct` to `partial`. It must be interpreted with Strict Accuracy, Soft Semantic Score, and label-transition counts.

The semantic measures and token measures answer different questions. A semantically correct paraphrase may receive a low token score. A response may also repeat reference terms while expressing a wrong relation. Chapter 5 therefore reports their agreement and disagreement rather than treating one family as the ground truth for the other.

4.6.3 Evidence-Access Metrics

Evidence evaluation compares annotated gold source turns with the canonical source identifiers linked to the retrieved EvidenceItems. Let G_i^E be the set of eligible annotated gold source identifiers for question i. Let R_i^E be the union of canonical source identifiers represented in its EvidenceBundle.

Per-question Evidence Recall is:

$$\text{EvidenceRecall}_i = \frac{|G_i^E \cap R_i^E|}{|G_i^E|}.$$

Any-hit records whether at least one annotated source is present:

$$\text{AnyHit}_i = \mathbf{I}\Big(|G_i^E \cap R_i^E| > 0\Big).$$

All-hit records whether every annotated source is present:

$$\text{AllHit}_i = \mathbf{I}\Big(G_i^E \subseteq R_i^E\Big).$$

Condition-level evidence metrics are macro-averaged over the 58 eligible questions. The underlying set operation removes repeated canonical identifiers. It does not treat duplicate evidence items as additional recall.

Evidence Recall measures source coverage rather than evidence precision or downstream use. A bundle can cover all annotated sources and still contain distracting material. The Answerer may also fail despite All-hit, or answer correctly from valid support not listed in the annotations. Evidence metrics are therefore used to diagnose access, not to replace answer evaluation.

Full Context receives the whole source history and does not perform retrieval. Its Evidence Recall, Any-hit, and All-hit values are `N/A`. Complete visibility by construction must not be reported as perfect retrieval.

4.6.4 Process and Efficiency Metrics

Process records distinguish retrieval, answering, fusion, recovery, and evaluation. For each selective answer, the study records Retrieval Agent model calls, filesystem or orchestration calls where applicable, input tokens, output tokens, total tokens, elapsed time, retrieval rounds or actions, failed attempts, and format-repair or retry activity. The Answerer has its own call and token record.

The accounting separates four cost classes:

1. **Memory-construction cost:** one-time work required to create S2 or S3 from the canonical source.
2. **Query-time cost:** retrieval, evidence packaging, and Answerer work required for one question.
3. **Failure and recovery cost:** failed requests, retries, protocol correction, and format repair.
4. **Evaluation-only cost:** token and call cost of the independent LLM judge.

Calls, tokens, monetary charges, and latency are not added into one synthetic scalar. Let d denote one additive cost dimension, such as model calls, input tokens, output tokens, total tokens, or monetary charge. For condition c and question i, each dimension is accounted for separately:

$$C_{i,c,d}^{\text{query}} = C_{i,c,d}^{\text{retrieval}} + C_{i,c,d}^{\text{packaging}} + C_{i,c,d}^{\text{answering}} + C_{i,c,d}^{\text{recovery}}.$$

The packaging term covers EvidenceBundle validation, ordering, merging, and budget enforcement. These operations are deterministic. Their value is therefore zero for model-call, model-token, and API-charge dimensions unless a deployment records a separate non-zero charge. Their wall-clock contribution remains part of the observed latency discussed below.

Judge cost is excluded from deployment query cost:

$$C_{i,c,d}^{\text{recorded}} = C_{i,c,d}^{\text{query}} + C_{i,c,d}^{\text{judge}},$$

with the query and judge components reported separately for every dimension.

For a store that requires one-time construction and is queried Q times, an amortised view is:

$$C_{c,d}(Q) = C_{c,d}^{\text{construction}} + Q\,\overline{C_{c,d}^{\text{query}}}.$$

This expression is used for scenario analysis rather than as a claim about a universal workload.

E7 requires an additional distinction. Its experiment-time record reuses two completed parent bundles. A realistic online E7 cost includes both parent retrieval paths and one fused Answerer call:

$$\begin{aligned} C_{i,E7,d}^{\text{online}} = {} & C_{i,E2,d}^{\text{retrieval}} + C_{i,E2,d}^{\text{packaging}} \\ & + C_{i,E6,d}^{\text{retrieval}} + C_{i,E6,d}^{\text{packaging}} \\ & + C_{i,E7,d}^{\text{fusion}} + C_{i,E7,d}^{\text{answering}} \\ & + C_{i,E2,d}^{\text{recovery}} + C_{i,E6,d}^{\text{recovery}} + C_{i,E7,d}^{\text{recovery}}. \end{aligned}$$

The fusion term covers deterministic deduplication and interleaving of the two parent bundles. As with packaging, it is zero for model-only cost dimensions unless separately charged. The parent Answerer calls are not required to construct a deployed E7 answer. They remain part of the experiment because E2 and E6 are independently evaluated conditions.

Latency is reported as an observed operational measure rather than inserted into the additive equations above. For a hypothetical online E7 deployment, latency depends on whether the two parent retrieval paths run sequentially or in parallel. Any scenario estimate states that scheduling assumption explicitly. Small observed latency differences may also reflect network and remote-service variation, so they are not automatically attributed to the architecture.

4.6.5 Metric Aggregation and Applicability

Table 4.6 summarises the calculation unit and interpretation of each metric family.

Table 4.6. Evaluation metrics, aggregation, and applicability.

Metric family	Per-question unit	Aggregate	Eligible denominator	Applicable paths	Analysis status	Main role
Token Precision	Normalised candidate-reference token overlap	Macro mean	60 E1–E7 and Full Context		Diagnostic answer metric	Lexical concision and extra text
Token Recall	Normalised candidate-reference token overlap	Macro mean	60 E1–E7 and Full Context		Diagnostic answer metric	Lexical answer coverage
Token F1	Category-aware per-question F1 from matched token overlap	Macro mean	60 E1–E7 and Full Context		Primary lexical metric	LoCoMo-aligned lexical answer quality
Strict Accuracy	`correct` indicator	Mean	60 E1–E7 and Full Context		Primary semantic metric	Complete semantic correctness
Partial Rate	`partial` indicator	Mean	60 E1–E7 and Full Context		Diagnostic semantic metric	Semantic label distribution
Soft Semantic Score	1, 0.5, or 0	Mean	60 E1–E7 and Full Context		Primary graded semantic metric	Graded semantic quality
Evidence Recall	Fraction of annotated source IDs retrieved	Macro mean	58 E1–E7 EvidenceBundles		Diagnostic evidence metric	Annotated-source coverage
Any-hit	At least one annotated source retrieved	Mean	58 E1–E7 EvidenceBundles		Diagnostic evidence metric	Minimum evidence access
All-hit	Every annotated source retrieved	Mean	58 E1–E7 EvidenceBundles		Diagnostic evidence metric	Complete annotated-source access
Calls, tokens, latency, rounds, and actions	Recorded process fields	Total, mean, median, or distribution as appropriate	All completed records with field availability	Path-dependent	Process and cost measure	Efficiency and process diagnosis
Memory construction	Store-level records	One-time total and amortised scenarios	Store or workload level	Relevant storage systems	Process and cost measure	Deployment trade-off

The main answer metrics are reported for every answer set. Evidence measures are reported for eligible questions with question-selected EvidenceBundles, including E7's fused parent bundles. They are `N/A` for Full Context. Process fields are not forced into one common unit when their mechanisms differ. An R1 round and an R2 action, for example, are described separately rather than treated as equivalent operations. Memory-construction effort is measured where records are available. Long-term maintenance is outside the executed experiment, so maintenance consequences are discussed qualitatively rather than reported as measured cost.

4.7 Statistical Analysis Plan

The analysis is primarily descriptive and paired. Main-60 contains the same questions under each compared condition. Preserving this pairing provides more useful evidence than treating condition means as unrelated samples. Confidence intervals and tests support selected comparisons; they do not organise the entire results chapter.

4.7.1 Unit of Analysis and Paired Design

The primary analysis unit is `question_id`. Let $m_i(A)$ and $m_i(B)$ be the value of metric m for question i under conditions A and B. The paired difference is:

$$d_i^{(m)}(A,B) = m_i(A) - m_i(B).$$

The average paired effect is:

$$\overline{d}^{(m)}(A,B) = \frac{1}{N}\sum_{i=1}^{N} d_i^{(m)}(A,B).$$

The median paired difference is reported beside the mean. The mean describes the average signed change. The median indicates whether the typical question moves in the same direction. Both are needed because a small number of large changes can shift the mean.

Architecture comparisons pair answers with the same question and backbone model. Backbone comparisons pair the same question and E-condition across Qwen and DeepSeek. E7 comparisons pair E7 with E2 or E6 for the same question and backbone. Full Context comparisons pair its answer with the selected DeepSeek condition for the same question.

4.7.2 Overall Paired Differences and Win/Tie/Loss

For each main comparison, metric direction is fixed before assigning wins and losses. Let $o_m = 1$ for higher-is-better metrics, such as F1 or Strict Accuracy, and let $o_m = -1$ for lower-is-better measures, such as calls, tokens, latency, or cost. The oriented difference is:

$$d_{i,+}^{(m)}(A,B) = o_m\, d_i^{(m)}(A,B).$$

A question is then a win, tie, or loss according to the sign of the oriented difference:

$$\text{state}_i(A,B) = \begin{cases} \text{win}, & d_{i,+}^{(m)}(A,B) > 0, \\ \text{tie}, & d_{i,+}^{(m)}(A,B) = 0, \\ \text{loss}, & d_{i,+}^{(m)}(A,B) < 0. \end{cases}$$

The report presents the mean difference, median difference, win/tie/loss counts, and the distribution of d_i. This combination distinguishes broad improvement from an average driven by a few large gains. It also exposes systems that repair some questions while harming others.

Win/tie/loss is metric-specific. A condition may win on Recall and lose on Precision for the same question. Semantic label transitions are analysed separately in Section 4.7.6.

4.7.3 Category-Level Analysis

The paired analysis is repeated descriptively for C1 multi-hop, C2 temporal, C3 open-domain, and C4 single-hop questions. The same estimators are used within each category, but category results are not treated as independent replications.

Category analysis supports mechanism interpretation. A retrieval change may primarily affect multi-hop evidence coverage. A curated representation may help temporal consolidation but lose local wording. Full Context may repair dispersed evidence while adding noise for simple questions.

The category groups are small and unequal. C3 contains seven questions. The report therefore emphasises effect direction, paired cases, and consistency with trace evidence rather than standalone category-level significance claims.

4.7.4 Storage–Retrieval Interaction and Backbone Stability

The per-question storage–retrieval interaction for metric m was defined in Section 4.3.2. Its aggregate form is:

$$\overline{DID}^{(m)} = \frac{1}{N}\sum_{i=1}^{N}\left[\big(m_i(E4) - m_i(E2)\big) - \big(m_i(E3) - m_i(E1)\big)\right].$$

For a direction-of-benefit summary, the report also uses:

$$\overline{DID}_{+}^{(m)} = o_m\, \overline{DID}^{(m)}.$$

This estimator asks whether the effect of changing from S1 to S2 differs between R1 and R2. It is reported with the two underlying storage contrasts. A DID value without those component effects can be misleading.

Backbone stability concerns the same architectural contrast under Qwen and DeepSeek. For conditions A and B, let:

$$\Delta_Q^{(m)}(A,B) = \overline{m_Q(A) - m_Q(B)},$$

and:

$$\Delta_D^{(m)}(A,B) = \overline{m_D(A) - m_D(B)}.$$

The contrast-of-contrasts is:

$$CoC^{(m)}(A,B) = \Delta_D^{(m)}(A,B) - \Delta_Q^{(m)}(A,B).$$

Stability is assessed from effect direction, magnitude, paired win/tie/loss patterns, and the contrast-of-contrasts. A result that crosses an arbitrary significance threshold under one backbone but not the other is not, by itself, evidence of model dependence. The difference between the effects must be examined directly.

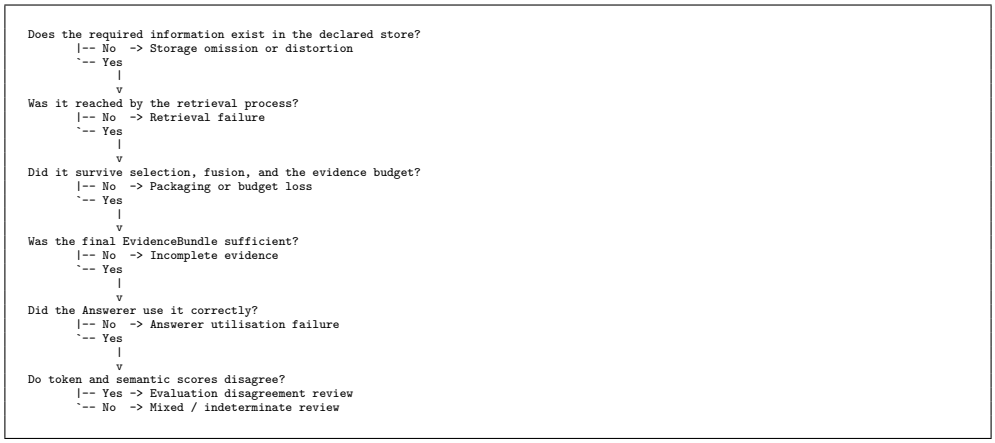


Figure 4.3. Stage-based error-attribution decision flow.

4.7.5 Auxiliary Paired Bootstrap Confidence Intervals

Paired bootstrap confidence intervals are used only for a limited set of central contrasts. In the implemented analysis, they are reported for the main raw-memory storage and retrieval comparisons. E7 and the principal Full Context comparisons remain paired central analyses, but their main evidence consists of descriptive paired effects and Strict-label transitions rather than additional bootstrap intervals.

For one bootstrap replicate, the analysis samples N question identifiers with replacement from the comparison set. All condition values associated with a sampled question move together. The statistic is then recomputed on the paired sample. If $\widehat{\delta}^{*}(1), \ldots, \widehat{\delta}^{*}(B)$ are the bootstrap estimates, the percentile interval is:

$$CI_{95\%} = \left[Q_{0.025}\left(\widehat{\delta}^{*}\right), Q_{0.975}\left(\widehat{\delta}^{*}\right) \right].$$

The interval describes resampling stability across the 60 observed Main-60 question units. It does not estimate variation across conversations because the study contains one conversation. Bootstrap results are supporting evidence. They do not replace the paired distributions, category patterns, or manual review.

The number of replicates, random seed, missing-record policy, and implementation details belong in Appendix H.

4.7.6 Strict-Label Transitions

Selected semantic comparisons use question-level transitions between Strict outcomes. Let $s_i(A)$ equal one when condition A is labelled `correct` and zero otherwise. Four transition counts are reported:

$$n_{ab} = \sum_{i=1}^{N} \mathbf{I}\big(s_i(B) = a \wedge s_i(A) = b\big), \qquad a, b \in \{0, 1\}.$$

Here, B is the baseline and A is the target condition. Therefore, n_{01} is an incorrect-to-correct gain and n_{10} is a correct-to-incorrect harm. The counts n_{00} and n_{11} represent unchanged incorrect and unchanged correct outcomes.

These transitions are central to E7 and Full Context diagnosis. They show whether a higher mean Strict Accuracy reflects a broad excess of repairs, a small net excess of repairs, or larger offsetting numbers of gains and harms. For a small number of central binary comparisons, an exact McNemar test may be reported using the discordant counts n_{01} and n_{10}. It remains supplementary.

4.7.7 Multiple Comparisons and Interpretation Boundaries

The study does not treat every combination of condition, metric, category, and backbone as an independent confirmatory test. Primary comparison families are defined by the research questions. Category, correlation, anomaly, and case analyses are exploratory.

Interpretation emphasises direction, magnitude, distribution, and cross-backbone consistency. The report does not rank systems by a large set of unadjusted p-values. If a multiple-comparison correction is applied to a formal test family, that family and procedure are stated explicitly in Appendix H.

Table 4.7. Principal comparison families and analysis methods.

Analytical purpose	Main contrasts	Primary unit	Main summaries	Supporting evidence
Directory organisation	E3−E1 and E4−E2	Question	Mean and median paired differences; win/tie/loss	Evidence coverage, traces, and selected bootstrap intervals
Retrieval system	E2−E1 and E4−E3	Question	Mean and median paired differences; win/tie/loss	Evidence coverage, calls, tokens, and traces
Storage–retrieval interaction	$(E4 - E2) - (E3 - E1)$	Question	DID and component contrasts	Category patterns and selected bootstrap interval
Curated-memory access	E6−E5	Question	Paired differences and label transitions	Fact/topic retrieval traces and cost
Backbone stability	Matched contrast under Qwen and DeepSeek	Question	Effect direction, magnitude, and contrast-of-contrasts	Question-level agreement and disagreement cases
Dual-source fusion	E7−E2 and E7−E6	Question	Paired differences and Strict transitions	Repair/harm review, evidence overlap, and online cost
Full Context diagnosis	Full Context versus DeepSeek selective conditions	Question	Paired differences and Strict transitions	Context scale, evidence state, latency, and manual review

4.8 Error Analysis Method

Answer scores show whether a candidate differs from the reference. They do not identify where information was lost or misused. The error analysis therefore follows the complete information path defined in Chapter 3.

4.8.1 Error Categories

The coding scheme is stage-based. A case is assigned to the earliest well-supported failure stage when the available records permit that decision. Cases with interacting causes may receive a mixed label. Cases without enough evidence remain indeterminate.

Table 4.8. High-level error taxonomy.

Error category	Operational question	Main evidence inspected
Storage omission or distortion	Was the required information absent, altered, stale, or incorrectly combined in the declared store?	Canonical source, S1/S2/S3 contents, and provenance locators
Retrieval failure	Was the required information present in the store but absent from the retrieved observations or selected evidence?	Store content, retrieval trace, ranked results, and EvidenceBundle
Incomplete multi-hop evidence	Was only part of the required annotated or inferential evidence retrieved?	Gold sources, retrieved source set, local context, and question requirements
Evidence-budget or packaging loss	Did relevant material appear upstream but disappear during selection, merging, ordering, or budget application?	Retrieval observations, accepted notes, merge records, and budget records
Answerer utilisation failure	Did the EvidenceBundle contain sufficient support while the candidate answer remained wrong or incomplete?	EvidenceBundle, candidate answer, citations, and reference answer
Fusion interference	Did E7 become worse after adding the second parent evidence source?	E2/E6 parent bundles, fused order, duplicate handling, and E7 answer
Long-context utilisation failure	Did Full Context fail despite complete visibility, especially where selective evidence succeeded?	Full Context input, paired selective evidence, answer, and source locations
Lexical–semantic evaluation disagreement	Did token and semantic measures disagree because of paraphrase, verbosity, omitted qualifiers, or keyword overlap?	Candidate, reference, token breakdown, and LLM judge label
Mixed or indeterminate	Are several stages implicated, or is the record insufficient for one supported attribution?	Complete case record and reviewer notes

Evidence coverage alone does not assign an error label. Low Evidence Recall with a wrong answer supports a retrieval hypothesis, but an alternative valid passage may exist. All-hit with a wrong answer supports an utilisation hypothesis, but the annotated source may still be insufficient for the full reasoning chain. Each attribution therefore requires inspection of the relevant artefacts.

Figure 4.3 presents the core decision sequence for selective-retrieval conditions. It is deliberately concise. E7 additionally requires a fusion-interference check, while Full Context bypasses the retrieval, EvidenceBundle, and evidence-budget stages and instead receives a long-context utilisation check. These condition-specific branches follow the definitions in Table 4.8.

The abbreviated labels in Figure 4.3 map to the taxonomy in Table 4.8: *Incomplete evidence* includes incomplete multi-hop evidence; *Packaging or budget loss* denotes evidence-budget or packaging loss; and *Evaluation disagreement review* denotes lexical–semantic evaluation disagreement. If the Answerer uses sufficient evidence correctly and the evaluation families agree, the case exits the error flow with no attributable error. *Mixed / indeterminate* is reserved for unresolved erroneous cases, interacting causes, or insufficient records.

4.8.2 Review Procedure

Manual review proceeds in a fixed order:

1. Read the question and reference answer.
2. Locate the annotated source turns, when available.
3. Check whether the required information exists in the declared memory representation.

4. Inspect the retrieval trace and returned observations, where applicable.
5. Inspect selected EvidenceItems, merging, order, and budget outcomes, where applicable. For Full Context, verify complete and chronological context assembly instead.
6. Compare the evidence input—an EvidenceBundle or the complete Full Context input—with the candidate answer and citations.
7. Inspect token scores, the LLM judge label, and any disagreement between them.
8. Assign a primary error category, a secondary category when justified, or `indeterminate`.

The review sample covers both backbone models, all condition families, and all question categories. It includes evidence–answer mismatches, E7 repairs and harms, Full Context repairs and failures, and token–semantic scoring disagreements. Cases are selected to explain aggregate patterns and important counterexamples. They are not presented as a random estimate of population-wide error frequencies unless a separate random-sampling procedure is applied.

The complete codebook records inclusion rules, exclusion rules, required artefacts, and decision examples. Reviewers must preserve uncertain cases rather than force one cause. Any agreement or adjudication statistic is reported only if more than one reviewer independently applies the codebook.

Chapter 5 reports the observed error distribution and representative cases. The complete statistical settings and error-analysis protocol belong in Appendix H. Extended per-question results and the full casebook belong in Appendix I.

This protocol links each research question to a controlled comparison, an applicable metric set, and an explicit interpretation boundary. It produces the answer, evidence, process, and review records analysed in Chapter 5.

Chapter 5. Results and Analysis

5.1 Analysis Scope and Research Questions

Chapter 3 defined the storage representations, retrieval systems, EvidenceBundle interface, and shared Answerer. Chapter 4 then specified the unified Main-60 test set and evaluation protocol. This chapter reports what the experiments show. It analyses 60 questions under seven selective evidence-access conditions for each of two backbone models, Qwen and DeepSeek, together with one DeepSeek-family Full Context diagnostic path.

Table 5.1 restates the condition mapping so that the experimental labels can be read without returning to Chapter 3. S1 is the flat raw-session store, S2 is the foldered raw-session store, and S3 is Agent-Curated Memory. R1 is Agentic Filesystem Retrieval, R2 is ReFind-style raw-chat retrieval, and R3 is retrieval adapted to the structure of S3.

Table 5.1. Experimental conditions used in Chapter 5. Full Context is a separate diagnostic path rather than E8.

Condition	Evidence source	Retrieval or processing path	Analytical role
E1	S1	R1	Flat raw-session filesystem baseline
E2	S1	R2	Enhanced raw-chat retrieval on S1
E3	S2	R1	Effect of folder organisation under R1
E4	S2	R2	Foldered raw sessions combined with R2
E5	S3	R1	Agent-Curated Memory accessed as a filesystem
E6	S3	R3	Retrieval adapted to Agent-Curated Memory
E7	E2 and E6 EvidenceBundles	Fixed deduplication and interleaving, followed by the shared Answerer	Dual-source complementarity and interference; no new retrieval
Full Context	All 30 S1 raw sessions	Complete chronological context sent directly to the shared Answerer	Diagnostic comparison without selective retrieval or a 16,000-character EvidenceBundle

The analysis is organised around six research questions. Table 5.2 provides the chapter-level roadmap. Each later section reports the relevant answer-quality, evidence-access, process, or cost evidence and states the boundary of the conclusion.

Table 5.2. Research questions and their locations in Chapter 5.

Research question	Main question	Section
RQ1	How do storage representation, retrieval design, and their interaction affect answer quality?	5.2
RQ2	Which architectural effects remain stable across Qwen and DeepSeek, and which depend on the backbone model?	5.3
RQ3	When does E7 dual-source fusion repair an answer, and when does it introduce interference?	5.4
RQ4	Does Full Context outperform selective retrieval, and which failure mechanisms remain under complete visibility?	5.5
RQ5	Where does the end-to-end pipeline fail: storage, retrieval, packaging, evidence utilisation, fusion or long-context processing, or evaluation?	5.6
RQ6	Which condition offers the most useful balance between answer quality, construction cost, query cost, and operational reliability?	5.7

Answer quality is reported through Precision, Recall, Token F1, Strict Accuracy, Partial Rate, and Soft Semantic Score. Evidence-access metrics apply to E1–E7 and use the 58 questions with valid source-turn annotations; all answer-quality metrics use the complete 60-question set. Full Context has no retrieval-level Evidence Recall because it bypasses selective evidence construction. The following sections present the detailed comparisons and mechanism analyses.

At a high level, the results show that system performance depends jointly on storage representation, retrieval design, and the backbone model; no single selective-retrieval condition dominates every metric. Dual-source fusion and Full Context provide benefits under different conditions, but neither eliminates evidence-utilisation errors. The preferred architecture therefore depends on both answer quality and operational cost.

5.2 Architecture Effects: Storage, Retrieval, and Their Interaction

Section 5.2 addresses RQ1: how do memory representation and retrieval design shape end-to-end answer quality? It asks three linked questions. First, does changing the storage layout help when the retrieval system is held fixed? Second, does changing the retrieval system help when the storage representation is held fixed? Third, do storage and retrieval interact, and how does Agent-Curated Memory behave as a complete system? Answering these questions separates the roles of storage, retrieval, and their combination before the report considers backbone robustness or evidence fusion.

The analysis uses six core effects: four raw-memory simple contrasts, one storage–retrieval interaction, and one access-system contrast on the frozen S3 store. It also uses two descriptive complete-system anchors. Every effect is estimated within one backbone model. E7 is not an RQ1 condition; its dual-source fusion effect is analysed under RQ3 in Section 5.4.1. Formal comparison of effect stability across backbone models is presented in Section 5.3.

All differences in this section are target minus baseline. Answer metrics use all 60 questions. Evidence metrics use the 58 questions with valid evidence annotations. The two excluded evidence cases are open-domain questions whose frozen gold records contain no annotated source-turn IDs; they remain in every answer-quality calculation and are excluded only from evidence metrics whose denominator would otherwise be zero. Multi-source evidence results use the corresponding 19-question subset. The category analysis uses C1/C2/C3/C4 answer denominators of 13/13/7/27 and evidence denominators of 13/13/5/27. These denominators are not pooled.

The main outcomes are token F1, Strict Accuracy, and Soft Semantic Score. Precision and Recall help explain the token result. Evidence Recall, Any-hit, and All-hit describe access to the annotated source turns. These evidence annotations identify valid source locations, but they are not guaranteed to be a minimal or exhaustive proof set. Process measures are used only to explain architectural behaviour. They do not determine the final system choice in this section.

Across all main simple comparisons and the storage–retrieval interaction, the paired median differences for F1, Strict Accuracy, and Soft Semantic Score are 0.0000. This reflects the large number of questions whose outcome is unchanged even when the mean changes. The tables therefore retain mean effects and question-level transitions. Partial Rate remains in the complete condition summary and the appendix diagnostics, but it is not oriented as higher-is-better: a change can reflect either movement from incorrect to partial or movement from correct to partial. The complete three-by-three LLM judge transition matrices are retained in Appendix I.

5.2.1 Effect of Storage Organisation under Each Retrieval System

The first comparison changes the raw-memory layout from S1 flat sessions to S2 foldered sessions. The conversation content remains raw. E3−E1 measures this change under R1, while E4−E2 measures it under R2.

Table 5.3. Storage-organisation effects on primary answer quality. All values are target minus baseline. Bracketed values are 95% paired-bootstrap intervals. F1 win/tie/loss and Strict repair/tie/harm counts use the 60 paired questions. The E1–E4 condition means underlying these comparisons are reported in Table 5.7.

Backbone	Contrast	ΔF1 [95% interval]	F1 W/T/L	ΔStrict [95% interval]	Strict R/T/H	ΔSoft
Qwen	E3−E1	**+0.0462** [−0.0160, +0.1109]	19/30/11	**+0.0333** [−0.0500, +0.1171]	5/52/3	**+0.0250**
DeepSeek	E3−E1	**−0.0041** [−0.0499, +0.0431]	17/25/18	**−0.0500** [−0.1333, +0.0167]	1/55/4	**−0.0500**
Qwen	E4−E2	**−0.0071** [−0.0678, +0.0533]	13/32/15	**+0.0333** [−0.0500, +0.1333]	5/52/3	**+0.0083**
DeepSeek	E4−E2	**−0.0141** [−0.0742, +0.0471]	20/20/20	**−0.0333** [−0.1000, +0.0333]	1/56/3	**0.0000**

Table 5.4. Storage-organisation lexical and evidence diagnostics. Precision, Recall, and Partial Rate use all 60 questions. Evidence Recall uses the 58 evidence-eligible questions. Full Soft transitions are retained in Appendix I.

Backbone	Contrast	ΔPrecision	ΔRecall	ΔPartial	ΔEvidence Recall
Qwen	E3−E1	+0.0378	+0.0527	−0.0167	+0.0072
DeepSeek	E3−E1	−0.0030	−0.0179	0.0000	−0.0302
Qwen	E4−E2	−0.0193	+0.0131	−0.0500	−0.0014
DeepSeek	E4−E2	−0.0090	−0.0519	+0.0667	−0.0287

The Qwen E3−E1 estimate provides the clearest positive storage-organisation result. It is the only storage contrast with positive point estimates for F1, Strict, and Soft: F1 rises by 0.0462, Strict by 0.0333, and Soft by 0.0250, with five Strict repairs against three harms. This is valid positive evidence within the observed Qwen/R1 setting. It is not, however, a statistically resolved or universal storage effect because the paired intervals include zero and the direction is not reproduced under DeepSeek. The answer gain also cannot be explained by a general increase in annotated evidence access: Evidence Recall rises by only 0.0072, Any-hit falls by 0.0172, and All-hit rises by 0.0172.

The category analysis shows where this aggregate pattern originates. For Qwen E3−E1, C2 temporal questions contribute ΔF1 = +0.1994, ΔStrict = +0.3077, and ΔSoft = +0.1923, even though C2 Evidence Recall falls by 0.0385. The other categories show little F1 gain, and C4 Strict falls by 0.0741. The result is therefore a concentrated temporal pattern, not a broad improvement across question types. Under DeepSeek, C2 F1 rises by 0.0452 while Strict and Soft each fall by 0.1538. This is a token–semantic disagreement rather than a consistent category-level repair.

The E4−E2 estimates are smaller. Under Qwen, the foldered layout slightly lowers F1 but raises Strict and Soft. Under DeepSeek, F1 and Strict fall while Soft is unchanged. No storage-effect interval for F1 or Strict excludes zero. The data therefore do not establish a general benefit from directory organisation.

The evidence-to-answer tables reinforce this limit. In Qwen E3−E1, none of the evidence-eligible questions combines higher Evidence Recall with a Strict repair, while five Strict repairs occur with unchanged Evidence Recall. Four questions gain Evidence Recall without gaining Strict. The corresponding DeepSeek counts are zero, one, and one. For E4−E2, only two Qwen questions and no DeepSeek questions combine higher Evidence Recall with a Strict repair. These selected diagnostic cells are not a complete partition of the 58 questions, but they show that annotated-source coverage and answer correctness are not interchangeable.

The bounded trace review provides three explanations for this separation. First, a flat-store bundle can already contain direct evidence even when it does not cover every annotated turn. Second, a foldered path can retain the same evidence but lead the Answerer to preserve a different level of detail. Third, some apparent repairs or harms arise near an evaluation boundary rather than from a clear storage change. These cases support a conditional interpretation: folder organisation can help an Agentic Filesystem Retriever under a particular backbone and question type, as observed for Qwen/R1 on temporal questions, but Main-60 does not establish an architecture-wide gain.

5.2.2 Effect of the Retrieval System under Each Raw Storage Representation

The second comparison holds the raw store fixed and changes the complete retrieval system. E2−E1 compares R2 with R1 on S1. E4−E3 makes the same comparison on S2. These are complete-system effects. They do not isolate BM25, query generation, local expansion, or one Retrieval Agent action.

Table 5.5. Retrieval-system effects on primary answer quality. All values are target minus baseline. Bracketed values are 95% paired-bootstrap intervals. F1 win/tie/loss and Strict repair/tie/harm counts use the 60 paired questions. Full Soft and Partial transitions are retained in Appendix I.

Backbone	Contrast	ΔF1 [95% interval]	F1 W/T/L	ΔStrict [95% interval]	Strict R/T/H	ΔSoft
Qwen	E2−E1	**+0.0082** [−0.0410, +0.0573]	13/27/20	**−0.0500** [−0.1333, +0.0333]	2/53/5	**−0.0500**
DeepSeek	E2−E1	**−0.0557** [−0.1189, −0.0002]	22/14/24	**−0.0500** [−0.1500, +0.0500]	4/49/7	**−0.1000**

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Backbone	Contrast	ΔF1 [95% interval]	F1 W/T/L ΔStrict [95% interval]	Strict R/T/H	ΔSoft
Qwen	E4−E3	**−0.0451** [−0.1096, +0.0161]	15/26/19 **−0.0500** [−0.1500, +0.0500]	4/49/7	**−0.0667**
DeepSeek	E4−E3	**−0.0657** [−0.1401, +0.0071]	22/12/26 **−0.0333** [−0.1333, +0.0667]	4/50/6	**−0.0500**

Table 5.6. Retrieval-system lexical, evidence, and query-cost diagnostics. Precision and Recall use all 60 questions. Evidence Recall uses the 58 evidence-eligible questions. Query-cost differences include retrieval and answering; negative values indicate fewer recorded calls or tokens under R2.

Backbone	Contrast	ΔPrecision	ΔRecall	ΔEvidence Recall	ΔBundle characters	ΔQuery calls	ΔQuery tokens
Qwen	E2−E1	+0.0225	−0.0154	−0.0330	+4,040.8	−7.57	−130,501.5
DeepSeek	E2−E1	−0.0625	+0.0020	−0.0805	+7,142.3	−7.98	−183,717.1
Qwen	E4−E3	−0.0346	−0.0550	−0.0417	+3,885.6	−6.68	−121,809.1
DeepSeek	E4−E3	−0.0685	−0.0320	−0.0790	+6,678.9	−7.70	−172,188.5

R2 does not retrieve more complete annotated evidence in these comparisons. It produces much larger EvidenceBundles, but Evidence Recall, Any-hit, and All-hit fall in all four backbone–store cells. The Any-hit/All-hit changes are −0.0690/−0.0172 for Qwen E2−E1, −0.0345/−0.0517 for Qwen E4−E3, −0.0862/−0.0862 for DeepSeek E2−E1, and −0.0517/−0.0862 for DeepSeek E4−E3. On the 19 multi-source questions, multi-source Evidence Recall also falls by 0.0482 and 0.0746 under Qwen and by 0.1930 and 0.1360 under DeepSeek for E2−E1 and E4−E3, respectively. More selected text therefore does not mean better coverage of the annotated source turns.

The semantic outcomes also favour caution. Strict and Soft both fall in all four R2 comparisons. Qwen E2−E1 shows the clearest lexical–semantic disagreement: F1 rises slightly alongside higher Precision and lower Recall, while Strict and Soft each fall by 0.0500. The only selected intervals that strictly exclude zero are DeepSeek E2−E1 for F1, `[-0.1189, -0.0002]`, and Soft, `[-0.2000, -0.0083]`. The F1 upper bound is very close to zero, so this is evidence of a negative observed-question effect, not a broad population claim.

The process results show a clear quality–efficiency trade-off rather than a uniformly heavier R2 pipeline. R2 uses fewer recorded model calls and fewer total query-time model tokens per question than R1. These totals include retrieval and answering rather than the retrieval stage alone. For E2−E1, the mean reductions are 7.57 calls and 130,501.5 tokens under Qwen, and 7.98 calls and 183,717.1 tokens under DeepSeek. For E4−E3, the corresponding reductions are 6.68 calls and 121,809.1 tokens, and 7.70 calls and 172,188.5 tokens. However, R2 sends substantially larger selected evidence to the Answerer and produces weaker annotated-evidence coverage and semantic outcomes. In the measured dimensions, R2 is therefore cheaper at query time but less effective. This use of *cheaper* refers only to recorded successful-query model calls and tokens; native latency and monetary cost are unavailable for E1–E6. Failed-attempt and recovery tokens are excluded from these successful-query totals and are reported separately in Section 5.7.3. Token usage is known for 39 of the 41 formal failed attempts; only two Qwen E5 DNS failures have unknown token usage. Across the 60 questions, E2 and E4 record no failed attempts; E1/E3 record 1/5 attempts under Qwen and 12/6 under DeepSeek. These are descriptive attempt counts, not independent failure-rate estimates. R1 provider rounds and R2 actions are different units and are not directly compared. Whether the lower recorded query-time totals compensate for the quality decline is deferred to RQ6.

The C2 temporal category is a notable weak point for R2 on S2. Qwen E4−E3 gives ΔF1 = −0.1373 and ΔStrict = −0.3077; DeepSeek gives ΔF1 = −0.2269 and ΔStrict = −0.1538. Some DeepSeek C3 token results are positive, but C3 has only seven answer questions and five evidence-eligible questions, and its evidence result still declines. These category values are descriptive only.

The reviewed traces locate different failure points. In one case, R2 reached the relevant session but failed to retain the adjacent decisive turn during local expansion. In another, R1 and R2 supplied the same annotated statement, but the final answers preserved different reference-level detail. A third case showed higher annotated coverage with no answer gain because the lower-coverage bundle already contained sufficient direct evidence. These cases show that R2 used fewer recorded query-time calls and tokens, produced larger EvidenceBundles, and was associated with lower semantic correctness in these comparisons.

5.2.3 Storage–Retrieval Interaction and Question-Category Heterogeneity

The interaction tests whether changing from S1 to S2 has a different effect under R2 than under R1. For metric m, it is computed per question as:

$$DID_i^{(m)} = \big(m_i(E4) - m_i(E2)\big) - \big(m_i(E3) - m_i(E1)\big)\,.$$

The four component contrasts remain necessary for interpretation. A negative F1 interaction means that the S1-to-S2 change was less favourable under R2 than under R1 for that backbone; it does not identify which internal retrieval action caused the difference.

Panel A. Qwen.

Table 5.7. Descriptive condition means underlying Figure 5.1. Answer metrics use all 60 questions. Evidence Recall uses the 58 evidence-eligible questions. Scores are reported on the 0–1 scale.

Condition	Storage + retrieval	Token F1	Strict Accuracy	Soft Semantic Score	Evidence Recall
E1	S1 + R1	0.3949	0.7167	0.7667	0.8147
E2	S1 + R2	0.4031	0.6667	0.7167	0.7816
E3	S2 + R1	0.4411	0.7500	0.7917	0.8218
E4	S2 + R2	0.3960	0.7000	0.7250	0.7802

Panel B. DeepSeek.

Condition	Storage + retrieval	Token F1	Strict Accuracy	Soft Semantic Score	Evidence Recall
E1	S1 + R1	0.3013	0.7500	0.8083	0.8779
E2	S1 + R2	0.2455	0.7000	0.7083	0.7974
E3	S2 + R1	0.2971	0.7000	0.7583	0.8477
E4	S2 + R2	0.2314	0.6667	0.7083	0.7687

Figure 5.1 visualises the same condition means as a 2 x 2 interaction profile. Under each backbone, the R1 line connects E1 to E3 and the R2 line connects E2 to E4. The figure therefore shows descriptive levels and slopes rather than paired effect estimates.

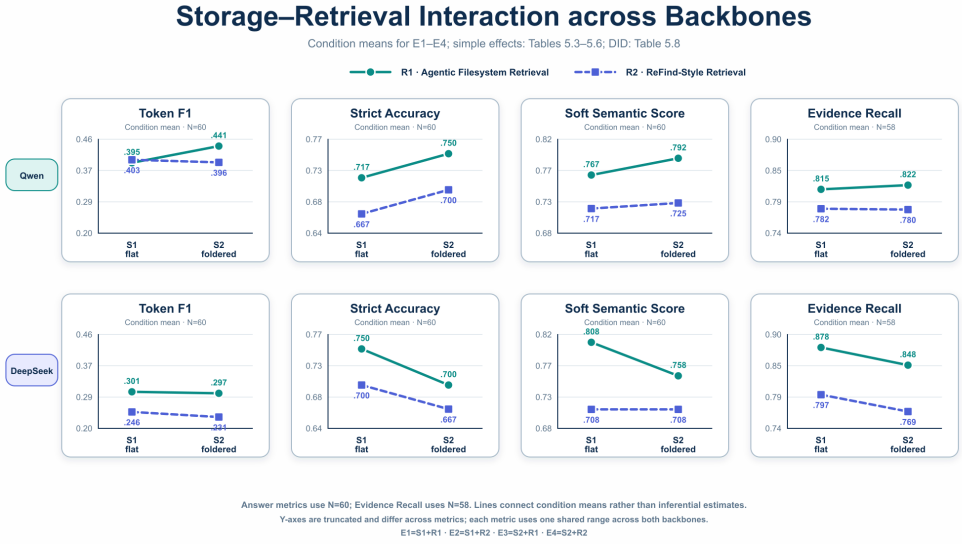


Figure 5.1. Storage–retrieval interaction across backbones. Each panel plots the E1–E4 condition means reported in Table 5.7. Answer metrics use N=60, and Evidence Recall uses N=58. The y-axes are truncated and differ across metrics; each metric retains one common range across the two backbones. Paired uncertainty and the DID estimates are reported in Table 5.8.

Table 5.8. Storage–retrieval interaction estimates. The final column reports positive/zero/negative question-level DID counts rather than repair/tie/harm transitions.

Backbone	ΔPrecision DID	ΔRecall DID ΔF1 DID (95% interval)	ΔStrict DID (95% interval)	ΔSoft DID (95% interval)	ΔEvidence Recall DID	Strict DID +/0/−
Qwen	−0.0571	−0.0396 **−0.0533** [−0.1251, +0.0152]	**0.0000** [−0.1500, +0.1500]	**−0.0167** [−0.1167, +0.0833]	−0.0086	5/49/6
DeepSeek	−0.0060	−0.0340 **−0.0100** [−0.0823, +0.0638]	**+0.0167** [−0.0833, +0.1167]	**+0.0500** [−0.0417, +0.1417]	+0.0014	5/51/4

All F1, Strict, and Soft interaction intervals in Table 5.8 include zero. The experiment therefore does not establish a reliable storage–retrieval interaction. The negative Qwen F1 point estimate is consistent with S2 being more favourable to R1 than to R2 in this observed set, but the interval and the semantic estimates do not support a firm general claim.

The absence of a resolved interaction does not make the four raw-memory configurations equivalent. Their descriptive rankings reveal a useful conditional pattern. For Qwen, E3 (S2+R1) is the strongest E1–E4 condition on F1 (0.4411), Strict (0.7500), and Soft (0.7917). For DeepSeek, E1 (S1+R1) is strongest on the same three measures, with F1 = 0.3013, Strict = 0.7500, and Soft = 0.8083. The common element is R1; the preferred raw layout changes from S2 under Qwen to S1 under DeepSeek. Thus, R1 is the stronger raw-memory retrieval family in Main-60, whereas the choice between S1 and S2 remains backbone-dependent. These are within-backbone descriptive rankings; formal cross-backbone effect stability is evaluated in Section 5.3.

The category results are heterogeneous. The Qwen C2 interaction is −0.1860 for F1 and −0.3846 for Strict, while its C4 Strict interaction is +0.1481. The DeepSeek C2 interaction is −0.1453 for F1, while C1 Strict is +0.1538 and C3 F1 is +0.1547. These values do not form one category-wide mechanism. In addition, C3 is too small to support independent inference.

The bounded cases further limit mechanistic attribution. One positive interaction case was associated with a different second query and a direct statement reached only by E4. Other apparent positive or negative interactions were sensitive to LLM judge treatment of near-equivalent temporal wording. Another case achieved complete annotated-turn recall but still lacked attributes required by the reference answer. Thus, official Evidence Recall can overstate evidence sufficiency, and individual DID changes can occur at query selection, evidence utilisation, or evaluation. The cases support interpretation of the aggregate effect, but they do not provide a frequency estimate of these mechanisms.



Figure 5.2. Agent-Curated Memory architecture-effect profile. Values are mean target-minus-baseline differences in percentage points. E6−E5 compares R3 with R1 on the same frozen S3 store; E5−E3 and E6−E4 are descriptive complete-system anchors. Answer metrics use N=60, and Evidence Recall uses N=58. No bootstrap interval was pre-declared for these S3 comparisons.

5.2.4 Complete-System Effect of Agent-Curated Memory

The S3 analysis contains one controlled core contrast and two broader descriptive anchors. E6−E5 holds the frozen S3 store fixed and changes its access system from R1 to R3. E5−E3 compares S3+R1 with S2+R1. E6−E4 compares S3+R3 with S2+R2. The latter two comparisons change several representation or retrieval properties together and must not be interpreted as isolated curation effects.

Table 5.9. Curated-memory effects on primary answer quality. All values are target minus baseline. F1 win/tie/loss and Strict repair/tie/harm counts use the 60 paired questions. No new bootstrap interval is introduced for these descriptive S3 comparisons.

Backbone	Contrast	Interpretation	ΔF1	F1 W/T/L	ΔStrict	Strict R/T/H	ΔSoft
Qwen	E6−E5	Same-S3 access effect	**−0.0135**	20/23/17	**−0.0500**	2/53/5	**−0.0583**
DeepSeek	E6−E5	Same-S3 access effect	**−0.0364**	16/15/29	**−0.0833**	2/51/7	**−0.0917**
Qwen	E5−E3	R1 system anchor	**−0.0463**	16/20/24	**+0.0333**	4/54/2	**+0.0250**
DeepSeek	E5−E3	R1 system anchor	**−0.0367**	22/13/25	**+0.0500**	5/53/2	**+0.0417**
Qwen	E6−E4	Adapted-system anchor	**−0.0147**	20/18/22	**+0.0333**	6/50/4	**+0.0333**
DeepSeek	E6−E4	Adapted-system anchor	**−0.0074**	14/17/29	**0.0000**	4/52/4	**0.0000**

Table 5.10. Curated-memory lexical, evidence, and bundle diagnostics. Precision and Recall use all 60 questions. Evidence Recall and bundle-character means use the 58 evidence-eligible questions. Full Soft and Partial transitions are retained in Appendix I.

Backbone	Contrast	ΔPrecision	ΔRecall	ΔEvidence Recall	ΔBundle characters
Qwen	E6−E5	−0.0114	−0.0736	**−0.0891**	+5,109.1
DeepSeek	E6−E5	−0.0300	−0.1306	**−0.2371**	+5,457.3
Qwen	E5−E3	−0.0436	+0.0380	**+0.1034**	+3,034.0
DeepSeek	E5−E3	−0.0385	+0.0566	**+0.1149**	+3,901.2
Qwen	E6−E4	−0.0204	+0.0194	**+0.0560**	+4,257.6
DeepSeek	E6−E4	+0.0000	−0.0420	**−0.0431**	+2,679.6

Figure 5.2 places the three S3 comparisons on a common percentage-point scale. It separates the controlled same-store access comparison, E6−E5, from the two descriptive complete-system anchors, E5−E3 and E6−E4.

On the same frozen S3 store, R3 underperforms R1 across every primary answer and evidence mean in both within-backbone analyses. Within Qwen, E6−E5 changes F1 by −0.0135, Strict by −0.0500, Soft by −0.0583, and Evidence Recall by −0.0891 while adding 5,109 characters to the mean eligible EvidenceBundle. Within DeepSeek, the corresponding changes are −0.0364, −0.0833, −0.0917, and −0.2371, while the bundle grows by 5,457 characters. Any-hit and All-hit fall by 0.0690 and 0.1207 under Qwen, and by 0.2069 and 0.2414 under DeepSeek. Multi-source Evidence Recall falls by 0.1140 under Qwen and 0.1974 under DeepSeek. Strict harms exceed repairs in both analyses. Evidence Recall falls in every C1–C4 category; the largest DeepSeek decreases occur in C3 and C2, at −0.4000 and −0.2692. The descriptive result is therefore stronger than a mixed or inconclusive pattern: representation-adapted retrieval did not improve access to S3 in Main-60. No bootstrap interval was pre-declared for E6−E5, so the claim remains bounded to the observed test set rather than extended to a general population.

The E5−E3 anchor shows a different pattern. F1 falls alongside lower token Precision despite higher Recall, while Strict, Soft, Evidence Recall, Any-hit, and All-hit rise in each separate within-backbone comparison. Multi-source Evidence Recall also rises by 0.1579 under Qwen and 0.1404 under DeepSeek. Under Qwen, Strict rises from 0.7500 to 0.7833 and Evidence Recall from 0.8218 to 0.9253. Under DeepSeek, Strict rises from 0.7000 to 0.7500 and Evidence Recall from 0.8477 to 0.9626. This is a clear token–semantic disagreement rather than a uniformly negative result.

Answer length moves in the direction expected under a broader-response explanation. From E3 to E5, mean normalised answer length rises from 11.97 to 16.13 tokens under Qwen (approximately 35%) and from 21.72 to 29.17 tokens under DeepSeek (approximately 34%). Mean answer length in characters also rises from 79.73 to 108.07 and from 143.45 to 190.73, respectively. Together with lower Precision and higher Recall, this pattern is consistent with S3+R1 producing broader or differently worded answers that receive higher semantic scores but lower lexical overlap. It does not establish answer length as the cause of the F1 decline. Section 5.6.5 analyses lexical–semantic disagreement directly, while Section 5.5 examines Full Context-specific examples of verbosity, paraphrase, omitted qualifiers, and substantive error. Nor does the E5−E3 contrast prove that curation alone caused the result, because S2-to-S3 also changes granularity, organisation, and provenance presentation.

The absolute condition means within the E1–E6 RQ1 scope further clarify E5's role. Under Qwen, E5 has the highest Strict Accuracy (0.7833), Soft Semantic Score (0.8167), and Evidence Recall (0.9253), while E3 has the highest token F1 (0.4411). Under DeepSeek, E5 has the highest Evidence Recall (0.9626) and ties E1 for the highest Strict Accuracy (0.7500), but E1 has the highest token F1 (0.3013) and Soft Semantic Score (0.8083). E5 is therefore the strongest curated-memory configuration and the semantic/evidence leader under Qwen, but it is not a universal winner on every metric and backbone. E7 is deliberately excluded from this formal RQ1 ranking; its fusion effect is analysed in Section 5.4.1, and practical system selection is deferred to Section 5.7.

The E6−E4 anchor is mixed. Its Qwen estimate raises Strict, Soft, and Evidence Recall while slightly lowering F1. Its DeepSeek estimate leaves Strict and Soft unchanged while F1 and Evidence Recall decline. On the 19 multi-source questions, however, multi-source Evidence Recall rises by 0.1184 under Qwen and 0.0789 under DeepSeek. These are two within-backbone complete-system descriptions. Formal assessment of whether the architecture effect is stable across backbones belongs to Section 5.3.

The query-time process results clarify the E6−E5 trade-off. R3 uses fewer recorded model calls and fewer total query-time model tokens per question than R1 on S3. These totals include retrieval and answering. The Qwen reductions are 3.22 calls and 244,976.5 tokens per question; the DeepSeek reductions are 3.90 calls and 339,242.3 tokens. E5 records 12 failed attempts under Qwen and five under DeepSeek, whereas E6 records none. Token usage is available for 15 of these 17 E5 failures; the remaining two are Qwen E5 DNS failures with unknown usage. Recovery cost is reported separately in Section 5.7.3 and is not included in the successful-query totals above. However, R3 supplies larger EvidenceBundles and produces lower answer and evidence scores. R1 provider rounds and R3 actions are not commensurate. No latency value is inferred where native latency is unavailable.

S3 also has a one-time construction cost that is separate from query-time evaluation. The frozen construction manifest records 85 build episodes, 842 LLM calls, 837 rounds, 1,350 tool calls, 29,669,563 prompt tokens, 1,195,395 completion tokens, and 30,864,958 total tokens. The resulting store contains two files, 37 headings, 472 lines, and references 555 of 568 source locators. Locator coverage is not a direct measure of factual fidelity. These figures describe the system's construction requirements; their amortisation and value are analysed in RQ6 rather than used to rank systems here.

The bounded S3 trace review does not support a simple storage-loss explanation. All six selected Section 5.2.4 cases preserved the relevant information in S3. One Qwen repair occurred because curated evidence co-located a goal with an adjacent acquisition-time fact that the raw-chat path had missed. One harm occurred after sufficient evidence had been retrieved and packaged, when broader context distracted the Answerer. A DeepSeek harm similarly retained an alternative direct statement but failed to use it inside a distractor-heavy bundle. Other cases showed answer saturation or LLM judge sensitivity despite higher annotated coverage. This review concerns selected explanatory cases; it does not estimate S3 fidelity or failure prevalence across Main-60.

Table 5.11. Evidence-to-answer diagnostic cells (eligible N=58). The table reports four selected cells from each complete Evidence Recall change × Strict change matrix. The cells are deliberately not exhaustive and must not be summed to 58; the complete three-by-three matrices are retained in Appendix I.

Backbone	Contrast	Evidence ↑ and Strict ↑	Evidence ↑, Strict not ↑	Evidence same, Strict ↑	Evidence ↓ without Strict harm
Qwen	E3−E1	0	4	5	3
DeepSeek	E3−E1	0	1	3	4
Qwen	E4−E2	2	2	3	2
DeepSeek	E4−E2	0	4	1	3
Qwen	E2−E1	0	9	2	6
DeepSeek	E2−E1	0	5	3	4
Qwen	E4−E3	1	6	3	4
DeepSeek	E4−E3	0	8	2	7
Qwen	E6−E5	0	1	2	5
DeepSeek	E6−E5	0	2	2	10
Qwen	E5−E3	1	8	3	1
DeepSeek	E5−E3	1	9	3	1
Qwen	E6−E4	2	5	4	2
DeepSeek	E6−E4	3	3	0	6

Table 5.12. RQ1 architecture evidence matrix.

Architectural question	Audited finding	Evidence supporting the finding	Boundary
Does folder organisation improve raw-memory performance?	A positive descriptive effect appears for Qwen with R1, mainly on temporal questions; the other storage estimates are small or negative.	Paired F1/Strict/Soft, selected intervals, category matrix, and trace review.	Every storage F1/Strict interval includes or touches zero. No universal folder benefit is established.
Does R2 improve retrieval on a fixed raw store?	R2 uses fewer recorded query-time calls and tokens, but selects more text while producing lower annotated-evidence coverage and weaker semantic outcomes in the four raw-store comparisons.	Process totals, Evidence Recall/Any-hit/All-hit, bundle length, Strict transitions, and paired intervals.	*Cheaper* refers only to recorded query-time calls and tokens. This is a complete R2-versus-R1 system effect, not an isolated BM25 effect.
Is there a storage–retrieval interaction?	No reliable DID interaction is established. Descriptively, E3 is the strongest raw condition for Qwen and E1 for DeepSeek; both use R1.	DID estimates, component contrasts, raw-condition means, paired intervals, and category values.	The shared R1 advantage is an observed Main-60 pattern; storage preference remains backbone-dependent. Category patterns are exploratory, and C3 is small.
Does R3 improve access to S3 relative to R1?	No. E6 has lower observed F1, Strict, Soft, and Evidence Recall means than E5 in both within-backbone analyses, despite larger bundles.	Same-store E6−E5 effects, Strict transitions, category evidence, and selected traces.	No bootstrap interval was pre-declared for this contrast; the underperformance claim is descriptive within Main-60.
Is Agent-Curated Memory useful as a complete system?	S3+R1 raises semantic and evidence measures relative to S2+R1 while lowering token F1. Within E1–E6, E5 is the Qwen semantic/evidence leader and the DeepSeek evidence leader, but not the universal F1 winner.	E5−E3 and E6−E4 anchors, absolute condition means, answer length, metric disagreement, and bounded cases.	These anchors do not isolate curation from other representation and retrieval changes. Formal E7 and final system-selection claims belong to later RQs.

The answer to RQ1 is therefore conditional but substantive. First, folder organisation provides a positive observed point estimate under Qwen with R1, especially on temporal questions, although the effect is not resolved statistically or reproduced as a general storage advantage. Second, R1 is the stronger raw-memory retrieval family in Main-60: Qwen performs best with S2+R1, whereas DeepSeek performs best with S1+R1. R2 lowers recorded query-time calls and tokens, but this efficiency gain is accompanied by larger EvidenceBundles, weaker annotated-evidence coverage, and lower semantic quality. Third, the experiment does not establish a reliable storage–retrieval DID interaction, so the raw-configuration rankings remain conditional rather than universal. Fourth, on the same frozen S3 store, R3 underperforms R1 across the main answer and evidence metrics under both backbones. Finally, S3+R1 is the strongest curated-memory path within the RQ1 conditions: it improves semantic correctness and annotated-evidence coverage relative to S2+R1 while lowering token F1, and E5 leads the Qwen semantic/evidence metrics and the DeepSeek evidence metric within E1–E6. The accompanying increase in answer length and Recall, together with lower Precision, motivates separate lexical and semantic evaluation rather than treating token F1 as a complete measure of answer quality. These findings establish useful architecture-level trade-offs without claiming a universal single winner. Formal cross-backbone stability, fusion, Full Context, global error location, and practical system selection are addressed in the following research questions.

5.3 Robustness across Backbone Models

Section 5.3 addresses RQ2: do the architecture conclusions from Section 5.2 remain consistent when the backbone model changes? This question distinguishes architecture effects that recur across models from effects that depend on one model's search and answering behaviour. The section asks two related questions: how Qwen and DeepSeek differ under the same E1–E7 condition, and which of the six principal RQ1 effects retain their direction across both pipelines.

The comparison uses the completed Qwen and DeepSeek pipelines. Each pipeline covers the same 60 questions under E1–E7, giving 420 condition-level answers per backbone and 840 answers in total. The question set, memory stores, retrieval definitions, evidence budget, Answerer objective, and evaluation procedure are held constant. Both result sets are evaluated by the same LLM judge.

The backbone model is used by both the Retrieval Agent and the Answerer. The results therefore compare two complete end-to-end pipelines rather than isolating either component. RQ2 first compares the two backbones under matched conditions. It then tests whether six architecture effects identified in RQ1 retain the same direction across the two pipelines. E7 is included only in the matched-condition overview; its fusion effect is analysed under RQ3. Full Context is excluded because a matched Qwen Full Context run is not available.

5.3.1 Matched Backbone Performance and the Token F1 Gap

Table 5.13, Panel A compares Qwen and DeepSeek under each matched E1–E7 condition. Token F1 measures lexical overlap with the reference answer. Strict Accuracy and Soft Semantic Score measure semantic correctness. All condition means use the same 60 questions.

Panel A. Matched performance under E1–E7.

Table 5.13. Cross-backbone performance and architecture robustness.

Condition	Qwen F1	DeepSeek F1	Qwen Strict	DeepSeek Strict	Qwen Soft	DeepSeek Soft
E1	0.3949	0.3013	0.7167	0.7500	0.7667	0.8083
E2	0.4031	0.2455	0.6667	0.7000	0.7167	0.7083
E3	0.4411	0.2971	0.7500	0.7000	0.7917	0.7583
E4	0.3960	0.2314	0.7000	0.6667	0.7250	0.7083
E5	0.3948	0.2604	0.7833	0.7500	0.8167	0.8000
E6	0.3813	0.2240	0.7333	0.6667	0.7583	0.7083
E7	0.3691	0.2854	0.7333	0.7500	0.7583	0.8000

Qwen has higher Token F1 in all seven conditions. Its equal-weight mean across E1–E7 is 0.3972, compared with 0.2636 for DeepSeek. The difference is 0.1336. The condition-level gap ranges from 0.0837 in E7 to 0.1646 in E4. These equal-weight means summarise repeated condition paths over the same questions; they are not independent-sample estimates.

The semantic difference is much smaller and changes direction across conditions. The equal-weight Strict Accuracy means are 0.7262 for Qwen and 0.7119 for DeepSeek. The corresponding Soft Semantic Score means are 0.7619 and 0.7560. DeepSeek has higher Strict Accuracy in E1, E2, and E7, and higher Soft Semantic Score in E1 and E7. Qwen therefore has a clear lexical-overlap advantage, but it is not semantically superior in every condition.

The Token F1 gap becomes clearer when Precision, Recall, and answer length are separated. Across E1–E7, Qwen has a mean Token Precision of 0.3411, whereas DeepSeek has a mean of 0.1957. DeepSeek has slightly higher mean Token Recall: 0.6884 compared with 0.6774. Its answers are also longer, with a mean normalised length of 25.35 tokens compared with 14.03 for Qwen. DeepSeek answers are therefore about 80.7% longer, while their mean Precision is about 42.6% lower.

Token Precision divides matched answer tokens by all tokens in the candidate answer. A correct fact embedded in additional explanation, qualification, or paraphrastic wording can therefore receive lower Precision than a short direct answer. Since Token F1 is the harmonic mean of Precision and Recall, DeepSeek's small aggregate Recall advantage cannot offset its much larger Precision deficit. Longer answers and systematically lower Precision are thus the main measured pattern behind its lower overall F1. This is an observed association rather than evidence that answer length alone causes the gap.

The condition-level results show why one explanation is insufficient. In E1, E2, and E7, DeepSeek has higher Recall and higher Strict Accuracy despite lower F1. Here, the lower F1 is consistent with a lexical or answer-style penalty: the required information is often present, but it is expressed with more surrounding text or different wording. E2 provides a finer distinction. DeepSeek has two more fully correct answers than Qwen, but its Soft Semantic Score is slightly lower. This reflects a small difference in the distribution of fully correct, partially correct, and incorrect labels rather than a uniform semantic improvement.

E3–E6 show a different pattern. DeepSeek has lower Precision, lower Recall, and lower semantic scores in all four conditions. Verbosity alone cannot explain these results. The joint decline points to additional differences in retrieval, evidence selection, or evidence utilisation within these architecture paths.

The condition rankings are also metric-dependent. Under Qwen, E3 has the highest Token F1, whereas E5 has the highest Strict Accuracy and Soft Semantic Score. Under DeepSeek, E1 has the highest Token F1 and Soft Semantic Score, while E1, E5, and E7 tie for the highest Strict Accuracy. E2 is second in Qwen Token F1 but lowest in both Qwen semantic measures. Token overlap and semantic correctness must therefore remain separate outcomes.

5.3.2 Stability of the Main Architectural Findings

The second analysis tests whether the main RQ1 architecture effects retain their direction across the two backbone pipelines. Each effect is the target-condition mean minus the baseline-condition mean. Token F1 and Strict Accuracy determine the primary robustness category, while Soft Semantic Score supports interpretation.

An effect is classified as *stable* when its F1 and Strict directions are retained across both pipelines. It is *partly stable* when one direction is retained and the other reverses. It is *backbone-dependent* when both primary directions reverse. These labels describe directional reproduction across the two tested pipelines. They do not imply equal effect sizes or universal invariance across other models.

Panel B. Stability of the six principal RQ1 architecture effects. Each effect cell reports `F1 / Strict / Soft`. The cross-backbone difference is the DeepSeek effect minus the Qwen effect.

Contrast	Qwen effect	DeepSeek effect	Cross-backbone difference	Classification
E3−E1	+0.0462 / +0.0333 / +0.0250	−0.0041 / −0.0500 / −0.0500	−0.0504 / −0.0833 / −0.0750	Backbone-dependent
E4−E2	−0.0071 / +0.0333 / +0.0083	−0.0141 / −0.0333 / 0.0000	−0.0070 / −0.0667 / −0.0083	Partly stable
E2−E1	+0.0082 / −0.0500 / −0.0500	−0.0557 / −0.0500 / −0.1000	−0.0640 / 0.0000 / −0.0500	Partly stable
E4−E3	−0.0451 / −0.0500 / −0.0667	−0.0657 / −0.0333 / −0.0500	−0.0206 / +0.0167 / +0.0167	Stable, negative
E6−E5	−0.0135 / −0.0500 / −0.0583	−0.0364 / −0.0833 / −0.0917	−0.0229 / −0.0333 / −0.0333	Stable, negative
E5−E3	−0.0463 / +0.0333 / +0.0250	−0.0367 / +0.0500 / +0.0417	+0.0096 / +0.0167 / +0.0167	Stable metric trade-off

Three of the six architecture profiles are stable, two are partly stable, and one is backbone-dependent.

The E3−E1 contrast shows that folder organisation provides a conditional advantage for the current Qwen+R1 pipeline rather than a model-independent benefit. Under Qwen, replacing S1 with S2 raises Precision by 0.0378, Recall by 0.0527, Token F1 by 0.0462, Strict Accuracy by 0.0333, and Soft Semantic Score by 0.0250. The mean answer becomes shorter, decreasing from 13.28 to 11.97 normalised tokens. The retrieval process also uses slightly fewer calls, actions, and tokens. S2 therefore matches the observed Qwen search–answer behaviour well: the pipeline searches somewhat more efficiently and produces more concise and precise answers.

DeepSeek does not reproduce this profile. Its Token F1 changes by −0.0041, while Strict Accuracy and Soft Semantic Score each decline by 0.0500. Precision and Recall also fall, and the mean answer becomes longer. The directory advantage is therefore specific to the tested Qwen+R1 path.

The Qwen gain should not be reduced to answer brevity alone because answer length was not independently controlled. Qwen Evidence Recall increases by only 0.0072. The improvement is therefore more consistent with better navigation or more effective use of retrieved evidence than with access to substantially more annotated evidence. The gain is also concentrated in temporal questions, for which Qwen E3−E1 raises Token F1 by 0.1994, Strict Accuracy by 0.3077, and Soft Semantic Score by 0.1923. Within the current experiment, S2 is particularly useful for the Qwen+R1 temporal-question path, but it is not a backbone-independent storage improvement.

The E4−E2 contrast is only partly stable. Folder organisation under R2 produces small negative F1 changes for both pipelines: −0.0071 for Qwen and −0.0141 for DeepSeek. However, Strict Accuracy rises by 0.0333 under Qwen and falls by 0.0333 under DeepSeek. Soft Semantic Score moves from a small positive value under Qwen to no change under DeepSeek. The shared F1 decrease is minor, and the semantic effect is not reproduced. The experiment therefore does not support a cross-backbone semantic benefit from adding the S2 directory structure to the current R2 path.

The E2−E1 contrast shows a different form of partial stability. On S1, replacing R1 with R2 lowers Strict Accuracy by 0.0500 under both backbones. Soft Semantic Score also falls under both, by 0.0500 for Qwen and 0.1000 for DeepSeek. Token F1, however, rises slightly under Qwen and falls under DeepSeek. The small Qwen F1 gain does not translate into better semantic correctness. Within S1, R1 is therefore the more consistent semantic-quality option, although the lexical result depends on the backbone.

The E4−E3 contrast supports a clearer cross-backbone conclusion. On the fixed S2 store, replacing R1 with R2 lowers Token F1, Strict Accuracy, and Soft Semantic Score under both pipelines. Evidence Recall falls by 0.0417 under Qwen and by 0.0790 under DeepSeek. Any-hit and All-hit also decline in both cases. Thus, under Main-60, the fixed S2 representation, the current R1 and R2 implementations, and the fixed evidence budget, R1 provides higher answer quality and better annotated-evidence coverage across both tested backbones.

R2 nevertheless has a clear efficiency advantage. Relative to R1 on S2, it uses 6.68 fewer recorded model calls and 121,809 fewer query-time tokens per question under Qwen. Under DeepSeek, it uses 7.70 fewer calls and 172,189 fewer tokens. At the same time, R2 produces larger EvidenceBundles but weaker evidence coverage and answer quality. This is a stable quality–efficiency trade-off: R1 is the quality-oriented choice on S2, while R2 is the lower-cost path. The final practical value of that trade-off is assessed under RQ6.

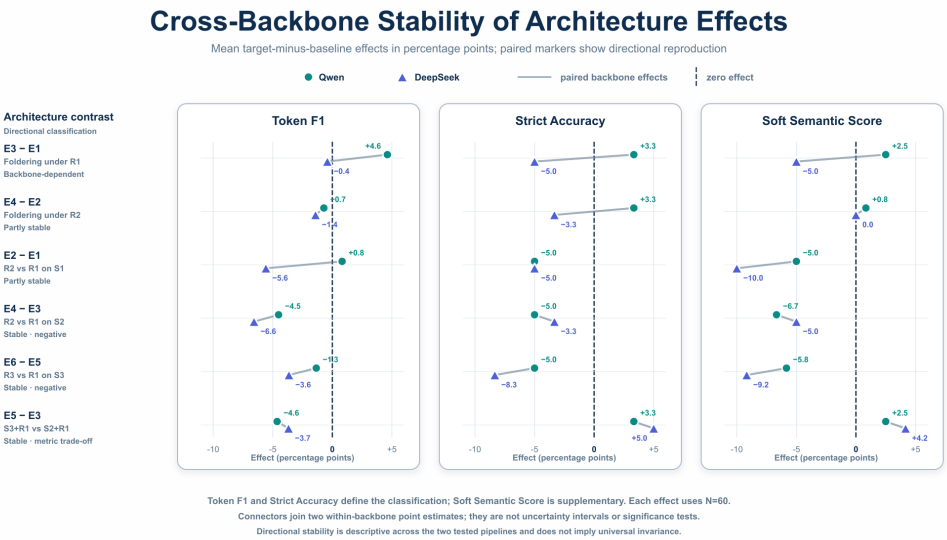


Figure 5.3. Cross-backbone stability of the six principal architecture effects. Each value is the target-minus-baseline mean difference over the same 60 questions, shown in percentage points; Table 5.13, Panel B reports the corresponding differences on the 0–1 scale. Circles denote Qwen and triangles denote DeepSeek. Each connector joins the Qwen and DeepSeek estimates for the same contrast; it is not a confidence interval or a significance test. Token F1 and Strict Accuracy determine the robustness classification, while Soft Semantic Score is supplementary.

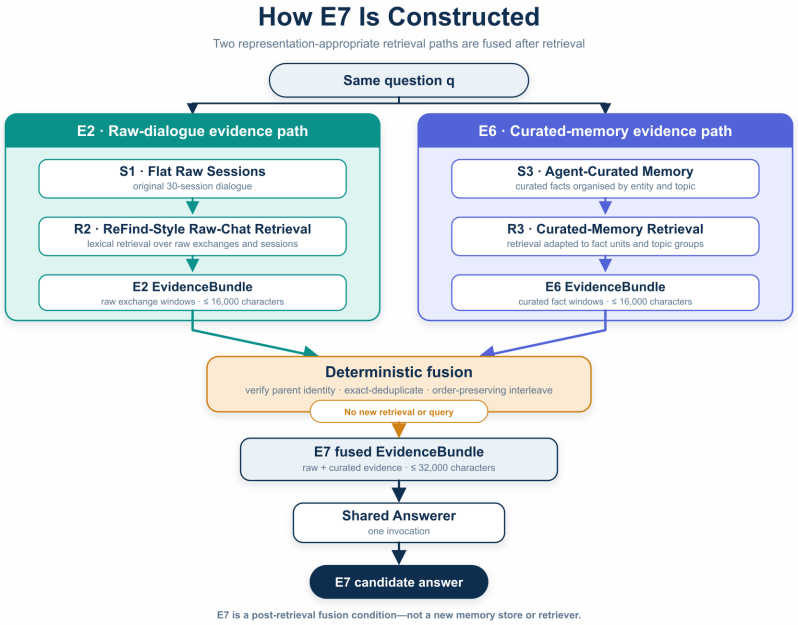


Figure 5.4. Composition of the E7 dual-source fusion condition. E2 combines S1 Flat Raw Sessions with R2 ReFind-Style Raw-Chat Retrieval, whereas E6 combines S3 Agent-Curated Memory with R3 Curated-Memory Retrieval. E7 validates and deterministically fuses the two completed EvidenceBundles, performs no new query or retrieval, and invokes the shared Answerer once.

The E6−E5 contrast is also stably negative. On the same frozen S3 store, replacing R1 with the current R3 path lowers all three answer metrics under both backbones. The decreases are larger under DeepSeek. Evidence Recall also falls by 0.0891 under Qwen and by 0.2371 under DeepSeek. The implemented R3 system therefore does not improve access to Agent-Curated Memory. Within the tested S3 setting, R1 is the stronger quality-oriented retrieval system for both backbone pipelines. This result applies to the current R3 implementation and does not imply that every retrieval method designed for curated memory would behave in the same way.

The E5−E3 comparison reproduces a stable metric trade-off. Under both backbones, S3+R1 lowers Token F1 but raises Strict Accuracy and Soft Semantic Score relative to S2+R1. Under Qwen, the changes are −0.0463, +0.0333, and +0.0250. Under DeepSeek, they are −0.0367, +0.0500, and +0.0417. Evidence Recall also rises by 0.1034 and 0.1149, respectively.

This result provides the strongest cross-backbone support for analysing lexical and semantic quality separately. Agent-Curated Memory produces answers that overlap less with the reference wording but are judged more semantically correct. The comparison is a complete-system anchor: S2 and S3 differ in representation, organisation, granularity, and provenance presentation. It therefore supports S3+R1 as a system configuration but does not isolate curation as the sole cause.

5.3.3 Answer to RQ2

RQ2 shows that the main architecture findings are conditionally, rather than universally, robust across backbone models.

First, three architecture profiles support clear cross-backbone conclusions within the tested setting. On S2, R1 provides higher answer quality and better annotated-evidence coverage than R2 under both Qwen and DeepSeek, although R2 uses fewer recorded calls and tokens. On S3, R1 also provides higher answer quality than the current R3 path under both backbones. Finally, S3+R1 consistently trades lower Token F1 for higher Strict Accuracy and Soft Semantic Score relative to S2+R1.

Second, two findings are only partly stable. On S1, R2 produces lower semantic scores than R1 under both backbones, but its F1 effect changes direction. Folder organisation under R2 produces a small negative F1 effect under both backbones, but its semantic effect is not reproduced.

Third, the main raw-storage benefit is backbone-dependent. Folder organisation improves lexical and semantic outcomes for Qwen+R1, shortens its answers, and slightly reduces its search effort. The effect is strongest on temporal questions. DeepSeek does not reproduce this pattern. S2 should therefore be treated as a useful configuration for the tested Qwen+R1 pipeline, not as a model-independent intervention.

The condition-level comparison also shows that Qwen's uniform Token F1 advantage does not constitute uniform semantic superiority. DeepSeek matches or exceeds Qwen's semantic scores in selected conditions, especially E1 and E7. Conversely, the stable S3+R1 result would be undervalued if only Token F1 were reported. Token overlap and semantic evaluation therefore answer different questions and must remain separate.

Overall, R1 is the more robust quality-oriented retrieval system in the tested S2 and S3 settings. R2 offers a cheaper but lower-quality alternative on S2, while the current R3 path does not improve S3. Agent-Curated Memory with R1 provides a reproducible semantic benefit despite lower lexical overlap. The preferred raw storage layout, however, depends on the backbone: S2 is beneficial for Qwen+R1, while this advantage does not transfer to DeepSeek. These findings support bounded architecture recommendations for the declared Main-60 setting rather than a single universal ranking of storage representations, retrieval systems, or backbone families.

5.4 Dual-Source Fusion

Section 5.4 addresses RQ3: does the fixed E7 condition benefit from combining evidence obtained through two different access paths? E2 retrieves evidence from raw dialogue, whereas E6 retrieves evidence from Agent-Curated Memory. E7 reuses both completed parent EvidenceBundles, performs no new search action, deterministically merges their evidence, and invokes the shared Answerer once on the fused bundle. It is therefore analysed as a fusion condition rather than as a third retrieval system.

The analysis forms 120 paired `(E2, E6, E7)` triples: 60 questions under each backbone model. All answer effects use the same 60 questions. Evidence Recall uses the 58 questions with valid evidence annotations. Category answer denominators are 13 multi-hop, 13 temporal, 7 open-domain, and 27 single-hop questions. The corresponding evidence denominators are 13, 13, 5, and 27. All effects in this section are descriptive paired changes. RQ3 introduces no additional confidence interval or significance test.

The analysis separates evidence-access complementarity from answer-level complementarity. Source-ID overlap concerns retrieved dialogue locations, not textual or semantic similarity. The descriptive parent oracle is a diagnostic oracle-selector benchmark rather than an executable system or a strict mathematical upper bound on fusion. Repair and Harm are defined using Strict Accuracy. They identify outcome patterns, not causal mechanisms. Causal attribution is deferred to RQ5, while the resource cost of executing both parent paths is deferred to RQ6.

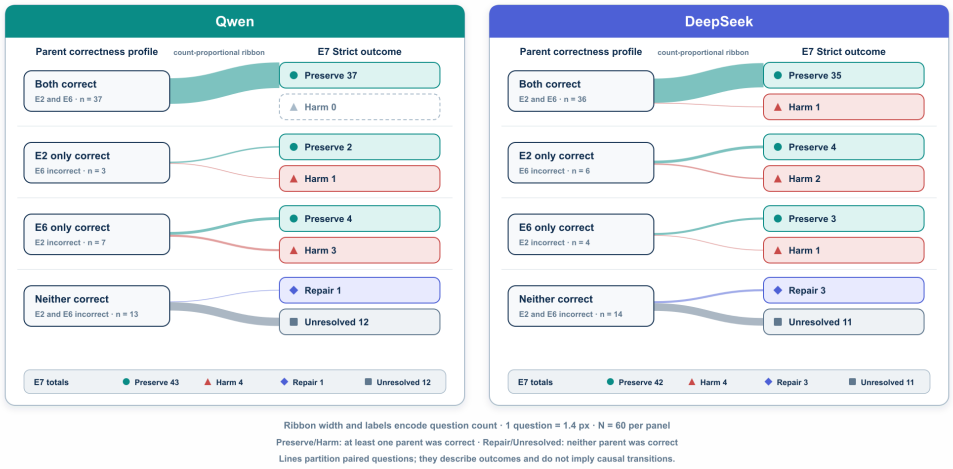


Figure 5.5. Parent Strict-correctness profiles and E7 fusion outcomes. Each panel maps the 60 questions from the four E2/E6 correctness profiles to Repair, Preserve, Harm, or Unresolved. Flow widths and labels represent question counts. Outcomes use Strict Accuracy only. Harm identifies a possible-interference case and does not establish a causal failure mechanism.

5.4.1 Does E7 Improve over Its Individual Parent Systems?

Table 5.14, Panel A reports paired E7-minus-parent effects. Positive values favour E7. Precision, Recall, F1, Strict Accuracy, and Soft Semantic Score use all 60 questions. Evidence Recall uses the 58 evidence-eligible questions.

Panel A. Paired E7-minus-parent effects.

Table 5.14. Dual-source fusion effects and diagnostics.

Backbone	Parent	ΔPrecision	ΔRecall	ΔF1	ΔStrict	ΔSoft	ΔEvidence Recall
Qwen	E2	−0.0425	+0.0194	−0.0340	+0.0667	+0.0417	+0.1178
Qwen	E6	−0.0028	−0.0131	−0.0122	0.0000	0.0000	+0.0632
DeepSeek	E2	+0.0408	+0.0341	+0.0398	+0.0500	+0.0917	+0.0977
DeepSeek	E6	+0.0498	+0.1280	+0.0614	+0.0833	+0.0917	+0.1695

Under Qwen, E7 provides a modest but meaningful semantic gain relative to E2. Strict Accuracy rises from 0.6667 to 0.7333, and Soft Semantic Score rises from 0.7167 to 0.7583. Evidence Recall increases from 0.7816 to 0.8994. Token F1 nevertheless falls from 0.4031 to 0.3691. Precision decreases by 0.0425, while Recall increases by only 0.0194. The Precision loss therefore dominates the lexical result.

The Qwen comparison with E6 is more limited. E7 and E6 have the same Strict Accuracy, 0.7333, and the same Soft Semantic Score, 0.7583. E7 raises Evidence Recall from 0.8362 to 0.8994, but its Token F1 is 0.0122 lower. Both Precision and Recall are slightly lower. Thus, E7 expands annotated-evidence coverage and matches the stronger Qwen parent semantically, but it does not improve Qwen answer quality on every metric.

The DeepSeek result is broader. Relative to E2, E7 raises Precision by 0.0408, Recall by 0.0341, Token F1 by 0.0398, Strict Accuracy by 0.0500, Soft Semantic Score by 0.0917, and Evidence Recall by 0.0977. Relative to E6, the corresponding gains are 0.0498, 0.1280, 0.0614, 0.0833, 0.0917, and 0.1695. E7 therefore exceeds both individual DeepSeek parents on all five reported answer-quality metrics and on annotated Evidence Recall.

The Strict transitions show that these mean changes contain both gains and losses. Against Qwen E2, E7 changes five non-correct answers to correct and one correct answer to non-correct. Against Qwen E6, the counts are three and three. Against DeepSeek E2, E7 produces six gains and three losses; against DeepSeek E6, it produces seven gains and two losses. These are parent-specific transitions. They are not the joint Repair and Harm states defined later, because a question that is wrong under one parent may still be correct under the other.

The lexical and semantic results must therefore remain separate. Under Qwen, fusion produces a semantic-and-coverage benefit mainly relative to E2, while Token F1 declines relative to both parents. Under DeepSeek, the fused condition improves every reported answer metric relative to both parents. This difference shifts the analysis from whether fusion is universally beneficial—it is not—to whether E2 and E6 provide complementary opportunities and how fully E7 converts those opportunities into correct answers.

5.4.2 Are the Two Parent Paths Complementary, and How Much of That Opportunity Does E7 Capture?

Complementarity is assessed at two levels. Answer-level complementarity asks whether E2 and E6 are Strict-correct on different questions. Evidence-access complementarity asks whether they retrieve different dialogue source IDs and contribute different annotated source locations. Neither measure guarantees that the shared Answerer will use the merged evidence correctly.

Panel B. Parent Strict-correctness profiles and descriptive oracle-selector benchmark.

Backbone	Both correct	E2 only	E6 only	Neither	Parent oracle correct	E7 correct
Qwen	37	3	7	13	47	44
DeepSeek	36	6	4	14	46	45

Panel B shows genuine answer-level complementarity. Under Qwen, E2 and E6 are both correct on 37 questions, only E2 is correct on three, only E6 is correct on seven, and neither is correct on 13. Under DeepSeek, the corresponding counts are 36, six, four, and 14. Exactly one parent is therefore Strict-correct on ten questions under each backbone. The two parent systems do not succeed and fail on identical question sets.

The parent oracle is defined as correct whenever either E2 or E6 is Strict-correct. It would select a correct parent answer for 47 Qwen questions and 46 DeepSeek questions. This is a descriptive selector that uses gold outcome labels after evaluation. It is not an implementable system. E7 answers 44 Qwen questions and 45 DeepSeek questions correctly. It therefore ties the best Qwen parent, E6, and exceeds the best DeepSeek parent, E2, by three questions. At the same time, it remains three Qwen questions and one DeepSeek question below the oracle-selector benchmark.

These statements are compatible. E7 can outperform an individual parent by recovering questions missed by that parent. It can also remain below the oracle selector if it fails to preserve every question answered correctly by either parent. In addition, E7 can repair a question that neither parent answers correctly. The parent oracle therefore measures available parent correctness, while the E7 total reflects both preservation and new repair.

Panel C. Evidence-source overlap and union coverage.

Backbone	E2 Evidence Recall	E6 Evidence Recall	Union/E7 Evidence Recall	Mean source-ID Jaccard	Either parent adds unique gold coverage	Union complete
Qwen	0.7816	0.8362	0.8994	0.1956	14/58	49/58
DeepSeek	0.7974	0.7256	0.8951	0.1587	19/58	49/58

The evidence-access results explain why fusion has room to help. The mean E2/E6 source-ID Jaccard is only 0.1956 under Qwen and 0.1587 under DeepSeek. The two paths therefore return substantially different source-ID sets. This measure does not compare evidence text and does not establish semantic conflict or semantic complementarity. It only shows that the access paths reach different recorded locations.

The combined bundle increases annotated-source coverage. Under Qwen, the union raises Evidence Recall from 0.7816 for E2 and 0.8362 for E6 to 0.8994. Under DeepSeek, it raises the values from 0.7974 and 0.7256 to 0.8951. The union reaches every annotated source on 49 of 58 eligible questions under each backbone.

The gold-source occurrence counts show contributions from both parents. Each backbone has 82 annotated-source occurrences. Under Qwen, 53 are covered by both parents, seven only by E2, 11 only by E6, and 11 by neither. Under DeepSeek, the partition is 48, 13, ten, and 11. At least one parent contributes unique annotated coverage on 14 Qwen questions and 19 DeepSeek questions. However, most eligible questions contain no marginal annotated source from either parent: 44 of 58 under Qwen and 39 of 58 under DeepSeek. The coverage gain is real, but it is not uniform across questions.

Because E7 performs no new retrieval, these results should not be described as an improvement in a new retriever. The improvement comes from combining the source sets already obtained by the two parent paths. The low source-ID overlap and the asymmetric parent-correctness cells together establish an opportunity for fusion. The remaining oracle gap shows that deterministic merging and one shared Answerer do not fully realise that opportunity.

5.4.3 When Does Fusion Repair Errors, and When Does It Introduce Interference?

The joint outcome analysis uses four mutually exclusive Strict-based states. **Repair** means that neither parent is Strict-correct but E7 is correct. **Preserve** means that at least one parent is correct and E7 remains correct. **Harm** means that at least one parent is correct but E7 is not. **Unresolved** means that neither parent nor E7 is correct. Here, *not Strict-correct* includes both partial and incorrect LLM judge labels.

Panel D. Overall Strict-based fusion outcomes.

Backbone	Repair	Preserve	Harm	Unresolved	Conditional repair	Conditional harm	Net versus parent oracle
Qwen	1	43	4	12	1/13 (7.69%)	4/47 (8.51%)	−3
DeepSeek	3	42	4	11	3/14 (21.43%)	4/46 (8.70%)	−1

Under Qwen, E7 repairs one of the 13 questions missed by both parents and preserves 43 questions answered correctly by at least one parent. It harms four oracle-correct questions and leaves 12 unresolved. The conditional Repair rate is 7.69%, and the conditional Harm rate is 8.51%. The net balance relative to the parent oracle is therefore `Repair − Harm = −3` questions.

Under DeepSeek, E7 repairs three of the 14 questions missed by both parents, preserves 42, harms four, and leaves 11 unresolved. The conditional Repair rate is 21.43%, while the conditional Harm rate is 8.70%. Its net balance relative to the parent oracle is −1 question. DeepSeek therefore converts more of the both-parent-wrong cases into correct fused answers, while the conditional Harm rate is similar under the two backbones.

Figure 5.5 makes the oracle gap explicit. Under Qwen, E7 preserves all 37 both-correct cases, two of three E2-only cases, and four of seven E6-only cases. Under DeepSeek, it preserves 35 of 36 both-correct cases, four of six E2-only cases, and three of four E6-only cases. Fusion can therefore preserve a large majority of available correct answers without preserving all of them. It can also generate a correct answer when neither parent is Strict-correct, as shown by the Repair flows.

The category analysis identifies where these aggregate outcomes occur. Table 5.14, Panel E reports every paired effect as `versus E2 / versus E6`. `R/P/H/U` denotes Repair, Preserve, Harm, and Unresolved. All category findings are exploratory.

Panel E. Category-level E7 effects and outcome states.

Backbone	Category	Answer/Evidence N	ΔF1	ΔStrict	ΔSoft	ΔEvidence Recall	R/P/H/U
Qwen	Multi-hop	13/13	−0.0578 / −0.0361	+0.1538 / +0.0769	+0.1154 / +0.0385	+0.1154 / +0.0641	1/9/0/3
Qwen	Temporal	13/13	−0.0497 / −0.0213	0.0000 / +0.0769	0.0000 / +0.0769	+0.2308 / +0.0385	0/9/1/3
Qwen	Open-domain	7/5	+0.0450 / −0.0050	+0.1429 / 0.0000	+0.1429 / 0.0000	+0.2667 / +0.2667	0/4/0/3
Qwen	Single-hop	27/27	−0.0356 / +0.0017	+0.0370 / −0.0741	0.0000 / −0.0556	+0.0370 / +0.0370	0/21/3/3
DeepSeek	Multi-hop	13/13	+0.0459 / +0.0818	+0.1538 / 0.0000	+0.2692 / −0.0385	+0.2179 / +0.0641	1/8/1/3
DeepSeek	Temporal	13/13	+0.0767 / +0.0534	+0.1538 / +0.2308	+0.1538 / +0.2308	+0.1538 / +0.1538	2/9/1/1
DeepSeek	Open-domain	7/5	+0.1316 / +0.0122	−0.1429 / 0.0000	−0.1429 / 0.0000	+0.1667 / +0.2000	0/2/1/4
DeepSeek	Single-hop	27/27	−0.0046 / +0.0682	0.0000 / +0.0741	+0.0370 / +0.1111	0.0000 / +0.2222	0/23/1/3

Qwen multi-hop questions show its clearest semantic fusion benefit. Strict Accuracy rises by 0.1538 relative to E2 and 0.0769 relative to E6. Soft Semantic Score also rises relative to both. The category contains one Repair and no Harm. Token F1 nevertheless falls by 0.0578 and 0.0361. The same fused answers can therefore improve semantic correctness while moving further from the reference wording.

Qwen temporal questions show a different pattern. Evidence Recall rises by 0.2308 relative to E2, but Strict Accuracy and Soft Semantic Score do not change. Relative to E6, the semantic gains are 0.0769. This pattern is consistent with a coverage-to-utilisation gap relative to E2, but it does not prove that the Answerer ignored useful evidence. That hypothesis requires the trace review in RQ5.

Qwen single-hop questions contain three of the four Qwen Harm cases and no Repair. E7 is slightly better than E2 on Strict Accuracy but worse than E6. It therefore does not fail against both parents uniformly. Instead, this category supplies the strongest possible-interference signal when the fused result is compared with the correctness available from either parent. The apparent Qwen open-domain gains remain exploratory because the category has only seven answer questions and five evidence-eligible questions.

DeepSeek temporal questions provide the clearest overall fusion success. Token F1, Strict Accuracy, Soft Semantic Score, and Evidence Recall all increase relative to both parents. The category contains two Repairs, nine Preserves, one Harm, and one Unresolved case. DeepSeek multi-hop questions also gain F1 relative to both parents, but their semantic result depends on the selected parent: Strict Accuracy improves relative to E2 and ties E6, while Soft Semantic Score improves relative to E2 and falls slightly relative to E6.

DeepSeek single-hop fusion is stronger than E6, but it ties E2 exactly on Strict Accuracy and Evidence Recall. The open-domain category shows the opposite movement of lexical and semantic scores. Token F1 rises relative to both parents, while Strict Accuracy and Soft Semantic Score fall relative to E2 and tie E6. Given the small category denominator, this is evidence of metric disagreement within the observed subgroup rather than a general open-domain result.

Several mechanisms could produce lower Precision or F1 after fusion: repeated information, longer answers, paraphrastic wording, irrelevant evidence, or competition between evidence items. The current RQ3 tables do not distinguish these mechanisms. Harm is therefore treated as a possible-interference candidate rather than proof of evidence conflict or noise. The audited analysis exports four Repair and eight Harm backbone-question records, corresponding to nine unique question IDs, for the unified RQ5 trace review.

5.4.4 Answer to RQ3

RQ3 shows that dual-source fusion is useful, but its value depends on the backbone and question type.

First, fusion increases annotated-evidence coverage. E2 and E6 retrieve substantially different source-ID sets. Their union raises Evidence Recall to 0.8994 under Qwen and 0.8951 under DeepSeek. These results establish different and partly complementary access paths. They do not show that the evidence texts are semantically different or conflicting.

Second, the answer-quality gain is backbone-dependent. Under Qwen, E7 improves Strict Accuracy and Soft Semantic Score relative to E2, ties E6 on both semantic measures, and lowers Token F1 relative to both parents. The Qwen result is therefore a modest, parent-dependent semantic-and-coverage benefit. Under DeepSeek, E7 improves Precision, Recall, Token F1, Strict Accuracy, Soft Semantic Score, and Evidence Recall relative to both parents. The observed DeepSeek result is a clear and broad descriptive improvement.

Third, complementarity creates an opportunity rather than a guarantee. E7 ties the best Qwen parent and exceeds both DeepSeek parents in Strict Accuracy. However, it remains below the descriptive oracle-selector benchmark by three Qwen questions and one DeepSeek question. It preserves most questions answered correctly by at least one parent and repairs some questions missed by both, while also producing four Harm cases under each backbone.

Finally, the benefit varies by question type. Qwen multi-hop questions show the clearest semantic gain, whereas Qwen single-hop questions contain the strongest possible-interference signal. DeepSeek temporal questions show the clearest overall fusion success. These category findings are descriptive, and the open-domain subgroup is too small to support an independent general claim.

Overall, E7 exploits part of the complementarity between raw-dialogue and Agent-Curated Memory access, especially under DeepSeek. It does not dominate both parents on every metric or preserve every available correct answer. Dual-source fusion should therefore be treated as a backbone- and question-type-dependent method rather than as a universally superior replacement for either parent system.

Having established that selective fusion expands evidence access without eliminating evidence-utilisation failures, Section 5.5 compares selective retrieval with Full Context. Full Context is treated as a diagnostic access condition, not as an additional E-series architecture.

5.5 Full Context versus Selective Retrieval

Section 5.5 addresses RQ4: does complete access to the raw conversation outperform selective retrieval, and what does each access regime reveal about the remaining system bottlenecks? Full Context supplies all 30 raw sessions to the Answerer in chronological order. It performs no retrieval and does not use the selective EvidenceBundle limit. It is therefore a diagnostic access condition rather than E8, a fourth storage representation, a fourth retrieval system, or an oracle.

The analysis compares Full Context with DeepSeek E1–E7 on the same 60 questions. This gives 420 question-condition pairs. The complete E1–E7 comparison is retained to avoid selecting only favourable baselines, while E1, E5, and E7 serve as the main interpretive anchors. E1 is the raw-session Agentic Filesystem condition, E5 is the Agent-Curated Memory condition accessed by R1, and E7 is the widest selective condition because it fuses the completed E2 and E6 EvidenceBundles.

All answer metrics use the same 60 questions. Full Context has complete raw-source visibility by construction, but it has no retrieval event and no retrieved source set. Evidence Recall, Any-hit, and All-hit are therefore not applicable to Full Context. Complete visibility must not be reported as Evidence Recall equal to one. The selective conditions were served by `deepseek-v4.1-flash`, whereas Full Context was served by `deepseek-v4.1-flash-1017`. The comparison is consequently a strong same-family diagnostic comparison, not a strict single-variable causal ablation. No Qwen Full Context run is available, so the observed Full Context pattern is not tested across backbone families.

5.5.1 Overall Quality and Lexical–Semantic Divergence

Table 5.15 reports the complete quality profile. Full Context has the highest Strict Accuracy, 0.7667, and the highest Soft Semantic Score, 0.8250. However, E1, E5, and E7 each reach Strict Accuracy of 0.7500. Full Context therefore answers 46 questions correctly, compared with 45 for each of these three conditions. Its semantic lead over the strongest selective conditions is one question rather than a broad separation.

Table 5.15. Overall answer quality under Full Context and DeepSeek E1–E7. Precision, Recall, F1, Strict Accuracy, Partial Rate, and Soft Semantic Score use all 60 questions. Evidence Recall uses the 58 evidence-eligible questions for E1–E7 and is not applicable to Full Context. Scores are macro-averages on the 0–1 scale.

Condition	Storage and access path	Precision	Recall	F1	Strict	Partial	Soft	Evidence Recall
Full Context	Complete S1 raw history; no retrieval	0.1680	0.7296	0.2445	**0.7667**	0.1167	**0.8250**	N/A
E1	S1 + R1	**0.2345**	0.7003	**0.3013**	0.7500	0.1167	0.8083	0.8779
E2	S1 + R2	0.1720	0.7023	0.2455	0.7000	0.0167	0.7083	0.7974
E3	S2 + R1	0.2315	0.6824	0.2971	0.7000	0.1167	0.7583	0.8477
E4	S2 + R2	0.1630	0.6504	0.2314	0.6667	0.0833	0.7083	0.7687
E5	S3 + R1	0.1930	**0.7389**	0.2604	0.7500	0.1000	0.8000	**0.9626**
E6	S3 + R3	0.1630	0.6084	0.2240	0.6667	0.0833	0.7083	0.7256
E7	Fused E2 + E6 evidence	0.2128	0.7364	0.2854	0.7500	0.1000	0.8000	0.8951

Full Context does not dominate the lexical metrics. Its F1 of 0.2445 is lower than E1, E3, E5, and E7, almost identical to E2, and higher only than E4 and E6. The contrast with E1 is the clearest lexical–semantic separation. Relative to E1, Full Context raises Recall by 0.0294 but lowers Precision by 0.0665 and F1 by 0.0567. Strict Accuracy and Soft Semantic Score each rise by only 0.0167.

This pattern should not be reduced to answer length. The mean Full Context answer contains 23.37 normalised scoring tokens. It is 3.15 tokens longer than E1, but it is shorter than E2, E4, E5, E6, and E7. The relevant difference is therefore not that Full Context always generates more tokens. It is that the generated tokens are often less concentrated on the short reference wording.

The LoCoMo reference answers in this sample are frequently minimal canonical statements, such as a yes/no response, one activity, one item, or a short list of locations. Full Context can support a broader response that includes a conclusion, its temporal basis, related facts, or a paraphrase of the same event. These additions can remain semantically valid while adding tokens that do not appear in the reference answer. Token Precision then falls even when the LLM judge regards the core answer as correct. In other cases, the model uses a correct synonym or standard spelling that does not reproduce the reference string exactly. Token F1 and semantic evaluation therefore reward different properties: the former rewards reference-wording concentration, whereas the latter evaluates whether the proposed answer is substantively correct.

Two observed cases illustrate this distinction. For a yes/no question about whether Calvin wished to become more popular, the one-token reference answer is much shorter than the explanatory Full Context response. The judge marks the response correct, but the token F1 remains low. In a multi-hop question about shared activities, the reference answer contains only working on cars, while the Full Context response also mentions music and later plans supported by the conversation. The judge accepts the broader answer, but the unmatched content lowers lexical Precision. Because a selective answer with similar additional content receives a Partial judgement on the same question, this case is sensitive to LLM-judge phrasing and is used only as an illustration, not as causal evidence. These cases show why a semantically valid response can receive a low token score. They do not imply that every low-F1 Full Context answer is correct.

Table 5.16, Panel A reports the same-question effects for every selective condition. All differences are Full Context minus the selective condition. The F1 win/tie/loss counts also take the Full Context perspective.

Panel A. Paired answer-quality effects.

Table 5.16. Paired Full Context-minus-selective effects and Strict outcome quadrants.

Selective condition	Delta Precision	Delta Recall	Delta F1	F1 W/T/L	Delta Strict	Delta Soft
E1	-0.0665	+0.0294	-0.0567	19/14/27	+0.0167	+0.0167
E2	-0.0040	+0.0273	-0.0010	22/17/21	+0.0667	+0.1167
E3	-0.0635	+0.0473	-0.0526	18/16/26	+0.0667	+0.0667
E4	+0.0050	+0.0793	+0.0131	20/15/25	+0.1000	+0.1167
E5	-0.0250	-0.0093	-0.0158	27/9/24	+0.0167	+0.0250
E6	+0.0050	+0.1213	+0.0206	30/15/15	+0.1000	+0.1167
E7	-0.0448	-0.0067	-0.0408	19/16/25	+0.0167	+0.0250

The paired counts reinforce the aggregate result. Full Context loses more F1 comparisons than it wins against E1, E3, and E7, despite its small semantic lead. Its largest observed semantic gains occur against E4 and E6, the weaker selective conditions in this sample. It is not a uniform replacement for the best selective paths.

5.5.2 What Complete Visibility Repairs and What It Does Not

The Strict outcome quadrants separate average gains from question-level asymmetry. A Full Context-only case is a question on which complete visibility is Strict-correct and the selective condition is not. A selective-only case shows the reverse. Neither label establishes the failure cause by itself.

Panel B. Strict outcome quadrants.

Selective condition	Both correct	Full Context only	Selective only	Neither correct
E1	42	4	3	11
E2	39	7	3	11
E3	41	5	1	13
E4	38	8	2	12
E5	43	3	2	12
E6	38	8	2	12
E7	42	4	3	11

The E7 comparison provides the most useful wide-evidence diagnosis. Full Context and E7 are both correct on 42 questions. Full Context alone is correct on four, E7 alone is correct on three, and neither is correct on 11. Complete visibility therefore produces a net gain of only one Strict-correct question over E7. The same net gain appears against E1 and E5.

The Full Context-only cases show two different repair patterns. The first is missing or incomplete evidence access. Against E7, one open-domain question has zero annotated Evidence Recall and omits the direct statement that Calvin wanted to grow his fanbase. Two multi-hop questions also show incomplete access: one omits Detroit from a list of cities, and another retrieves only one of two mishaps. Full Context contains the missing source material and produces the complete semantic answer. These cases support the direct role of complete visibility in repairing a missed source, a missing answer item, or one absent hop in a cross-session answer.

The second pattern occurs even when the selective bundle covers all annotated source turns. In one temporal case, E1 has Evidence Recall of 1.0 but gives only the month. Full Context combines the statement made on 31 May that the event happened “last week” with the surrounding timeline and returns the last week of May. Two E5 comparisons provide further examples. For one single-hop question, E5 has Evidence Recall of 1.0 but declines to connect a tour discussion with skiing; Full Context uses the broader narrative and answers *skiing*. For one temporal question, E5 also has Evidence Recall of 1.0 but remains cautious about whether references to a red car and a blogged car denote the same entity; Full Context follows the chronology and answers *late October 2023*. These cases are consistent with complete visibility supporting relation resolution after the required source turns are already available. They suggest that complete visibility can repair both evidence omission and some cases of incomplete evidence utilisation. The latter interpretation remains a mechanism candidate until the unified trace review in RQ5.

Complete visibility does not remove the remaining errors. Full Context itself produces 46 correct, seven partial, and seven incorrect judgements. Against E7, the 11 neither-correct cases show that access to every raw session is not sufficient for all questions. The three E7-only cases are equally important because they show that selective evidence can outperform complete visibility.

The observed E7-only cases illustrate three focusing effects. First, E7 combines two March purchases correctly, whereas Full Context becomes over-cautious about whether the mansion was explicitly bought. Second, E7 uses a return-from-San-Francisco statement to infer Dave’s location in the final two weeks of August, whereas Full Context focuses on nearby dates that do not state the location directly and concludes that the answer is unavailable. Third, E7 preserves the modifier *gold* in a necklace description, whereas Full Context retains the necklace and diamond pendant but drops the material and receives a partial judgement. A separate E5 comparison cannot support a competing-state interpretation: trace review found that the annotated evidence referred to a different speaker from the person named in the question. That case is therefore treated as a source–question entity mismatch with an indeterminate mechanism and is not used as evidence for a focusing effect.

These examples provide concrete support for evidence focus. Selection can remove unrelated dialogue, reduce competition between repeated or temporally distinct facts, expose a decisive temporal anchor, and preserve a fine-grained attribute. However, RQ4 does not claim that a specific deleted passage caused each change. The defensible inference is narrower: Full Context contains the required source information in these cases, yet a smaller selective input produces the correct answer. Detailed causal labels are assigned only after the RQ5 trace review.

The scale of the compression strengthens this interpretation. Full Context supplies 108,960 raw-context characters per question. E1 supplies a mean EvidenceBundle of 1,720.9 characters, or 1.58% of the Full Context size. E7 supplies 19,758.6 characters, or 18.13%. Despite this reduction, E1 and E7 each answer only one fewer question correctly than Full Context. Selective retrieval therefore does more than reduce the Answerer input. It changes the set of facts presented simultaneously to the Answerer, which is consistent with a focusing effect. This observation concerns the Answerer input only; total retrieval and model-call cost is analysed under RQ6.

5.5.3 Evidence Focus and Question-Category Differences

Table 5.17 compares Full Context with E7 by question category. E7 is used here because it is the widest selective EvidenceBundle and combines raw-dialogue and Agent-Curated Memory access. The category results are exploratory. In particular, the open-domain subgroup contains only seven questions.

Panel A. Answer-quality results.

Table 5.17. Category-specific comparison of Full Context and E7.

Category	Path	N	Precision	Recall	F1	Strict	Partial	Soft
Multi-hop	E7	13	0.2201	0.7417	0.2908	0.6923	0.2308	0.8077
Multi-hop	Full Context	13	0.1520	0.7026	0.2366	**0.8462**	0.0769	**0.8846**
Temporal	E7	13	0.2186	0.6231	0.2636	**0.8462**	0.0000	**0.8462**
Temporal	Full Context	13	0.2166	0.6590	0.2636	0.7692	0.0769	0.8077
Open-domain	E7	7	0.1692	0.5000	0.1909	0.2857	0.0000	0.2857
Open-domain	Full Context	7	0.0320	0.6429	0.0608	**0.4286**	0.1429	**0.5000**
Single-hop	E7	27	0.2179	**0.8496**	**0.3177**	**0.8519**	0.1111	**0.9074**
Single-hop	Full Context	27	0.1876	0.7992	0.2868	0.8148	0.1481	0.8889

Panel B. Strict outcome quadrants for E7 versus Full Context.

Category	Both correct	Full Context only	E7 only	Neither correct
Multi-hop	8	3	1	1
Temporal	10	0	1	2
Open-domain	2	1	0	4
Single-hop	22	0	1	4

Multi-hop questions show the clearest benefit from complete visibility. Full Context reaches Strict Accuracy of 0.8462 and Soft Semantic Score of 0.8846, compared with 0.6923 and 0.8077 for E7. Three multi-hop questions are Full Context-only, while one is E7-only. This pattern is consistent with complete visibility avoiding the loss of one required fact from a cross-session list or reasoning chain.

The same category also shows the strongest lexical–semantic disagreement. Full Context F1 is 0.2366, below E7 at 0.2908, even though its semantic scores are higher. This is not a simple answer-length effect: the mean Full Context multi-hop answer contains 27.00 normalised tokens, compared with 28.69 for E7. The difference lies in answer content and reference-wording concentration. When Full Context recovers a missing city or a second mishap, it often also provides explanatory context. The LLM judge rewards the completed answer, but the extra or paraphrased content can reduce token Precision. In addition, a reference answer may contain only one canonical activity even when the broader conversation supports several. Full Context can then be semantically acceptable yet lexically distant from the reference.

Temporal questions favour a more specific selective design. E7 and Full Context have almost identical F1, 0.2636, but E7 has higher Strict Accuracy, 0.8462 versus 0.7692, and higher Soft Semantic Score, 0.8462 versus 0.8077. E7 is composed of E2 and E6 evidence: E2 applies R2 to S1 raw sessions, while E6 applies R3 to S3 Agent-Curated Memory. The fused bundle can place original time expressions and structured event facts in the same focused input. This concentration can make a return date, relative-time phrase, or event order more salient than it is inside all 30 sessions. The advantage belongs to the complete E7 access path; it is not evidence that every selective retriever improves temporal reasoning. E2, E4, and E6 individually remain below Full Context on the semantic measures for this category.

Single-hop questions provide the broadest evidence for focused retrieval, especially R1 over raw sessions. E1, which uses S1+R1, reaches F1 of 0.3960 and Strict Accuracy of 0.8519. E3, which uses S2+R1, reaches 0.3690 and 0.8519. E7 reaches 0.3177 and 0.8519. Full Context reaches only 0.2868 and 0.8148. A single-hop question often depends on one item, location, hobby, or attribute. R1 can inspect a small region of the raw session store and pass the local fact to the Answerer without exposing many later states or related attributes. The result is a more concentrated answer and, in the observed E1 and E3 conditions, substantially higher token F1. This is not a universal selective-retrieval result: E6 is weaker than Full Context on Single-hop Strict Accuracy. The strongest claim is therefore specific to R1 on S1 or S2, with E7 providing a smaller supporting focusing effect.

The open-domain result points in the opposite direction but remains too small for an independent conclusion. Full Context improves Strict and Soft relative to E7, while its Precision and F1 are much lower. Some questions in this category require a judgement over the absence or presence of support across the whole conversation. Full visibility can help such a judgement, but a long explanatory answer is poorly aligned with a one-token reference. With only seven questions, this pattern is retained as an exploratory mechanism candidate rather than a category-wide finding.

5.5.4 Answer to RQ4

RQ4 does not identify one access regime that dominates on every outcome. Full Context has the highest overall Strict Accuracy and Soft Semantic Score in the tested DeepSeek-family comparison, but its lead over E1, E5, and E7 is one correct question. It also has lower Precision and F1 than several selective conditions. Complete visibility therefore yields the highest semantic means in this same-family diagnostic comparison without guaranteeing a concise reference-aligned answer.

The observed Full Context gains are concentrated in cases where the selective path misses information or leaves a relation unresolved. These include an omitted source, a missing member of a list, an incomplete multi-hop chain, and a relation that becomes clearer in the complete chronology. Relative to E7, this pattern is strongest in the Multi-hop category in this sample. However, 14 Full Context answers remain non-correct under the Strict metric, and some selective conditions solve questions that Full Context misses. Complete source visibility raises the lower bound of information availability, but it does not remove long-context localisation, temporal reasoning, competing-state, evidence-utilisation, or answer-calibration problems.

Selective retrieval retains an independent focusing value. It can compress the Answerer input to a small fraction of the raw history while preserving almost the same overall semantic accuracy. The most useful selective path depends on the question structure. Temporal questions show the clearest semantic benefit from E7’s dual-source focused evidence. Single-hop questions show the clearest lexical and Strict benefit from R1 operating on S1 or S2 raw sessions. Full Context is stronger for observed Multi-hop aggregation, whereas selective evidence is stronger when one temporal anchor or one fine-grained fact must be isolated from competing history.

The practical conclusion is conditional. Full Context is a useful diagnostic and a possible recovery path when selective evidence appears incomplete. It is not an oracle and does not make retrieval unnecessary. Conversely, selective retrieval should not be viewed only as a token-saving mechanism. It also controls the evidence that competes for the Answerer’s attention. The asymmetric cases identified here are carried into RQ5 for unified trace-level error attribution. Their total resource implications, including Full Context input cost and selective retrieval cost, are evaluated under RQ6.

5.6 RQ5: Error Chain and Bottleneck Diagnosis

RQ5 asks where information is lost or misused between the memory store and the evaluated answer. The analysis follows the implemented information path rather than treating every incorrect answer as the same type of failure. Section 5.6.1 defines the common error population and attribution rules. Section 5.6.2 then begins with the storage layer and tests whether S1–S3 preserve the required information and whether their organisation creates a plausible access burden. Sections 5.6.3–5.6.5 examine retrieval, evidence packaging, Answerer utilisation, Full Context processing, and evaluation disagreement. Section 5.6.6 integrates these findings into the final answer to RQ5.

5.6.1 Root-Cause Model and Analysis Design

RQ5 uses one common diagnostic model across all tested conditions. The purpose is to identify the first supported break in the information chain rather than to infer a cause directly from a low final score. Source integrity and data normalisation serve only as eligibility checks. They are not treated as an independent empirical result.

Two linked populations are used. The complete outcome population contains all 900 frozen answers: 420 Qwen E1–E7 records, 420 DeepSeek E1–E7 records, and 60 DeepSeek-family Full Context records. The primary error-attribution population contains the 250 Strict-noncorrect answer paths. These comprise 205 E1–E6 selective paths, 31 E7 fusion paths, and 14 Full Context paths. Of the 250 paths, 233 are evidence-eligible: 191 E1–E6 paths, 29 E7 paths, and 13 Full Context paths. The remaining 17 comprise 14 E1–E6 paths, two E7 paths, and one Full Context path associated with the two questions without official source annotations. They remain in answer and evaluation analysis but are excluded from locator-based attribution.

The unit of analysis is an answer path under one condition and backbone model. Related paths are also grouped by `backbone x question`. This grouping allows an incorrect path to be compared with a correct counterfactual path for the same question. It also prevents E1–E7 records from being interpreted as unrelated examples. Answer outcomes use all 60 questions, whereas annotated-evidence metrics use the fixed 58-question eligible set.

For E1–E6, the diagnostic chain contains seven ordered states: store availability, retriever reach, model observation, evidence retention, final-input sufficiency, Answerer utilisation, and evaluation agreement. E7 enters after the E2 and E6 EvidenceBundles have been produced because it performs no new retrieval. Full Context enters at final-input availability because it bypasses both selective retrieval and EvidenceBundle construction.

The earliest broken state supported by the trace is treated as the primary cause. A later state is recorded as a secondary mechanism only when it is independently visible and materially affects the outcome. A low answer score is not evidence of a storage failure when the required fact is present. Low Evidence Recall alone is not sufficient to establish retrieval failure because a valid alternative source may be available. An Answerer failure is assigned only when the final input is sufficient for the required answer. Cases remain `mixed` when two mechanisms cannot be separated and `indeterminate` when the available record does not support a defensible attribution.

This common contract keeps the following analyses comparable. Section 5.6.2 audits storage fidelity and organisation. Section 5.6.3 reconstructs retrieval and evidence-access loss. Section 5.6.4 examines final-input sufficiency, packaging, Answerer utilisation, E7, and Full Context. Section 5.6.5 tests whether lexical and semantic evaluation disagreement changes the architectural interpretation. Section 5.6.6 assigns mutually exclusive primary causes, records co-occurring mechanisms, and gives the final answer to RQ5.

5.6.2 Storage Organisation, Fidelity, and Failure Attribution

This section examines whether S1–S3 preserve the required information, how their organisation changes the access burden, and whether any observed error can be attributed to a store-specific omission. No S1-, S2-, or S3-specific decisive omission was verified in Main-60. The stores differ more clearly in organisation and discoverability than in fact survival. The remaining errors require downstream retrieval, packaging, and utilisation analysis.

The storage audit separates three questions. First, does the required information survive in the declared store? Second, is the stored statement faithful to its source and sufficiently detailed for the benchmark reference? Third, does the representation make the information easier or harder to locate? These questions must remain separate. A source locator can be present while the associated statement is too coarse, and a fully preserved fact can remain difficult to retrieve.

The primary unit is `question x store`: 60 questions evaluated once under each of S1, S2, and S3, giving 180 storage records. Official source locators are available for 58 questions, giving 174 evidence-eligible records. The two questions without source annotations remain in answer-quality analysis but are excluded from locator-based storage analysis. Structural and semantic storage labels were fixed before they were linked to the 720 E1–E6 outcomes. An incorrect answer was therefore not used as evidence that its store was defective.

5.6.2.1 S1 Preserves Content but Leaves Cross-Session Composition

Question. Does S1 preserve the declared memory content, and what evidence-access burden remains?

Finding. S1 is lossless within the declared text-plus-caption renderer, but it leaves a substantial cross-session composition burden.

Seventeen of 58 questions require cross-session evidence, and 12 of 13 eligible Multi-hop questions do so. S1's 30 session files contain 568 turns. All 568 rendered records preserve the declared speaker, text, BLIP caption, source locator, session order, and chronology.

This fidelity claim does not cover every canonical metadata field. The renderer excludes image URLs, image-search queries, and download metadata. Most exclusions do not imply semantic loss because all 99 image-bearing turns retain a BLIP caption. One question, however, requires a precise detail that appears only in an excluded query field and outside the official gold locators. The same rendered input is inherited by S1, S2, S3, E7, and Full Context. This case is therefore a shared source-scope limitation rather than an S1-specific omission.

The main S1 weakness is access structure. Seventeen of the 58 evidence-eligible questions require evidence from more than one session, and 19 require more than one source locator. Cross-session evidence spans a mean of 116.65 turns and 50.53 days. The burden is concentrated in Multi-hop questions: 12 of the 13 eligible Multi-hop questions require cross-session evidence. By contrast, none of the 27 eligible Single-hop questions does so.

Table 5.18. Panel A. S1 evidence-dispersion diagnostics. Rates use the 58 evidence-eligible questions unless a category-specific denominator is shown.

Diagnostic	Result
Questions with multiple gold locators	19/58 (32.76%)
Questions with cross-session gold evidence	17/58 (29.31%)
Cross-session turn span	Mean 116.65; median 103; range 8–256
Cross-session date span	Mean 50.53 days; median 41; range 3–113
Multi-hop questions with cross-session evidence	12/13 (92.31%)
Temporal questions with cross-session evidence	3/13 (23.08%)
Open-domain questions with cross-session evidence	2/5 (40.00%)
Single-hop questions with cross-session evidence	0/27 (0.00%)

These results identify a structural burden, not a storage omission. S1 retains the information, but a retriever may need to traverse distant sessions and combine several observations. Whether that burden produces an error depends on the actual R1 or R2 trace and is analysed in Section 5.6.3.

5.6.2.2 S2 Preserves Content but Provides an Imperfect Routing Cue

Question. Does S2 improve navigation without changing the underlying memory content?

Finding. S2 is content-equivalent to S1 and can narrow navigation, but its folder labels are imperfect routing cues.

Oracle-known folders would reduce the mean candidate set by 57.36%, but 18 of 79 local answer-bearing session units carry a misleading folder cue. All 30 S2 files are byte-identical to their S1 counterparts. The two stores contain the same 568 turns, the same 125 captions, and the same 114,456 bytes. Any S2 effect must therefore arise from directory organisation rather than from a changed session body.

The four folders provide useful structural concentration. Fifty of the 58 eligible questions have all gold-bearing sessions in one folder. If the correct folder were known in advance, the mean candidate set would fall from 30 sessions to 12.79, a theoretical reduction of 57.36%. This is an oracle concentration measure, not observed retriever performance.

The routing structure is also imperfect because one complete session is assigned to one dominant topic even when it contains several local topics. The outcome-blind review examined 79 `question x gold-bearing session` units. Fifty-five units have an aligned folder label, six are partially aligned, and 18 are misleading for the local fact. At the question level, 17 of 58 questions contain at least one misleadingly located gold-bearing session.

Table 5.18 (continued). Panel B. S2 semantic-routing audit. Session-unit labels evaluate whether the folder name is a useful cue for the local answer-bearing content; they do not measure whether R1 actually used that cue.

Session-unit state	Units	Unit rate	Question-level aggregation	Questions	Question rate
Aligned	55	69.62%	All gold-bearing sessions aligned	36	62.07%
Partially aligned	6	7.59%	Partial alignment, no misleading session	5	8.62%
Misleading	18	22.78%	At least one misleading session	17	29.31%

The direct navigation contrast, E3 minus E1, does not show a monotonic relation between routing quality and answer change. Qwen has small positive point estimates under both fully aligned and misleading placements, whereas DeepSeek has small negative estimates. The pattern changes with the backbone and does not become consistently worse in the misleading group.

Table 5.18 (continued). Panel C. Descriptive E3-minus-E1 outcomes by S2 routing class. Positive values favour S2+R1 over S1+R1. Strict W/T/L counts use the target perspective.

Backbone	S2 routing class	N	Delta Strict	Strict W/T/L	Delta Soft	Delta F1
Qwen	All aligned	36	+0.028	3/31/2	+0.028	+0.060
Qwen	Partial, no misleading session	5	0.000	0/5/0	0.000	+0.005
Qwen	At least one misleading session	17	+0.059	2/14/1	+0.029	+0.031
DeepSeek	All aligned	36	-0.056	0/34/2	-0.069	-0.018
DeepSeek	Partial, no misleading session	5	0.000	0/5/0	0.000	+0.008
DeepSeek	At least one misleading session	17	-0.059	1/14/2	-0.029	+0.021

S2 is therefore content-lossless but not a semantically reliable index for every local fact. Its directories create a potential navigation shortcut and a potential misrouting risk. Neither effect is causal at this stage because the storage audit does not show whether R1 followed, ignored, or recovered from the folder cue. E4 minus E2 is not treated as a folder effect because R2 uses the same raw exchange index and does not consume the S2 directory path.

5.6.2.3 S3 Preserves Provenance but Does Not Guarantee Answer-Sufficient Facts

Question. Does S3 preserve source meaning while converting distributed evidence into answer-sufficient memory?

Finding. S3 is an expanded, source-faithful reorganisation that improves thematic co-location, but it does not guarantee one answer-ready fact.

S3 is 2.662 times the size of S1, contains all 68 Main-60 gold locators, and shows no curator-introduced decisive loss. However, only two of 19 multi-locator questions place all gold locators in one fact unit, and benchmark-reference support is sufficient for 50 of 58 eligible questions.

S3 contains two files, 35 topic groups, and 390 fact units. Its total size is 304,640 bytes. Agent-Curated Memory is an expanded, topic-organised representation with repeated provenance.

Provenance coverage is high. S3 contains all 68 unique Main-60 gold locators, so all 58 evidence-eligible questions are covered at the locator level. This does not mean that each Multi-hop answer has been collapsed into one fact. Among the 19 questions with multiple gold locators, only two place all locators in one fact unit. Fifteen place them in one topic group, and all 19 place them in one of the two S3 files. S3 usually narrows navigation to a common topic region, while leaving the final composition task to retrieval and answering.

The semantic audit uses two independent axes. Source-to-store fidelity asks whether S3 faithfully represents its declared source. Reference-support sufficiency asks whether that source is detailed and unambiguous enough to support the benchmark answer. Table 5.18, Panel D shows why these axes cannot be merged.

Table 5.18 (continued). Panel D. S3 source fidelity and benchmark-reference support. Rates use the 58 evidence-eligible questions.

Audit axis	State	Questions	Rate
Source-to-store fidelity	Faithful to source	51	87.93%
Source-to-store fidelity	Compressed but source-faithful	5	8.62%
Source-to-store fidelity	Source conflict faithfully preserved	1	1.72%
Source-to-store fidelity	Shared renderer detail loss	1	1.72%
Source-to-store fidelity	Curator-introduced decisive loss	0	0.00%
Reference support	Sufficient	50	86.21%
Reference support	Partial	4	6.90%
Reference support	Available but conflicting	1	1.72%
Reference support	Source/reference entity mismatch	3	5.17%

No curator-introduced decisive omission or distortion was found. Three of the four partial-support cases faithfully preserve an under-specified source, while the fourth inherits the shared renderer limitation. The conflict case preserves competing source accounts rather than creating a new contradiction. These cases should not be counted as failures introduced by S3 curation.

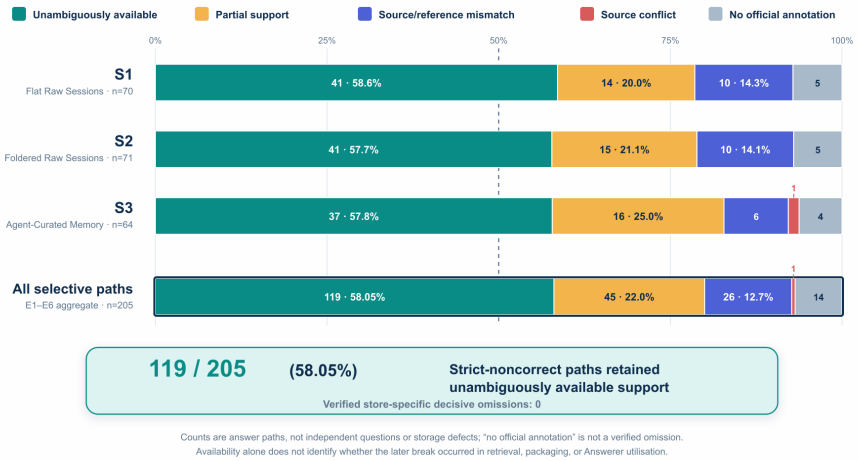


Figure 5.6. Stored-support states across the 205 E1–E6 Strict-noncorrect answer paths. Each 100% bar is normalised within its store; labels report path counts and, where space permits, within-store percentages. The aggregate bar pools S1–S3. Counts are answer paths rather than independent questions or storage defects, and "No official annotation" is not a verified omission. Storage labels were assigned before linkage to answer outcomes.

The answer linkage nevertheless shows that locator coverage is not equivalent to semantic sufficiency. Across E5 and E6, the four partial-support questions produce 16 backbone-condition answer paths, and none is Strict-correct. Some of these paths still have high locator-based Evidence Recall. In contrast, the 50 sufficiently supported questions achieve substantially higher Strict Accuracy under both retrieval systems and both backbones.

Table 5.19. E5 and E6 outcomes conditional on S3 reference support. Evidence Recall is locator-based. The Partial rows contain four questions per backbone-condition cell. The table reports the 50 Sufficient and four Partial questions; the one Available-but-conflicting case and three Source/reference entity-mismatch cases are excluded because they do not define a comparable support stratum.

Backbone	Condition	Reference support	N	Strict	Soft	F1	Evidence Recall
Qwen	E5	Sufficient	50	0.860	0.880	0.424	0.913
Qwen	E6	Sufficient	50	0.840	0.860	0.419	0.860
DeepSeek	E5	Sufficient	50	0.820	0.860	0.272	0.957
DeepSeek	E6	Sufficient	50	0.740	0.770	0.233	0.742
Qwen	E5	Partial	4	0.000	0.250	0.082	1.000
Qwen	E6	Partial	4	0.000	0.125	0.115	0.750
DeepSeek	E5	Partial	4	0.000	0.250	0.081	1.000
DeepSeek	E6	Partial	4	0.000	0.125	0.088	0.750

The association in Table 5.19 is not a curator-effect estimate. All four partial-support gaps originate in the source, reference relationship, or shared renderer rather than in S3 curation. The result instead establishes a narrower point: source-ID coverage can remain high when the available statement lacks the entity, relation, time, or granularity required by the reference answer.

5.6.2.4 Storage Omission Is Not the Dominant Observed Bottleneck

Question. Are most E1–E6 Strict-noncorrect paths explained by missing information in the declared store?

Finding. No. Storage omission is not the dominant observed bottleneck: 119 of the 205 error paths (58.05%) retain unambiguous support in the declared store.

The full error set comprises 70 S1 paths, 71 S2 paths, and 64 S3 paths. Forty-one S1 errors, 41 S2 errors, and 37 S3 errors occur with unambiguous store support. These cases require downstream retrieval, packaging, and utilisation analysis.

Table 5.20. Storage state of the E1–E6 Strict-noncorrect paths. Counts are answer paths, not independent storage defects; the same question can contribute more than one condition or backbone path.

Store	Strict-noncorrect paths	Unambiguously available	Partial support	Source/reference mismatch	Conflict	No gold evidence
S1	70	41	14	10	0	5
S2	71	41	15	10	0	5
S3	64	37	16	6	1	4

Figure 5.6 shows how the 205 selective-path errors are distributed across the audited storage states.

Only one question has a directly verified representation insufficiency, and it is shared by every tested path because it originates in the common rendered input. It appears in 12 of the 205 selective error paths, but these are repeated exposures to one upstream limitation rather than 12 separate storage defects. Moreover, one E7 path answers the question correctly despite the missing precise fact. The limitation is real, but its causal sufficiency for every affected answer is not established.

Section conclusion. The stores differ mainly in how they organise and expose information. S1 preserves raw content but leaves cross-session composition. S2 adds a potentially useful but imperfect navigation layer. S3 improves thematic co-location without guaranteeing answer-ready semantic sufficiency. Because no store-specific decisive omission was verified and 119 of 205 error paths retain unambiguous support, Section 5.6.3 examines retrieval, evidence retention, and EvidenceBundle construction as the next candidate bottlenecks.

5.6.3 Retrieval Bottlenecks and Task-Specific System Choice

This section asks two related questions. First, do most remaining errors arise because retrieval fails to deliver the annotated evidence? Second, which storage–retrieval pairing is most suitable for each question type? The answer is conditional. Measured annotated-locator loss is not the dominant error state across the complete selective-retrieval system, but it remains important for particular retrievers and question structures. R1 usually moves more cases into downstream review with complete annotated evidence. R2 trades evidence coverage for much lower cost. R3 reaches annotation-bearing curated groups reliably but does not always expose or retain every official locator. These differences motivate a task-specific choice rather than one universal retriever.

The analysis covers all 720 E1–E6 retrieval paths. Of these, 696 paths correspond to the 58 questions with official evidence annotations. The E1–E6 error set contains 205 Strict-noncorrect paths, of which 191 are evidence-eligible. E7 is excluded because it reuses the E2 and E6 EvidenceBundles without running a new retriever. Full Context is excluded because it has no selective retrieval stage.

5.6.3.1 Where Does the Pipeline Actually Fail?

Question. Are most evidence-eligible errors caused by retrieval omission?

Finding. No. Of the 191 evidence-eligible Strict errors, 111 contain every annotated locator in the final EvidenceBundle, 73 show retrieval-stage attrition, and seven lose selected evidence at the bundle boundary; the dominant stage nevertheless differs by retriever.

Table 5.21. Location of annotated-evidence loss among evidence-eligible E1–E6 Strict errors. Retrieval-stage attrition includes failure to reach, observe, completely observe, or select the official annotated locators. Bundle-boundary loss means that all annotated locators were selected but did not all survive in the final EvidenceBundle.

Retriever	Eligible Strict errors	Complete annotated bundle	Retrieval-stage attrition	Selected-to-bundle loss
R1	85	68 (80.0%)	17 (20.0%)	0 (0.0%)
R2	72	28 (38.9%)	44 (61.1%)	0 (0.0%)
R3	34	15 (44.1%)	12 (35.3%)	7 (20.6%)
Overall	**191**	**111 (58.1%)**	**73 (38.2%)**	**7 (3.7%)**

Figure 5.7 compares the same annotated-evidence states after grouping the eligible errors by retriever.

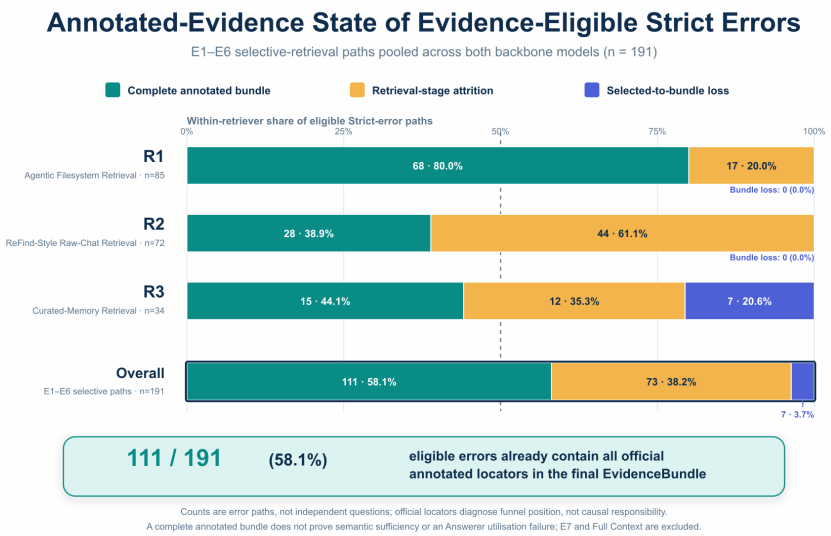


Figure 5.7. Annotated-evidence state of the 191 evidence-eligible E1–E6 Strict-error paths. Each 100% bar is normalised within its retriever, and the aggregate bar pools R1–R3. Retrieval-stage attrition combines failure to reach, observe, completely acquire, or select all official annotated locators. Selected-to-bundle loss means that all annotated locators were selected but did not all survive in the final EvidenceBundle. Counts are error paths rather than independent questions. Official locators diagnose funnel position rather than causal responsibility, and a complete annotated bundle does not by itself establish semantic sufficiency or an Answerer utilisation failure. E7 and Full Context are excluded because neither executes an independent selective retriever in this comparison.

The aggregate result should not be read as evidence that retrieval no longer matters. It shows that annotated-locator attrition is not the most common measured diagnostic state among the remaining errors. For R1, 80.0% of eligible errors already contain a complete annotated bundle, so these cases move into downstream review after evidence acquisition. For R2, 61.1% of eligible errors still contain measurable retrieval attrition. R3 has a mixed profile: 35.3% of its eligible errors lose evidence during retrieval, and a further 20.6% lose selected evidence at the EvidenceBundle boundary.

A complete annotated bundle is not yet proof of an Answerer utilisation failure. LoCoMo locators identify source turns, not a complete reasoning chain. The bundle may still contain ambiguity, competing states, excessive surrounding text, or insufficiently explicit relations. The correct conclusion is therefore narrower: the majority of eligible errors require downstream analysis of evidence sufficiency, packaging, reasoning, and Answerer utilisation. Section 5.6.4 separates these possibilities.

5.6.3.2 Why Do R1, R2, and R3 Fail Differently?

Question. Do the three retrievers fail at the same stage and offer the same quality–cost profile?

Finding. No. R1 provides the strongest annotated-evidence funnel at the highest retrieval cost; R2 is much cheaper but loses more evidence before complete observation; R3 reaches relevant curated groups but is less reliable at locator exposure and final retention.

Table 5.22. Aggregate retrieval, answer-quality, and cost profiles. Evidence metrics use evidence-eligible paths. Strict Accuracy uses all paths. Retrieval tokens are descriptive means per executed path; native filesystem observations and ranked-search actions are not interchangeable units. These are descriptive profiles across the conditions assigned to each retriever, not controlled estimates of a retriever-only effect.

Retriever	Eligible paths	Annotated source reached	Observed all	Final All-hit	Evidence Recall	Strict Accuracy	Mean retrieval tokens
R1	348	100.0%	92.8%	**83.6%**	**0.875**	**74.2%**	220,205.7
R2	232	92.7%	75.0%	73.7%	0.782	68.3%	**14,050.2**
R3	116	100.0%	82.8%	73.3%	0.781	70.0%	30,830.6

R1: high coverage moves more cases into downstream review. R1 reaches an annotated source on every eligible path and obtains the highest Final All-hit rate and Evidence Recall. Its main measurable retrieval loss occurs after observation, when the agent decides which notes to retain. No R1 path loses an already selected annotated locator at the final bundle boundary. Among R1 errors, 68 of 85 already contain a complete annotated bundle. This pattern indicates that simply adding more search rounds is unlikely to resolve most remaining R1 errors. More promising targets are evidence selection, evidence organisation, and the Answerer's use of temporal, entity, and multi-source relations. The strength of R1 comes with substantial cost: its mean retrieval-token use is approximately 15.7 times that of R2.

The relation between All-hit and correctness supports this interpretation. For R1, Strict Accuracy is 76.6% when all annotated evidence is present and 70.2% when it is not. The small gap does not make annotated evidence unimportant. It is consistent with some answers using partial or unannotated support, but it does not identify the source of that support. It also shows that R1 can still fail after retrieving every official locator.

R2: low cost is obtained by accepting a retrieval bottleneck. R2 uses only 14,050.2 retrieval tokens per path on average. It is therefore the most efficient retriever in this comparison. Its lower cost is accompanied by lower complete-evidence coverage. Across all eligible R2 paths, 17 do not reach an annotated session, another 17 reach a relevant session without exposing an annotated locator, and 24 expose only part of the required evidence. The fixed ranked result and local expansion can therefore locate an annotation-bearing session without exposing the official annotated turn or every annotated part of a multi-source answer.

For R2, complete evidence acquisition is strongly associated with answer correctness. Strict Accuracy is 83.6% under All-hit but only 27.9% without All-hit. In addition, 44 of 72 eligible R2 errors contain retrieval-stage attrition. R2's main opportunity is therefore better query formulation, ranking, session fusion, or context expansion rather than a larger Answerer prompt alone.

R3: topic-group reach does not guarantee annotated-locator delivery. R3 reaches an annotation-bearing S3 topic group on every eligible path, yet complete observation falls to 82.8% and Final All-hit to 73.3%. This gap shows that reaching an annotation-bearing thematic region is easier than exposing every official locator inside the returned local window. R3 also has the clearest final-bundle pressure: seven of its 34 eligible errors select all annotated evidence but lose part of it before the final EvidenceBundle.

R3 does not outperform R1 on the same Agent-Curated Memory. Relative to E5, E6 reduces Evidence Recall by 0.089 for Qwen and 0.237 for DeepSeek. The corresponding Final All-hit transitions contain no repairs and seven harms for Qwen, and one repair and fifteen harms for DeepSeek. Strict Accuracy is 82.4% with All-hit and 38.7% without it. These results locate R3's measured losses at annotated-locator exposure and evidence retention rather than topic-group reach alone.

Across R1–R3, 359 paths reach their retriever-specific round or action cap, and 234 of them remain Strict-correct. This evidence is sufficient to show that a cap hit is a process state rather than an error label. It does not estimate what the same paths would have done under a higher cap. The caps also govern different actions and should not be compared as if they measured the same amount of work.

5.6.3.3 Which Storage–Retrieval Pairing Fits Each Question Type?

Question. Should all question types use the same storage–retrieval combination?

Finding. No. S3+R1 provides the strongest stable observed coverage for Multi-hop and Temporal questions, whereas raw-chat R2 is a lower-cost candidate for Single-hop questions. Open-domain evidence is too limited and partly external to support a strong default.

Table 5.23. Task-specific evidence-access recommendations. Final All-hit and Evidence Recall use the evidence-eligible denominator. These are descriptive coverage and retrieval-cost recommendations, not final end-to-end system rankings.

Question type	Evidence-access choice	Qwen evidence result	DeepSeek evidence result	Interpretation
Multi-hop	**E5: S3+R1**	All-hit 10/13; Recall 0.833	All-hit 10/13; Recall 0.872	E5 provides the strongest stable observed coverage for distributed annotated sources.
Temporal	**E5: S3+R1**	All-hit 13/13; Recall 1.000	All-hit 13/13; Recall 1.000	E5 provides complete official-locator coverage; downstream sufficiency and use remain separate questions.
Single-hop	**E2: S1+R2 for efficiency; E5 for maximum coverage**	E2: 25/27, 0.926; E5: 26/27, 0.963	E2: 26/27, 0.963; E5: 27/27, 1.000	R2 achieves high observed coverage for lexically anchored facts; E5 adds little coverage relative to its much higher cost.
Open-domain	**No strong default; E5 for dialogue-evidence coverage**	All-hit 3/5; Recall 0.767	All-hit 4/5; Recall 0.900	Dialogue retrieval alone may not supply the required external knowledge; only five questions per backbone are evidence-eligible.

Multi-hop questions show the strongest stable coverage under E5. Twelve of the 13 evidence-eligible Multi-hop questions draw support from more than one session. E5 produces the strongest stable observed profile: both backbones obtain Final All-hit on 10 of 13 questions, and both reach Strict Accuracy of 0.769. The combination of thematic co-location and iterative browsing is a plausible explanation because it can bring distributed sources into one search process, but the experiment does not isolate those mechanisms from each other. R2 is less suitable in the observed runs because a short lexical query and fixed neighbourhood may expose only one part of the annotated chain.

Temporal questions require high-recall access to updates, followed by correct temporal reasoning. E5 reaches every annotated locator for all 13 Temporal questions under both backbones. Its Evidence Recall is 1.000 in both cases. Raw-session R1 conditions also achieve high temporal coverage, so S1+R1 remains a reasonable lower-construction alternative. Within the Temporal subset, E3 minus E1 changes Evidence Recall by -0.038 for Qwen but +0.038 for DeepSeek. Directory navigation should therefore not be treated as a backbone-independent temporal solution.

Perfect official-locator coverage does not guarantee a correct Temporal answer. E5 Strict Accuracy is 0.769 for Qwen and 0.692 for DeepSeek despite perfect annotated coverage. These failures therefore enter downstream review of semantic sufficiency, temporal ordering, state updates, and Answerer utilisation. Temporal performance illustrates the difference between retrieving an annotated event and establishing that its time relation was used correctly.

Single-hop questions make R2 a strong low-cost candidate. E2 obtains Final All-hit on 25 of 27 Single-hop questions for Qwen and 26 of 27 for DeepSeek. E5 raises these values only to 26 and 27, respectively. On this category, E2 uses a mean of 14,594 retrieval tokens for Qwen and 12,937 for DeepSeek, compared with 245,372 and 309,595 for E5. A Single-hop question often contains a distinctive entity, activity, object, or location that supports direct lexical retrieval.

The small evidence-coverage gap does not imply equal final-answer quality. For Qwen, Strict Accuracy rises from 0.741 under E2 to 0.852 under E5, a difference of three questions. For DeepSeek, both conditions reach 0.852. E2 should therefore be described as the simpler low-retrieval-cost candidate under a coverage–cost objective, not as an unconditional end-to-end winner. RQ6 evaluates the complete quality–cost decision.

E4 provides another R2 result, but it does not establish a benefit from S2. R2 uses the same raw exchange index in E2 and E4 and does not consume the S2 folder hierarchy. If a separate curated store is not already required, E2 is therefore the simpler operational choice. If S3 already exists and maximum evidence coverage is more important than retrieval cost, E5 remains the coverage-first option.

Open-domain questions remain retrieval-limited only in part. E5 gives the highest dialogue-evidence coverage in this small group, but higher locator coverage does not produce a stable semantic advantage. Some questions require a judgement that combines dialogue evidence with outside knowledge. Moreover, only five Open-domain questions per backbone have official evidence annotations. E5 can be described as the coverage-first pairing, but the data do not justify a universal deployment recommendation for this category.

The category results also clarify the role of storage–retrieval matching. S2 is potentially useful only when the retriever consumes its navigation structure, and even then its effect is backbone-dependent. R2 gains no structural benefit from S2 because it searches the same raw exchange index. On S3, R1 is consistently more successful than R3 at converting curated organisation into complete final annotated evidence. Storage structure therefore creates an opportunity, not an automatic gain; the retriever must exploit that structure without losing official locator coverage.

5.6.3.4 Section Synthesis

Section conclusion. Three conclusions answer the retrieval component of RQ5.

First, retrieval omission is not the dominant measured system-wide state among the remaining selective-path errors. A complete annotated EvidenceBundle is already present in 111 of 191 evidence-eligible Strict errors. More cases therefore require downstream review than exhibit measured annotated-locator attrition. This statement is strongest for R1. It does not apply uniformly to R2, for which retrieval-stage attrition still appears in 61.1% of eligible errors.

Second, no single retriever is best for every question structure. R2 is a low-retrieval-cost candidate for Single-hop questions under a coverage–cost objective. S3+R1 is the strongest stable pairing for Multi-hop and Temporal evidence access. Open-domain performance depends partly on information outside the memory store and remains underdetermined by the current sample.

Third, storage value depends on retrieval compatibility. S2 does not provide a stable R1 gain and is not consumed by R2. S3 provides the best evidence-access results when paired with R1, but its adapted ranked retriever R3 loses more annotated-locator coverage and faces greater final-bundle pressure. E5 is therefore the strongest general coverage configuration in these runs. E2 is the simplest low-retrieval-cost candidate for simple factual questions under a coverage–cost objective; the final end-to-end recommendation is deferred to RQ6.

These conclusions localise the next analysis step. More eligible errors enter downstream review with complete official locators than exhibit measured locator attrition. Measuring retrieval recall again cannot distinguish the remaining mechanisms. Section 5.6.4 examines whether the evidence is semantically sufficient, whether packaging introduces ambiguity or noise, and whether the Answerer uses the available evidence correctly.

5.6.4 Evidence Conversion, Answerer Utilisation, Fusion, and Full Context

This section follows the evidence after retrieval. It asks whether the final input is semantically sufficient, whether packaging removes relevant content, whether the shared Answerer uses sufficient evidence correctly, and which failures remain when selective retrieval is removed. The primary selective-path population contains the 205 E1–E6 Strict-noncorrect answer paths. Of these, 191 have official evidence locators. These are error-conditioned populations and should not be interpreted as error rates over all 720 E1–E6 paths. E7 is analysed separately through 120 matched E2/E6/E7 triples, including 31 E7 Strict-noncorrect paths. Full Context contributes 60 unique answers, of which 14 are Strict-noncorrect.

5.6.4.1 Evidence Sufficiency and Packaging

Question. Do selective-path errors mainly arise from insufficient final input or from relevant evidence being lost during packaging?

Finding. Insufficient final input is common, whereas demonstrated packaging loss is limited. Official locator coverage is informative, but it does not establish semantic sufficiency.

Of the 205 Strict-noncorrect paths, 88 (42.9%) contain sufficient final input, 111 (54.1%) contain insufficient input, and six (2.9%) remain indeterminate. The sufficient group contains 16 complete-and-focused bundles and 72 complete-but-noisy bundles. The insufficient group contains 70 partial bundles and 41 irrelevant or empty bundles. The pattern is similar across backbones: 43 of 99 Qwen errors and 45 of 106 DeepSeek errors have sufficient final input.

Table 5.24. Panel A. Final-input sufficiency among E1–E6 Strict-noncorrect paths. Shares use the 205-path error-conditioned denominator.

Final-input judgement	Qwen	DeepSeek	Total	Share
Sufficient	43	45	88	42.9%

Continued from the previous page

Final-input judgement	Qwen	DeepSeek	Total	Share
Insufficient	52	59	111	54.1%
Indeterminate	4	2	6	2.9%
Strict-noncorrect paths	**99**	**106**	**205**	**100.0%**

Sufficiency also varies by condition. E1 has sufficient input in 17 of 32 errors, E2 in 14 of 38, E3 in 18 of 33, E4 in 10 of 38, E5 in 17 of 28, and E6 in 12 of 36. These fractions describe the composition of each condition's errors; they are not condition-level success rates.

Official locator coverage is an informative but incomplete proxy for usable evidence. Among the 191 evidence-eligible errors, 111 have an All-hit EvidenceBundle. Only 63 of these 111 bundles are semantically sufficient, while 48 still omit a required relation, time anchor, entity link, or usable context. Conversely, 19 of the 80 non-All-hit paths contain sufficient alternative support, while 61 are insufficient. All-hit is therefore neither a necessary nor a sufficient semantic test. It establishes locator inclusion, not whether the final input expresses the answer-bearing relation clearly enough for the task.

Demonstrated packaging loss is real but small. Nine of the 205 paths (4.4%) lose relevant content during packaging. Six lose the relevant material completely, and three retain it only partially. A further 42 paths (20.5%) show possible noise or ordering pressure, but the evidence does not establish a specific loss. In 148 paths (72.2%), no packaging failure is demonstrated; six remain indeterminate. Seventeen paths contain at least one skipped item, but only nine establish that a skipped item was relevant to the required answer. No EvidenceItem is truncated internally in any of the 205 paths.

Table 5.24 (continued). Panel B. Demonstrated packaging loss by condition. The numerator is the number of Strict-noncorrect paths with directly supported relevant loss at the EvidenceBundle boundary.

Condition	Strict-noncorrect paths	Demonstrated relevant loss	Rate within condition errors
E1	32	1	3.1%
E2	38	1	2.6%
E3	33	2	6.1%
E4	38	1	2.6%
E5	28	0	0.0%
E6	36	4	11.1%
Overall	**205**	**9**	**4.4%**

The concentration in E6 is consistent with the R3 funnel result in Section 5.6.3: R3 can reach a relevant curated region and still fail to expose or retain every required locator. However, even in E6, demonstrated packaging loss explains only four of 36 errors. Across the full selective error set, the larger problems occur before packaging, through insufficient evidence acquisition, or after packaging, through failure to use evidence that survives.

5.6.4.2 Shared Answerer Utilisation

Question. When sufficient evidence reaches the shared Answerer, how often does the remaining error arise from evidence utilisation or unsupported abstention?

Finding. Answerer utilisation becomes a major confirmed downstream bottleneck once the final input is sufficient: 56 of 88 sufficient-input errors contain an explicit Answerer failure.

Answerer attribution is gated on final-input sufficiency. The 111 insufficient and six indeterminate paths are not labelled as Answerer failures because the available input does not support that causal claim. Among the 88 sufficient-input errors, 36 misuse the available evidence and 20 abstain despite adequate support. The other 32 paths are sensitive to the reference or evaluation boundary, or remain indeterminate at answer conversion.

The result is stable at the aggregate backbone level. Qwen has 26 explicit Answerer failures among 43 sufficient-input errors (60.5%), while DeepSeek has 30 among 45 (66.7%). The similarity does not imply identical behaviour. The failure form changes with the question type.

Table 5.25. Answerer outcomes within the 88 sufficient-input selective errors. "Explicit failure" contains evidence misuse and unsupported abstention. The residual column contains reference-sensitive, evaluation-sensitive, and indeterminate conversions.

Question category	Sufficient-input errors	Explicit Answerer failure	Main explicit mechanism	Residual conversion cases
Single-hop	43	20	Evidence misuse	23
Multi-hop	11	5	Evidence misuse	6
Temporal	14	11	Evidence misuse	3
Open-domain	20	20	Unsupported abstention	0
Overall	**88**	**56**	**36 misuse; 20 abstention**	**32**

Temporal errors most often occur after the relevant events are already present. Eleven of 14 sufficient-input Temporal errors misuse the available evidence, which is consistent with failures in ordering, state updates, or selection of the current fact rather than simple source absence. Open-domain errors show a different mechanism: all 20 sufficient-input cases are unsupported abstentions. Here, the Answerer is too cautious or fails to combine dialogue evidence with available background knowledge. Single-hop and Multi-hop cases are less uniform because concise reference answers make some otherwise defensible responses sensitive to answer scope or wording.

This finding does not justify labelling 56 of all 205 errors as the only downstream problem. It establishes a conditional result: when sufficient evidence reaches the Answerer, incorrect utilisation or unsupported abstention is the dominant directly supported failure state. When the final input is insufficient, retrieval and evidence construction remain the earlier bottlenecks.

5.6.4.3 E7 Dual-Source Fusion

Question. Does deterministic fusion convert the complementary E2 and E6 evidence into reliable repairs without introducing new failures?

Finding. E7 provides limited but genuine complementarity, while most outcomes are preserved rather than changed. Repair is uncommon, and an observed Harm outcome does not by itself prove fusion interference.

All 120 E7 paths exactly reproduce the deterministic fusion of their E2 and E6 parents. Every parent item is preserved, the expected interleaving order is maintained, no new retrieval occurs, and no parent item is removed through deduplication or truncation. E7 outcomes can therefore be interpreted without an unobserved third retriever or an implementation-level loss of parent evidence.

Table 5.26. E7 outcome states by backbone. Repair means that E7 is Strict-correct while neither parent is correct. Preserve means that at least one parent is Strict-correct and E7 is also correct. Harm means that at least one parent is Strict-correct while E7 is not. Unresolved means that neither parent nor E7 is correct.

E7 outcome	Qwen	DeepSeek	Total	Share of 120
Preserve	43	42	85	70.8%
Repair	1	3	4	3.3%
Harm	4	4	8	6.7%
Unresolved	12	11	23	19.2%

The four repairs are concentrated in composition-heavy questions: two occur in Multi-hop and two in Temporal. One Temporal case provides direct evidence that complementary parent information enables the correct answer. One Multi-hop case recovers the answer without a uniquely demonstrated fusion mechanism. The remaining two repairs cannot be assigned a single causal explanation because the parent-sufficiency states conflict. No repair occurs in Single-hop or Open-domain questions. The observed value of dual-source fusion is therefore selective rather than universal.

The error-side evidence also limits the interpretation of Harm. Among the 31 E7 errors, 17 fused inputs are semantically sufficient and 14 are insufficient. Nine contain explicit Answerer failures: six unsupported abstentions and three evidence-misuse cases. Of the eight Harm cases, two have insufficient fused input, two have explicit Answerer failures, and four are sensitive to the reference or evaluation threshold. Harm cannot therefore be treated automatically as fusion noise. Some failures arise because the combined input still lacks a required relation; others arise after sufficient fused evidence has already reached the Answerer.

E7 demonstrates that E2 and E6 are not fully redundant, but it does not achieve parent-oracle behaviour. It can use complementary raw-dialogue and curated-memory evidence to repair a small number of Multi-hop and Temporal cases. It does not guarantee that E7 will be correct whenever either parent contains useful information, and the current deterministic interleaving does not decide which parent should dominate when the two sources compete.

5.6.4.4 Full Context and Selective-Focus Diagnostics

Question. Which failures remain when the Answerer receives the complete rendered history, and do evidence position or context length explain them?

Finding. Full Context bypasses selective retrieval but does not remove localisation, evidence-utilisation, or Answerer failures. The position results do not support a generic lost-in-the-middle account, and selective retrieval retains an independent focusing benefit.

Full Context supplies all 30 rendered raw sessions to the shared Answerer: 568 turns and 108,960 evidence characters, with no retriever and no 16,000-character EvidenceBundle limit. It remains subject to the source scope and rendering choices shared by the other conditions. It achieves 46 correct answers out of 60, giving Strict Accuracy of 0.7667 and Soft Semantic Score of 0.8250. Its macro Precision is 0.1680, Recall is 0.7296, and F1 is 0.2445.

Table 5.27. Full Context results by question category. All metrics use the complete 60-question test set and are reported on the 0–1 scale.

Question category	Correct / N	Strict Accuracy	Soft	Precision	Recall	F1
Single-hop	22/27	0.8148	0.8889	0.1876	0.7992	0.2868
Multi-hop	11/13	0.8462	0.8846	0.1520	0.7026	0.2366
Temporal	10/13	0.7692	0.8077	0.2166	0.6590	0.2636
Open-domain	3/7	0.4286	0.5000	0.0320	0.6429	0.0608
Overall	**46/60**	**0.7667**	**0.8250**	**0.1680**	**0.7296**	**0.2445**

Nine of the 14 Full Context errors occur despite sufficient complete-history input. Five misuse available evidence, and four abstain despite adequate support. The other five are ambiguity-limited at this downstream stage because the source, question, reference, or task boundary does not support a clean model-failure label; the integrated analysis in Section 5.6.6 assigns them to the evaluation-sensitive primary category. A separate long-context view identifies three missed-relevant-content cases, two distractor-confusion cases, one temporal-confusion case, and three cases in which the relevant material is located but the answer still fails. These views are complementary rather than additive: one describes how the long input is processed, while the other describes the final Answerer decision.

Open-domain is the clearest weak point. Four of seven Full Context answers are Strict-noncorrect, and three of these errors combine sufficient input with unsupported abstention. Complete dialogue visibility cannot solve a calibration problem in which the model declines to use the evidence or the background knowledge allowed by the task. More generally, Full Context replaces selective omission with a different burden: the Answerer must locate, compare, and compose facts across the full history without an explicit evidence filter.

The answer profile is consistent with this burden. Correct Full Context answers contain 20.78 normalised tokens on average, compared with 31.86 for Strict-noncorrect answers. Correct answers have Precision 0.2031, Recall 0.8414, F1 0.2929, and Soft 1.000. Error answers have Precision 0.0529, Recall 0.3624, F1 0.0857, and Soft 0.250. The error forms comprise five under-answers, two over-answers that still miss the required relation, one incorrect temporal or state selection, one fine-grained attribute omission, and five ambiguity-limited cases later classified as evaluation-sensitive. More text can therefore coexist with omission of the target entity, relation, state, or attribute. This is a descriptive post-outcome association; it does not establish that length itself causes failure.

Evidence position does not provide a general explanation for these errors. The analysis uses 58 questions because two Open-domain questions have no official source locator. Position is defined by the zero-based global turn ordinal divided by 567. It is not a token or character position. The midrange is the midpoint between the earliest and latest annotated turns, not the mean of all source positions.

Table 5.28. Full Context evidence position by Strict outcome. Positions are normalised global turn ordinals on the 0–1 scale.

Full Context outcome	N	Mean earliest position	Mean latest position	Mean midrange position
Strict-correct	45	0.5115	0.5846	0.5480
Strict-noncorrect	13	0.4029	0.4193	0.4111

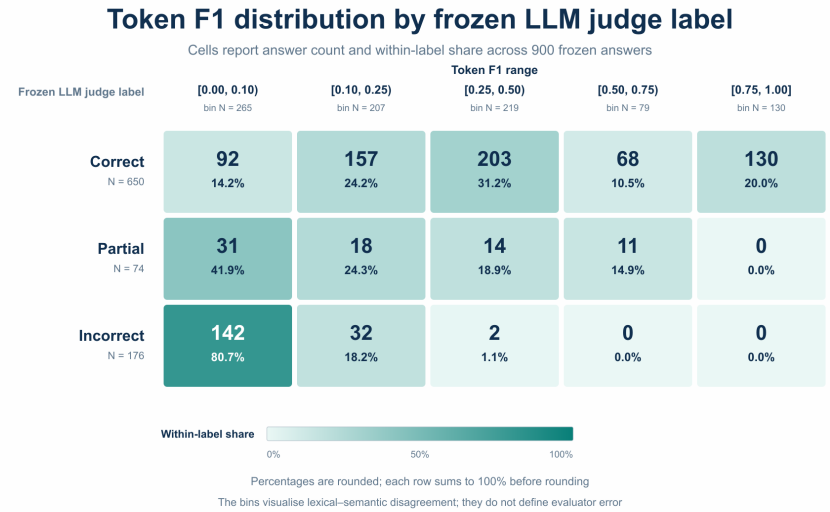


Figure 5.8. Token F1 distribution within each frozen LLM judge label across 900 answers. Cell shading represents the within-label share, cell annotations report the answer count and row percentage, and the column headers report the total number of answers in each F1 interval. Rows contain 650 Correct, 74 Partial, and 176 Incorrect answers. Bins are left-closed and right-open, except the final bin, which includes 1.00. Percentages are rounded to one decimal place. The figure visualises lexical–semantic disagreement; the bins do not define evaluator error.

The annotated evidence for the errors occurs earlier on average than the evidence for the correct answers. The experiment therefore provides no support for one general claim that middle-position evidence explains Full Context failure. This result is descriptive, comes from one fixed conversation, and does not exclude position effects within individual categories or cases.

Selective retrieval also solves cases that Full Context misses. In the oracle-style `any-selective-correct` view, 46 questions are correct under both Full Context and at least one selective condition, none is correct only under Full Context, six are correct only under at least one selective condition, and eight remain incorrect under both. This oracle is not deployable because it chooses among seven selective systems after observing their outcomes. It nevertheless shows that selective retrieval can add value beyond token reduction by suppressing competing facts and focusing the Answerer on a smaller evidence set. In the deployable pairwise comparisons, Full Context has between three and eight exclusive successes against each individual E1–E7 condition, so the oracle result should not be interpreted as universal dominance by one selective system.

Section conclusion. The results identify two material bottleneck families. First, evidence acquisition and conversion remain incomplete: 117 of 205 selective errors do not have demonstrated sufficient final input, comprising 111 insufficient and six indeterminate paths, and nine paths have directly supported packaging loss. Second, sufficient evidence does not guarantee a correct answer: 56 of 88 sufficient-input selective errors contain an explicit Answerer failure, while nine of 14 Full Context errors occur despite sufficient complete-history input. E7 shows genuine but infrequent complementarity, with four repairs against eight harms and 23 unresolved cases. Full Context removes the selective-retrieval stage, but it retains localisation, competition, temporal reasoning, composition, and calibration failures, and remains exposed to source and task ambiguity.

The evidence therefore does not support one universal downstream explanation. Retrieval quality and evidence sufficiency remain important, while Answerer utilisation becomes the main confirmed failure state once sufficient evidence is available. The independent primary-cause decomposition in Section 5.6.6 reconciles these stages without double-counting the same error.

5.6.5 Evaluation Disagreement and Metric Sensitivity

Token F1 and the LLM judge observe different properties of an answer. Token F1 measures lexical overlap with a short canonical reference. The LLM judge evaluates semantic task completion against the reference. This section asks why the two evaluations disagree and whether those disagreements change the architectural conclusions. It analyses all 900 frozen answers: 420 Qwen E1–E7 answers, 420 DeepSeek E1–E7 answers, and 60 DeepSeek-family Full Context answers. No answer is rescored, and the official evaluation remains unchanged.

5.6.5.1 Do Token F1 and Semantic Evaluation Measure the Same Outcome?

Question. Are the lexical and semantic evaluations sufficiently aligned to support one common interpretation?

Finding. They are positively associated but not interchangeable. Higher semantic labels correspond to higher F1 on average, yet low F1 does not imply that an answer is semantically wrong.

Table 5.29. Token-score distributions by frozen LLM judge label. Means and medians use answer-level scores over the complete 900-answer population.

LLM judge label	N	Mean Precision	Mean Recall	Mean F1	Median F1
Incorrect	176	0.0253	0.2339	0.0427	0.0000
Partial	74	0.1483	0.4824	0.2060	0.1483
Correct	650	0.3386	0.8317	0.4145	0.3333
Overall	**900**	**0.2617**	**0.6860**	**0.3247**	**0.2222**

Figure 5.8 shows the complete cross-tabulation behind the distributional comparison. Each row is normalised within one frozen LLM judge label. Cell annotations report the exact answer count and the within-label percentage, while the column headers report the number of answers in each Token F1 interval.

The Pearson correlation is 0.4576 between Token F1 and Strict Accuracy and 0.4737 between Token F1 and the Soft Semantic Score. The association is therefore meaningful but moderate. The asymmetry is important: 92 answers labelled Correct by the LLM judge have F1 below 0.10, whereas no answer labelled Incorrect reaches F1 0.50. Token F1 is more likely to under-credit a semantically correct answer than to assign a high score to a semantically incorrect one.

5.6.5.2 Why Do Semantically Correct Answers Receive Very Low F1?

Question. Do Correct/low-F1 cases reflect an overly permissive LLM judge, or does lexical scoring miss valid answer forms?

Finding. Lexical under-credit is the dominant explanation. Of the 92 Correct answers below F1 0.10, 87 (94.6%) are semantically correct answers whose wording is poorly represented by token overlap. The other five are concentrated in reference- or boundary-sensitive cases.

Table 5.30. Reviewed explanations for the four diagnostic score tails. Thresholds select cases for diagnosis; they do not define evaluator error. Repeated condition-level cases were reviewed as question families before row counts were reported.

Diagnostic set	Set N	Reviewed explanation	Rows	Share within set
Correct, F1 < 0.10	92	Supported elaboration around a short reference	41	44.6%
		Date or temporal normalisation	24	26.1%
		Semantic paraphrase or morphology	22	23.9%
		Reference- or boundary-sensitive	5	5.4%
Partial, F1 < 0.10	31	Genuine incomplete or non-responsive answer	18	58.1%
		Reference- or boundary-sensitive	13	41.9%
Partial, F1 >= 0.50	11	Genuine incomplete answer	4	36.4%
		Reference-scope or granularity-sensitive	7	63.6%
Incorrect, F1 >= 0.10	34	Lexical over-credit	18	52.9%
		Source- or reference-relation sensitive	16	47.1%

The largest under-credit mechanism is supported elaboration. A short reference may contain only a fact, date, item, or yes/no decision, while the candidate states the same answer and explains it. The extra supported text raises the Precision denominator without making the answer less correct. Date normalisation creates a second failure mode: an ISO date, a natural-language date, and a relative-time expression can denote the same time while sharing few or no normalised tokens. The third mechanism is semantic paraphrase, where synonyms or morphological variants preserve the answer without reproducing the reference wording.

Answer and reference length support this explanation. Across the 92 Correct/low-F1 rows, the candidate contains a median of 22 normalised tokens, compared with three in the reference; the median length ratio is 6.0. DeepSeek contributes 50 of these cases, Qwen contributes 36, and Full Context contributes six. Their median answer lengths are 25.5, 19, and 25 tokens, respectively. This pattern is consistent with DeepSeek receiving a larger lexical penalty because it more often gives an explanatory answer. It does not establish answer length as an independent cause, because verbosity was not experimentally controlled.

5.6.5.3 Why Can Partial or Incorrect Answers Retain Token Credit?

Question. Does non-trivial or high Token F1 guarantee that the required answer is complete and correct?

Finding. No. Token overlap can preserve credit when a required component, predicate, entity, time relation, polarity, or final decision is wrong.

The 42 reviewed Partial extremes contain 22 genuinely incomplete answers. These answers omit a required list item, relation, date component, attribute, or explicit conclusion. Four of them still reach F1 0.50 because they reproduce most of a short reference while missing one required part. The other 20 Partial extremes are sensitive to reference scope or decision boundaries. Typical cases include a plural question with a non-exhaustive reference, a candidate that adds source-supported specificity, and a relative-time expression that does not map exactly to the reference's calendar convention.

Among the 34 Incorrect answers with F1 at least 0.10, 18 are clear lexical over-credit. Thirteen repeat the reference option inside an abstention: the correct term is present, but the answer explicitly refuses the required conclusion. The remaining cases include an unresolved geographic inference, a wrong relation, a wrong count, an incorrect polarity, and incidental keyword overlap. The other 16 cases expose a source–reference entity, predicate, or temporal conflict. Token F1 cannot represent these relational distinctions. Nevertheless, over-credit remains bounded in this experiment: the maximum F1 among answers labelled Incorrect is 0.4444, and none reaches 0.50.

These cases also limit the interpretation of Evidence Recall. Gold source locators identify where annotated support appears; they are not a complete semantic rubric. A candidate may use alternative valid support, while an All-hit EvidenceBundle may still fail to express the required relation clearly. Evaluation disagreement therefore cannot be resolved by substituting Evidence Recall for answer-level assessment.

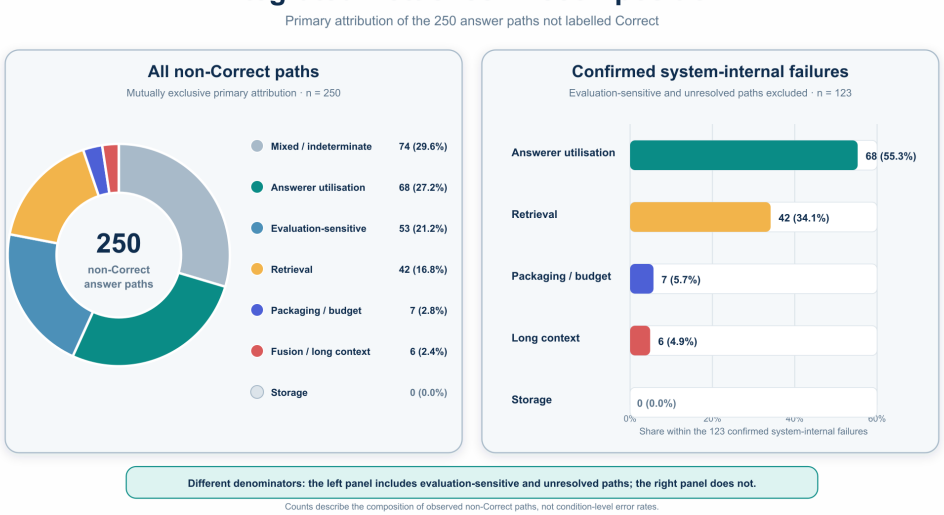


Figure 5.9. Integrated primary-cause decomposition of the 250 answer paths not labelled Correct. The left panel includes all mutually exclusive primary attributions. The right panel restricts the denominator to the 123 confirmed system-internal failures by excluding evaluation-sensitive and mixed-or-indeterminate paths. Counts describe the composition of observed non-Correct paths, not condition-level error rates.

5.6.5.4 How Sensitive Is Semantic Judging Near the Decision Boundary?

Question. Is metric disagreement mainly caused by unstable LLM judge labels?

Finding. No broad instability is observed, although a small number of near-boundary cases require caution.

The dataset contains 80 exact duplicate-input groups covering 305 LLM judge records. None has a label conflict. This supports exact-input consistency within the frozen run, but it does not establish stability for semantically similar answers. The clearest boundary sensitivity is concentrated in two question families and 18 reviewed rows. One concerns relative-week interpretation; the other concerns whether omission of a secondary qualifier should separate Correct from Partial. The LLM judge's reported confidence is not treated as a calibrated probability, and one frozen run does not support an inter-judge reliability claim.

5.6.5.5 Do Evaluation-Sensitive Cases Change the Architectural Conclusions?

Question. Are the main storage, retrieval, fusion, and Full Context findings artifacts of a small number of evaluation-sensitive questions?

Finding. The central conclusions remain stable, but two small semantic margins should remain qualified.

A conservative stress test removes all nine questions flagged as evaluation-sensitive from every principal paired comparison. The denominator falls from 60 to 51. This is a diagnostic exclusion, not a corrected score. Token F1 direction is preserved in all 17 checked contrasts, and Strict Accuracy direction is preserved in 15.

The main architecture findings survive. E4−E3 and E6−E5 remain negative in both Token F1 and Strict Accuracy for both backbones. The E5−E3 lexical–semantic trade-off also remains: Token F1 decreases while Strict Accuracy increases under both backbones. DeepSeek E7 remains better than both E2 and E6 on F1 and Strict Accuracy. Qwen E7 remains lexically weaker than its parents while retaining a more favourable semantic profile. The backbone-dependent E3−E1 result also keeps its direction: positive under Qwen and negative under DeepSeek.

Two small Strict differences are sensitive to the exclusion. Qwen E7−E6 changes from a tie to a small positive difference, while Full Context−E5 changes from a one-question positive margin over 60 questions to a one-question negative margin over the remaining 51. The latter comparison cannot support a strong claim that Full Context is semantically better than E5.

Section conclusion. Token F1 and the LLM judge provide complementary evidence. Token F1 is useful for comparison with LoCoMo and for measuring lexical precision and recall, but it systematically under-credits some correct dates, paraphrases, and explanatory answers. The LLM judge better captures semantic completion in those cases, while remaining sensitive to a small number of reference and decision-boundary ambiguities. These disagreements do not overturn the principal architectural conclusions. They explain why Chapter 5 reports lexical and semantic metrics separately and qualifies conclusions that depend on one-question semantic margins.

5.6.6 Integrated Bottleneck Decomposition and Answer to RQ5

The preceding sections examined storage, retrieval, evidence packaging, answer generation, and evaluation separately. This section combines those findings into one mutually exclusive decomposition of the 250 answers that were not labelled Correct by the frozen LLM judge. Each path is assigned to the earliest stage at which the trace provides sufficient evidence of failure. Secondary mechanisms are then recorded separately, because one answer may exhibit more than one weakness.

5.6.6.1 Primary Root-Cause Distribution

Question. Which stage is the primary bottleneck after storage contents, retrieval traces, final EvidenceBundles, candidate answers, and evaluation records are considered together?

Finding. Among resolved system-internal cases, Answerer utilisation is the largest confirmed bottleneck, followed by retrieval. However, the overall error population also contains a substantial mixed or indeterminate group and a separate evaluation-sensitive group.

The attribution applies a source–reference consistency guard before declaring retrieval failure. If the question or reference conflicts with the annotated source, failure to retrieve that locator is not sufficient evidence of a retrieval defect. Such cases are treated as evaluation-sensitive unless another earlier break is independently supported. This prevents annotation conflicts from being counted as system failures.

Table 5.31. Panel A. Mutually exclusive primary attribution of the 250 non-Correct answer paths. Percentages use all 250 paths as the denominator.

Primary root cause	N	Share
Storage omission or distortion	0	0.0%
Retrieval failure	42	16.8%
Packaging or evidence-budget loss	7	2.8%
Answerer utilisation failure	68	27.2%
Fusion or long-context failure	6	2.4%
Evaluation-sensitive	53	21.2%
Mixed or indeterminate	74	29.6%
Total	**250**	**100.0%**

Figure 5.9 visualises both the complete 250-path attribution and the 123 confirmed system-internal failures.

Of the 250 paths, 176 receive a resolved primary label, while 55 remain mixed and 19 remain indeterminate. Within the 176 resolved paths, Answerer failure accounts for 38.6%, evaluation sensitivity for 30.1%, and retrieval failure for 23.9%. If evaluation-sensitive cases are excluded to isolate confirmed system-internal failures, 123 paths remain: Answerer utilisation accounts for 68 (55.3%), retrieval for 42 (34.1%), packaging for seven (5.7%), and long-context processing for six (4.9%).

The evidence-strength register makes this separation explicit. The 123 confirmed system-internal paths comprise 81 direct attributions and 42 convergent retrieval attributions. The 53 evaluation-sensitive paths require bounded manual review, while the remaining 74 paths retain indeterminate attribution strength. The figure therefore does not treat reviewed evaluation ambiguity or unresolved mechanisms as confirmed component failures.

The distribution is architecture-dependent. The 205 selective-retrieval errors contain 41 retrieval, seven packaging, 56 Answerer, 39 evaluation-sensitive, and 62 mixed or indeterminate cases. The 31 E7 errors contain one retrieval, nine Answerer, nine evaluation-sensitive, and 12 mixed cases. The 14 Full Context errors contain three Answerer failures, six long-context failures, and five evaluation-sensitive cases. All six cases in the combined fusion-or-long-context category are Full Context long-context failures; no E7 path receives fusion as its primary cause. Together, the three Answerer and six long-context cases are the nine Full Context errors with sufficient complete-history input identified in Section 5.6.4; the remaining five are evaluation-sensitive. Full Context therefore bypasses the selective-retrieval stage but does not remove evidence localisation, cross-event composition, or answer-generation failure.

Retriever-level profiles further separate the mechanisms. Among selective-retrieval error paths, R1 has 93 cases: 34 Answerer, three packaging, 19 evaluation-sensitive, and 37 mixed or indeterminate. R2 has 76 cases, including 33 primary retrieval failures. R3 has 36 cases, including eight retrieval, three packaging, and eight Answerer failures. These are compositions of observed error paths, not condition-level error rates. In particular, the absence of a clean R1 primary retrieval case does not prove that R1 never misses evidence; it means that no R1 miss remained uniquely attributable after the source-consistency guard and competing explanations were applied.

Interpretation. The main confirmed bottleneck is not a universal retrieval problem. R2 is distinctly retrieval-heavy, whereas R1 errors concentrate downstream or remain unresolved. R3 distributes errors across retrieval, packaging, and utilisation. No path is assigned to a verified storage omission or distortion, but this does not establish that all memory representations are lossless. It only means that storage loss was not the earliest supported cause among these 250 paths.

5.6.6.2 Co-occurring Mechanisms

Question. Which secondary mechanisms commonly accompany the primary failures?

Finding. Retrieval attrition, Answerer failure, and evaluation sensitivity are the most frequent co-occurring mechanisms. Possible evidence competition and ordering recur across 42 paths, although they cannot always be isolated as the primary cause.

Table 5.31 (continued). Panel B. Non-exclusive mechanisms observed across the 250 non-Correct paths. A path may appear in several rows; shares therefore do not sum to 100%.

Co-occurring mechanism	N	Share
Annotated-evidence retrieval attrition	76	30.4%
Demonstrated packaging or budget loss	9	3.6%
Possible evidence noise or ordering pressure	42	16.8%
Explicit Answerer utilisation failure	74	29.6%
Long-context localisation or composition failure	6	2.4%
Explicit evaluation sensitivity	74	29.6%
Shared-representation insufficiency	12	4.8%
E7 harm with exact parent-item preservation	8	3.2%

The non-exclusive view explains why a single-stage label is insufficient. For example, nine paths show demonstrated packaging loss, but only seven receive packaging as the primary label because retrieval had already failed in two paths. Likewise, eight E7 harm cases preserve every parent evidence item. Their degradation therefore cannot be explained by item loss during fusion; the added evidence more plausibly changed the Answerer's selection, weighting, or confidence. This is evidence of interference under a larger bundle, not proof of a defective fusion operation.

Interpretation. Primary attribution identifies the earliest supported break, while co-occurring labels retain downstream effects. The two views are complementary: the first avoids double-counting, and the second shows that retrieval, evidence presentation, answer generation, and evaluation can interact within the same failed path.

5.6.6.3 Representative Cases and Final Answer to RQ5

Question. Do individual traces support the aggregate decomposition, and what is the direct answer to RQ5?

Finding. Reviewed traces confirm distinct failure signatures for R1, R2, R3, E7, and Full Context. They also show why source–reference conflicts must be separated from system failures.

Table 5.32. Representative trace-reviewed cases. The descriptions report the diagnostic mechanism without reproducing question or answer text.

Path	Case	Trace evidence	Primary interpretation
R1 / Qwen E1	q037	The final bundle is focused, contains the complete annotated evidence, and uses only 373 characters, but the answer returns only part of the required date.	Answerer utilisation failure after sufficient evidence
R2 / Qwen E2	q022	BM25 query and session ranking do not reach either annotated session before the four-action cap; the 9,896-character bundle has Evidence Recall 0.	Retrieval failure caused by query/ranking and the action limit
R3 / Qwen E6	q052	The trace finds and saves the key fact, but three selected items are removed by the evidence budget; the final 14,584-character bundle has Evidence Recall 0.5.	Packaging and budget loss after successful discovery
R1 / Qwen E1	q089	The question and reference target one entity, while the annotated source describes another.	Evaluation-sensitive source–reference mismatch, not a clean retrieval defect
E7 / Qwen	q051	E2 is Correct, but E7 is not; E7 preserves all parent items, has Evidence Recall 1, and expands the bundle to 17,515 characters.	Additional evidence changes Answerer selection despite no fusion loss
Full Context / DeepSeek	q019	All 30 sessions are visible, but the answer combines only one of two relevant events distributed across three sessions.	Long-context localisation and cross-event composition failure

Interpretation. RQ5 has a conditional rather than a single-stage answer. Once evaluation-sensitive and unresolved cases are separated, the largest confirmed internal bottleneck is the Answerer's use of available evidence. Retrieval is the second-largest bottleneck and is concentrated in R2, with a smaller but clear contribution from R3. Packaging loss is less frequent but directly observable, especially when relevant R3 items exceed the common evidence budget. E7 can exploit complementary evidence, but additional preserved evidence can also change the Answerer's decision. Full Context removes selective retrieval and packaging, yet introduces its own localisation and cross-event composition burden.

The final conclusion is therefore that failure location depends on the evidence-access architecture. Selective retrieval remains useful for focused evidence access, but better retrieval alone cannot resolve most confirmed downstream failures. Improving long-term memory QA requires both stronger evidence acquisition and more reliable evidence utilisation, while evaluation-sensitive and mixed-or-indeterminate cases must remain explicit rather than being forced into a system component.

5.7 RQ6: Quality, Cost, and Practical System Choice

RQ6 asks which system should be used when answer quality, construction cost, online query cost, and operational reliability cannot all be maximised at once. The analysis combines the frozen Main-60 quality results with the recorded construction, successful-query, and recovery costs. Token usage is the primary cost proxy. Model-call count is reported separately because one long-context call is not equivalent to one short call. LLM judge usage is an evaluation-only cost and is excluded from deployment comparisons. Comparable latency and monetary prices are unavailable, so no latency or currency ranking is claimed.

E7 requires a special accounting rule. Its online cost is reconstructed as E2 retrieval plus E6 retrieval plus one fused Answerer call. The two parent Answerer calls are excluded because they are not part of a deployed E7 request. Construction, successful-query, and recovery costs remain separate throughout the analysis.

5.7.1 Is the Cheapest System Also the Best System?

Question. Does the condition with the lowest online token cost also provide the strongest answer quality?

Finding. No. E2 is the lowest-token condition under both backbone models, but this saving is accompanied by lower semantic accuracy than the stronger R1 or fused alternatives.

Table 5.33. Panel A. Qwen answer quality and successful-query cost. Quality metrics use all 60 questions. Query cost includes retrieval and the shared Answerer, but excludes construction, recovery attempts, and LLM judge usage. E7 uses reconstructed online cost.

Condition	Token F1	Strict Accuracy	Soft Semantic Score	Mean query tokens	Mean model calls
E1	0.395	0.717	0.767	148,920	12.50
E2	0.403	0.667	0.717	**18,419**	4.93
E3	**0.441**	0.750	0.792	140,288	11.67
E4	0.396	0.700	0.725	18,479	4.98
E5	0.395	**0.783**	**0.817**	281,354	8.05
E6	0.381	0.733	0.758	36,378	**4.83**
E7	0.369	0.733	0.758	53,637	8.77

Table 5.33 (continued). Panel B. DeepSeek answer quality and successful-query cost. Full Context is included as a same-family diagnostic. It used `deepseek-v4.1-flash-1017`, whereas E1–E7 used `deepseek-v4.1-flash`; it is therefore not an exact served-model ablation.

Condition	Token F1	Strict Accuracy	Soft Semantic Score	Mean query tokens	Mean model calls
E1	**0.301**	**0.750**	**0.808**	201,030	12.98
E2	0.246	0.700	0.708	**17,313**	5.00
E3	0.297	0.700	0.758	189,646	12.70
E4	0.231	0.667	0.708	17,458	5.00
E5	0.260	**0.750**	0.800	374,700	8.90
E6	0.224	0.667	0.708	35,457	5.00
E7	0.285	**0.750**	0.800	51,581	9.00
Full Context*	0.245	**0.767**	**0.825**	36,646	**1.00**

E2 uses 87.6% fewer query tokens than E1 under Qwen and 91.4% fewer under DeepSeek. However, its Strict Accuracy is lower by 0.050 under both backbones, and its Soft Semantic Score is lower by 0.050 under Qwen and 0.100 under DeepSeek. R2 therefore provides a clear cost-first option, not a cost-free replacement for stronger evidence-access paths.

Interpretation. Online efficiency and answer quality form a trade-off. E2 is appropriate when query cost is the binding constraint and its accuracy is sufficient. A quality-sensitive application should instead choose a higher point on the quality–cost frontier.

5.7.2 Which System Provides the Best Quality–Cost Balance?

Question. Which non-dominated condition gives the most useful balance for each backbone model?

Finding. The best balance is backbone-dependent. E3 provides the strongest general Qwen balance, while E7 provides the strongest DeepSeek selective balance when S3 already exists or can be reused at sufficient scale.

Table 5.34. Pareto-efficient conditions using mean successful-query tokens as the cost axis. Conditions are ordered from lower to higher token cost. A condition appears on a frontier when no cheaper condition is at least as good on the named quality metric. Full Context is shown only in the diagnostic same-family scope.

Scope	Strict Accuracy frontier	Soft Semantic Score frontier	Token F1 frontier
Qwen, E1–E7	E2 -> E4 -> E6 -> E3 -> E5	E2 -> E4 -> E6 -> E3 -> E5	E2 -> E3
DeepSeek, E1–E7	E2 -> E7	E2 -> E7 -> E1	E2 -> E7 -> E3 -> E1
DeepSeek family, with Full Context*	E2 -> Full Context	E2 -> Full Context	E2 -> E7 -> E3 -> E1

Figure 5.10 visualises the Soft Semantic Score frontier. The horizontal axis uses a logarithmic scale because successful-query cost spans more than one order of magnitude. Solid points are Pareto-efficient within the stated panel, while pale points are dominated. The Full Context point is shown separately because it belongs to a same-family diagnostic comparison rather than an exact served-model ablation.

Qwen. E3 improves on E1 by 0.046 in Token F1, 0.033 in Strict Accuracy, and 0.025 in Soft Semantic Score. It also uses 5.8% fewer successful-query tokens. E3 is therefore a stronger balance than the flat-session R1 baseline. E5 reaches the highest Qwen Strict Accuracy (0.783) and Soft Semantic Score (0.817), but it uses about twice the query tokens of E3 and reduces Token F1 by 0.046. E5 is a semantic-quality maximum, not the default efficient choice.

DeepSeek. E7 matches E1's 0.750 Strict Accuracy, remains only 0.008 lower in Soft Semantic Score and 0.016 lower in Token F1, and uses 74.3% fewer successful-query tokens. Relative to E2, E7 gains 0.050 Strict Accuracy, 0.092 Soft Semantic Score, and 0.040 Token F1, while using 2.98 times as many query tokens. E7 is therefore a meaningful quality step above the cost floor. It is not a free improvement, and its recommendation assumes that the S3 asset is already available or sufficiently reused. If no curated store exists, E1 remains the simpler no-build DeepSeek quality baseline.

Interpretation. One global ranking would hide the main result. The useful Qwen path is an inexpensive structural intervention: S2 improves R1 without changing the raw session contents. The useful DeepSeek path is a selective fusion strategy: E7 recovers much of E1's quality at a substantially lower online token cost, but depends on the expensive S3 store. Lexical and semantic frontiers also differ, so Token F1 and LLM judge scores should not be collapsed into one utility value.

5.7.3 Do Construction and Recovery Costs Change the Recommendation?

Question. Do one-time store construction and failed-attempt recovery alter the query-time ranking?

Finding. Yes. S2 is inexpensive enough to support E3 after modest reuse. S3 requires far more reuse and never becomes cheaper than E2. R1 also incurred all observed formal recovery attempts in these runs.

Table 5.35. Panel A. One-time storage-construction cost. S1 has no model-based construction cost. Its deterministic normalisation and file-processing cost was not measured and must not be interpreted as zero total system cost.

Store	Representation	Model calls	Tool calls	Total tokens	Deployment implication
S1	Flat raw sessions	0	0	0	No model-based build; raw baseline
S2	Foldered raw sessions	12	62	395,749	Small one-time organisation cost
S3	Agent-Curated Memory	842	1,350	30,864,958	Large one-time curation cost; update cost unmeasured

Table 5.35 (continued). Panel B. Selected token-accounting break-even points against E1. The threshold is the first whole query after cumulative construction plus query tokens fall below E1. It does not assume equal quality.

Backbone	Target condition	Required store	First query after break-even	Quality qualification
Qwen	E3	S2	46	Better than E1 on F1, Strict, and Soft
Qwen	E6	S3	275	Lower F1 and Soft, but higher Strict than E1; not quality-equivalent
Qwen	E7	S3	324	Lower F1, higher Strict and lower Soft than E1
DeepSeek	E3	S2	35	Lower than E1 on F1, Strict, and Soft
DeepSeek	E6	S3	187	Lower than E1 on all three quality metrics
DeepSeek	E7	S3	207	Same Strict; slightly lower F1 and Soft than E1

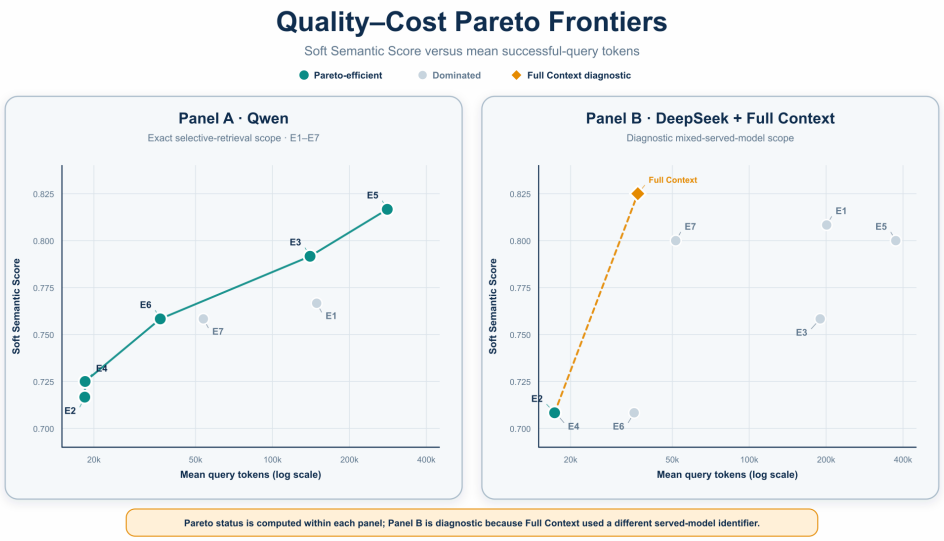


Figure 5.10. Quality–cost Pareto frontiers under Soft Semantic Score. Panel A compares the exact Qwen E1–E7 scope. Panel B adds Full Context to the DeepSeek conditions as a diagnostic mixed-served-model point. Mean query tokens include retrieval and the shared Answerer; they exclude store construction, formal recovery attempts, and LLM judge evaluation. E7 uses reconstructed online cost comprising E2 retrieval, E6 retrieval, and one fused Answerer. Pareto membership is computed separately within each panel.

E6 and E7 never become token-saving alternatives to E2 under the observed query rates. E5 has no token break-even against E1 because both its construction cost and its query cost are higher. The 35-query DeepSeek E3 crossover is also not a recommendation because E3 is lower quality than E1. By contrast, the 46-query Qwen E3 crossover combines lower cumulative token cost with higher observed quality.

Table 5.35 (continued). Panel C. Formal failed-attempt and recovery cost. Recovery tokens are kept separate from successful-query cost. Two Qwen E5 DNS failures have unknown token usage and are not assigned zero cost.

Backbone	Condition	Failed attempts	Usage-known attempts	Usage-unknown attempts	Known recovery tokens
Qwen	E1	1	1	0	87,303
Qwen	E3	5	5	0	702,070
Qwen	E5	12	10	2	2,245,594
DeepSeek	E1	12	12	0	1,973,003
DeepSeek	E3	6	6	0	707,690
DeepSeek	E5	5	5	0	4,410,698
Total		**41**	**39**	**2**	**10,126,358**

All observed formal failures occurred in R1 conditions E1, E3, and E5. This is an observed-run operational result, not proof that R1 is universally unstable. The known 10.13 million recovery tokens are a lower bound because two DNS attempts have no usage record. The result nevertheless strengthens the practical case for R2 or R3 when their lower quality is acceptable.

Interpretation. Query-time tables alone favour systems with prebuilt structure. Cold-start deployment changes that conclusion. S2 has a modest and recoverable build cost. S3 should be treated as a reusable infrastructure investment, not as a free input to E5–E7. Recovery burden also matters: an expensive agentic path can consume substantial resources before it produces the final successful record.

5.7.4 Does Full Context Replace Selective Retrieval?

Question. Does the DeepSeek-family Full Context result justify removing selective retrieval?

Finding. No. Full Context is a strong diagnostic and escalation path, but the current experiment does not establish it as a universal replacement.

Full Context reaches the highest observed DeepSeek-family Strict Accuracy (0.767) and Soft Semantic Score (0.825). It uses 36,646 mean query tokens, which is 29.0% fewer than E7, and requires one model call. However, its Strict advantage over E7 is one question, while its Token F1 is 0.041 lower. The experiment also uses a different served-model identifier and a fixed 30-session context. These constraints prevent a strict claim that complete visibility is superior to retrieval.

RQ5 provides a second reason for caution. Full Context removes retrieval omission, but it does not remove evidence localisation, temporal ordering, cross-event composition, or Answerer utilisation errors. Six of its non-Correct paths were primarily attributed to long-context processing, and three to Answerer utilisation. Complete visibility therefore changes the failure mechanism rather than eliminating failure.

Interpretation. Full Context is most useful when retrieval omission is suspected or when a difficult question requires broad cross-session inspection. It should be used as a targeted diagnostic or recovery path. Selective retrieval remains preferable when provenance, focused evidence, exact-model comparability, or predictable scaling is required.

5.7.5 Practical Selection and Direct Answer to RQ6

Question. Which condition should be selected in practice?

Finding. No condition is best under every objective. The evidence supports conditional selection by backbone, available memory assets, and the application's quality threshold.

Table 5.36. Scenario-based system selection. Recommendations use observed Main-60 quality and cost. They do not imply unmeasured latency or monetary advantages.

Deployment objective	Recommended condition	Evidence	Main qualification
Minimum online token cost	E2 under either backbone	Lowest mean query tokens: 18,419 for Qwen; 17,313 for DeepSeek	Lower semantic quality than stronger R1 or fused paths
Best Qwen quality–cost balance	E3	F1 0.441, Strict 0.750, Soft 0.792; 5.8% fewer query tokens than E1	Requires S2; build amortises after 46 queries against E1
Maximum Qwen semantic quality	E5	Highest Qwen Strict (0.783) and Soft (0.817)	Very high S3 construction, query, and recovery cost
Best DeepSeek selective balance	E7	Matches E1 Strict at 0.750 with 74.3% fewer query tokens	Requires S3 and two online retrieval paths
No-build, provenance-oriented DeepSeek baseline	E1	Strict 0.750, Soft 0.808, F1 0.301; direct raw-source traceability	High query and observed recovery cost
Diagnostic or failure-recovery path	Full Context; E7 when complementary evidence is needed	Complete visibility or dual-source evidence can repair retrieval omissions	Targeted escalation only; Full Context is not an exact-model oracle

The decision rule is therefore compact. Use E2 when online cost dominates. Use E3 as the default Qwen balance once S2 can be amortised. Use E5 only when the highest Qwen semantic score justifies an expensive curated store. For DeepSeek, use E7 when S3 already exists and selective quality matters; otherwise use E1 as the no-build quality baseline. Reserve Full Context for diagnosis or recovery.

This recommendation also follows the bottleneck evidence in RQ5. Among the 123 confirmed system-internal failures, 68 (55.3%) are Answerer-utilisation failures and 42 (34.1%) are retrieval failures. More retrieval work is therefore not automatically the most efficient correction. The final answer to RQ6 is conditional routing by objective, not one global champion.

5.8 Chapter Summary and Consolidated Answers to RQ1–RQ6

This chapter examined how memory representation, retrieval design, backbone model, evidence fusion, and complete-context access affect long-term memory question answering. It also traced observed errors through the information pipeline and compared answer quality with construction and query costs. Table 5.37 consolidates the direct answers to the six research questions.

5.8.1 Consolidated Answers to RQ1–RQ6

Table 5.37. Consolidated answers to RQ1–RQ6.

RQ	Direct answer	Strongest evidence	Main boundary
RQ1	Storage and retrieval must be considered jointly. R1 is the stronger quality-oriented retriever in the tested raw and curated stores. S3+R1 improves semantic quality and evidence access over S2+R1, but lowers Token F1.	E4−E3 is negative on Token F1, Strict Accuracy, and Soft Semantic Score under both backbones. E6−E5 is also negative under both. E5−E3 raises Strict Accuracy and Soft Semantic Score while lowering Token F1 in both pipelines.	No reliable storage–retrieval interaction is established. The preferred raw layout is backbone-dependent, and complete-system contrasts do not isolate one component as the sole cause.
RQ2	Some architectural findings transfer across backbones, but storage effects do not transfer uniformly. R1 remains stronger than R2 on S2 and stronger than R3 on S3, while the S2 benefit under R1 appears only for Qwen.	Three of the six principal effects are directionally stable, two are partly stable, and E3−E1 is backbone-dependent.	The backbone model controls both retrieval and answering. The results compare two end-to-end pipelines rather than independently isolating either component.
RQ3	E7 exploits genuine complementarity between raw-dialogue and curated-memory evidence, especially under DeepSeek, but fusion does not preserve every correct parent outcome.	The fused union reaches Evidence Recall of 0.8994 under Qwen and 0.8951 under DeepSeek. E7 improves all main answer metrics over both parents under DeepSeek, but produces four Harm cases under each backbone.	Fusion effects depend on the backbone model and question type. Category results are descriptive, and the open-domain subgroup is small.
RQ4	Full Context achieves the highest DeepSeek-family semantic means, but it does not dominate selective retrieval on every metric or remove all errors.	Full Context reaches Strict Accuracy of 0.7667 and Soft Semantic Score of 0.8250, only one correct answer above E1, E5, and E7. Fourteen Full Context answers remain non-correct.	Full Context uses a different served-model identifier, has no retrieval-level Evidence Recall, and is evaluated on a fixed 30-session context. It is a diagnostic path rather than an oracle.
RQ5	The main confirmed internal bottleneck is evidence utilisation by the Answerer, followed by retrieval. Failure location nevertheless varies by architecture.	Among 123 confirmed system-internal failures, 68 (55.3%) are attributed to Answerer utilisation, 42 (34.1%) to retrieval, seven (5.7%) to packaging, and six (4.9%) to long-context processing.	Fifty-three paths are evaluation-sensitive and 74 remain mixed or indeterminate. These counts describe the observed non-Correct paths, not general component failure rates.
RQ6	No condition is best under every objective. Practical selection should depend on the backbone model, existing memory assets, quality threshold, and available query budget.	E2 has the lowest online token cost. E3 provides the strongest general Qwen quality–cost balance. E7 provides the strongest DeepSeek selective balance when S3 is available, while E1 remains the simpler no-build quality baseline.	Construction and recovery costs materially affect deployment choice. Comparable monetary prices and latency measurements are unavailable, and Full Context is not an exact served-model ablation.

5.8.2 Overall Synthesis and Transition

Three conclusions connect the six research questions.

First, architecture effects are conditional. No memory representation or retrieval system is uniformly best across backbone models, metrics, and question types. R1 provides the most consistent quality-oriented retrieval, but the preferred raw storage differs by backbone model. Agent-Curated Memory is most useful with R1, while the current R3 implementation does not

realise its intended advantage. Token F1 and semantic evaluation must also remain separate because longer or more explanatory answers can improve semantic correctness while reducing lexical overlap.

Second, evidence access and evidence utilisation are distinct bottlenecks. Better evidence coverage creates an opportunity for a correct answer, but does not guarantee one. E7 increases coverage without preserving every correct parent outcome. Full Context removes selective retrieval without eliminating localisation and composition failures. The root-cause analysis confirms that the Answerer often fails even when sufficient evidence is available. Improving retrieval alone is therefore insufficient.

Third, deployment requires conditional routing rather than one global winner. E2 is appropriate when online cost is the main constraint. E3 is the strongest general Qwen balance once the modest S2 construction cost is amortised. E5 is justified only when its higher Qwen semantic quality warrants the cost of S3. For DeepSeek, E7 is preferable when S3 already exists and selective quality matters; otherwise, E1 is the simpler quality-oriented baseline. Full Context is better treated as a diagnostic or recovery path than as a universal replacement for selective retrieval.

The next chapter interprets these findings in relation to prior work, examines the limitations of the experimental design, and derives broader implications for long-term memory system design.

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